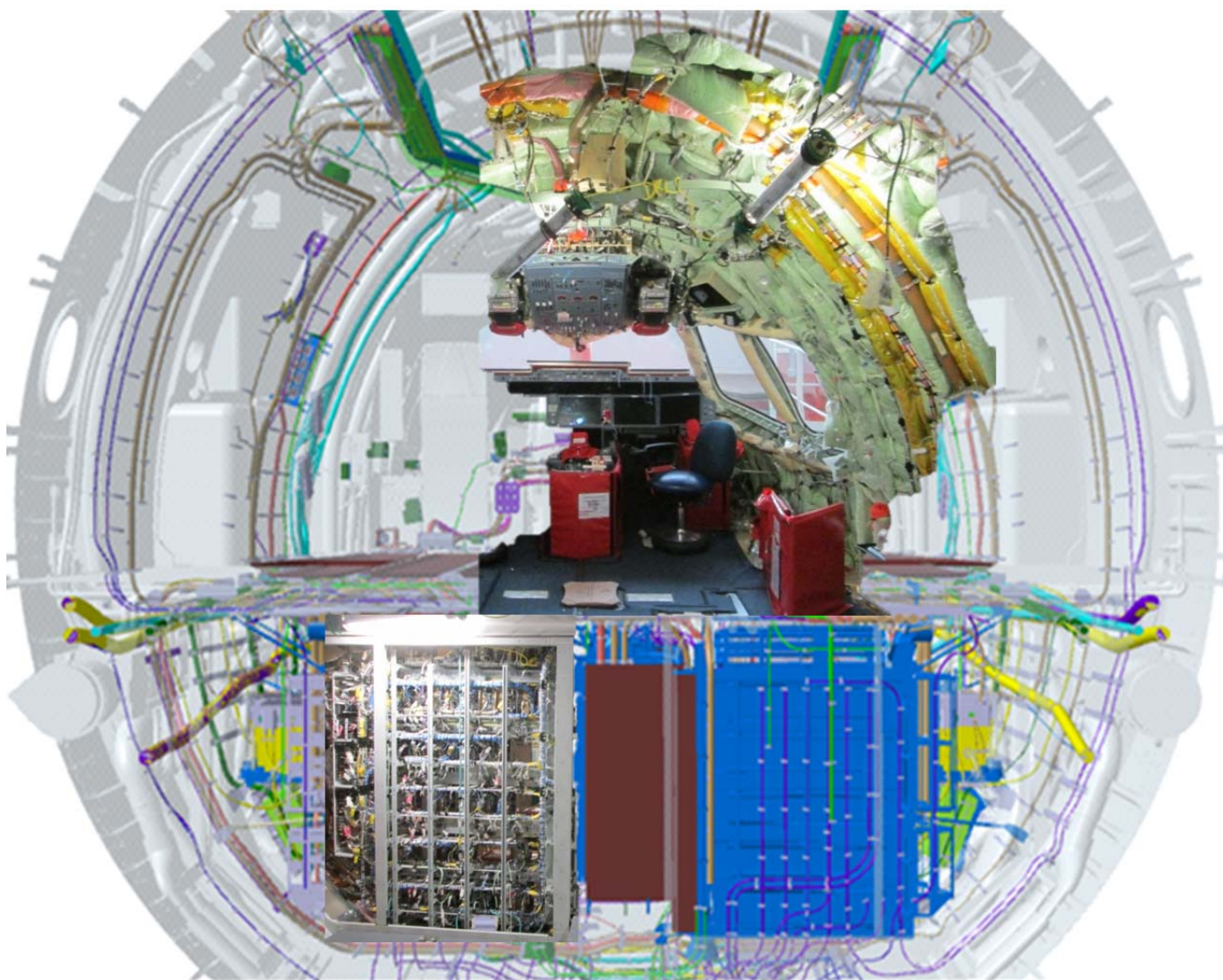


Compendium Electric



A Guide To Understand Electrics

3rd Edition

Published by ESDCY & ESORA



**Technical Authority
Electrics**



Preface & Acknowledgement

During the last years the Electrics Community was influenced by extensive changes of the engineering organization, processes, methods and tools. And even if the Electrics Community has been spread all over the Airbus Design Organization it became a more transnational and well harmonized Team.

The Engineering Airframe Faculty and the Electrics Design Authority are proud of presenting the *3rd edition* of the “Compendium Electric” which has completely been reworked and updated in accordance to the changes the electrics domain is facing to. Furthermore this booklet is the result of an excellent cooperation between the Airbus Design and Airframe Business Management domain.

The “Compendium Electric” is designed to provide comprehensive explanation, background information and guidance on the most relevant processes in order to put the complexity of airplane electrics design into simple terms.

Since the finalization of its first edition in 2007 and its first rework in 2010 the “Compendium Electric” has been distributed a thousand times to all Airbus design offices as printed as well as electronic version. The positive feedback from the readers pushed us to update the document again in order to cover the latest Airbus process and method evolution.

Even if this booklet will not replace any Airbus official procedure it will further support the deployment of common processes at all Airbus sites. As learning guide it facilitates different trainings related to electrics.

This document does not claim to be exhaustive because it will not be updated regularly. Nevertheless it is planned to be further improved with coming editions. Therefore we would like to encourage you to let us know in case you have any comments, questions or suggestions.

Have fun reading the *3rd edition* of the “Compendium Electric”!

Hamburg, June 2014

Peter Bersuch
Thorsten Soltow
Ulrich Thoele

Contents

1	Introduction.....	11
1.1	Abstract / Overview of the Field of Application.....	11
1.2	Selected “Way of Working” in this Document.....	11
1.2.1	Approach/Intended Audience.....	11
1.2.2	Software- /Documentation-Related Issues.....	12
1.2.3	What This Document Doesn’t Cover	13
1.2.4	How This Document Is Organized	13
1.2.5	Recommended Environment for the Reader.....	15
1.3	Aircraft Construction at AIRBUS	15
2	Introducing a Change into Aeronautical Products	17
2.1	Considerations as to the Nature of a “Change”.....	17
2.2	Authority & Quality Management Relations	18
3	Configuration Management (CM)	19
3.1	Embodiment of CM Inside AIRBUS	19
3.2	CM Disciplines	21
3.2.1	“Helper Concept” Basic Approved ATA Breakdown Index (BAABI).....	22
3.3	Sub-Processes of CM	23
3.3.1	P1 – Product Structure.....	24
3.3.2	P2 – AIRBUS Offer Management	27
3.3.3	P3 – Change Process	28
3.3.4	P4 – Baseline Creation (by MSN (Manufacturer Serial Number)).....	28
3.3.5	P5 – P6 – P7 – Configuration Conformity Management and Documentation (CCMD)	28
3.3.6	P8 – CM Steering Process.....	28
4	Development Domain – The Current Development Process.....	29

5	Overview of Authority Documentation & Requirements	32
5.1	European Aviation Safety Agency (EASA).....	32
5.1.1	Document Structure & Naming Conventions	32
5.1.2	Implementing Rules (IR) & Certification Specifications (CS)	32
5.1.3	Notice of Proposed Amendments (NPA) & Comment Response Document (CRD)	33
5.1.4	Acceptable Means of Compliance (AMC) & Guidance Materials (GM).....	33
5.1.5	Airworthiness Directives (AD) & Certification Review Items (CRI).....	33
5.2	Related Authority Organizations	34
5.2.1	Federal Aviation Agency (FAA).....	34
5.2.2	Civil Aviation Agency (CAA).....	35
5.2.3	Other Related Authorities.....	35
5.2.4	International Civil Aviation Organization (ICAO)	35
5.2.5	Airlines for America (A4A).....	36
6	Organizational Requirements for the Company	37
6.1	Organization Approvals.....	37
6.2	Considerations as to Privileges Granted.....	38
6.3	The Modification System & Related Top-Level Documents.....	38
6.4	Overview of the AIRBUS Document Structure	39
6.5	Dossiers Types	41
6.6	Certification vs. Attestation.....	41
7	Approval of Changes to the Type Design.....	43
7.1	Type Certification (TC)	43
7.2	Classification of Changes.....	43
7.3	Minor vs. Major Changes	43
8	Introduction to Responsibilities.....	44

8.1	Compliance with Part 21	44
8.2	Document Type Level	46
8.3	Design Organization Signatories (DOS) Database	47
8.4	External delegation	47
8.5	Design Function	49
8.5.1	Technical Authority (TA).....	49
8.5.2	1 st Independent Technical (Technological Expert) Signatories	49
8.5.3	2 nd Independent Technical (Technological Expert) Signatories	49
8.5.4	Specialists	50
8.6	Airworthiness Function.....	50
8.6.1	Compliance Verification Engineer (CVE)	50
8.6.2	Designated Certification Specialist (DCS).....	51
8.6.3	Designated Airworthiness Engineer (DAE)	51
8.7	Independent Monitoring Function.....	51
8.8	Chief Engineer (CE)	51
9	Begin the Design with the End in Mind.....	52
9.1	Continued Airworthiness	52
9.2	Electrical Wiring Interconnection System (EWIS)	53
9.2.1	EWIS (ICA) Change to Type Design & Continued Airworthiness	54
9.2.2	EWIS Type Certification	55
10	System Design-Related Requirements	56
10.1	“Helper Concept” Functional Item Number (FIN)	56
10.2	Architect's Point of View.....	57
10.3	Requirement-Based Engineering (RBE)	57
10.4	Validation & Verification (V&V).....	58
10.5	Top-Level Requirements Dossier (TLxxRD)	60

10.6	System Definition & System Requirement Dossiers	62
11	Type of Change Processes	63
11.1	Requirement Change Note (RCN) Process	63
11.2	Change Note Process	63
11.3	Full Change Process	63
12	Phase Concepts in the Different Change Processes	65
12.1	Overview	65
12.2	Initialization	65
12.3	Evaluation	65
12.4	Investigation	65
12.5	Implementation	65
12.6	Conclusions Regarding Change Note/Full Change Process	66
13	Origin of a Change	67
13.1	Nature of Changes in the ATA 92/87 Domain	67
13.2	Changes Introduced by the Authorities	67
13.3	Customer-Initiated Changes	67
13.3.1	Request for Change (RFC)	67
13.3.2	Customer Query (CQ)	68
13.4	Changes Initiated within the Organization / Request for Work (RFW) Series	68
14	Change Items & Documents	69
14.1	Change Items	69
14.1.1	Change Request	69
14.1.2	Modification Proposal	69
14.2	Change Documents	70
14.2.1	Repercussion Sheets	70
14.2.1.1	Technical Repercussions (TRS)	70

14.2.1.2	Production & Point of Embodiment (PRS/ERS)	70
14.2.1.3	Costs & Weights (CRS / WRS)	70
14.2.2	Technical Dossier (TD)	71
14.2.3	Modification Approval Sheet (MAS)	71
15	System Design	72
15.1	(Non-)Specific Design Work (NSDW / SDW)	72
15.2	Electric Schematic Diagrams (ESD) System Design	72
15.3	Block Diagrams (BD).....	73
15.4	Functional Diagrams (FD)	73
15.5	Equipment-Related Issues	74
16	Functional Definition	75
16.1.1	Overall Workflow	75
16.2	Electric Schematic Diagrams (ESD) Functional Design.....	76
16.2.1	Principle Diagrams (PD).....	77
16.2.2	Extended Principle Diagrams (EPD)	77
16.2.3	Wiring Diagram (WD)	78
16.3	Release of ESD (PD, EPD, WD)	79
16.3.1	Engineering Change Note (ECN).....	79
16.3.2	Release Process of ESD (PD, EPD, WD).....	79
17	Harness Pre-Definition	81
17.1	Topologies.....	81
17.2	Electrical Routing Harness Concept Dossier (ERHCD)	81
17.3	Referential Concept / Referential in Configuration (RiC)	82
18	Installation Definition	84
18.1	Space Allocation (SAM)	85
18.2	System Frontier Model (SFM) / Geometrical Reference Model (GRM).....	85

18.3	Configured Digital Mock-Up (cDMU).....	86
18.4	ATA92 related definition dossier elements.....	86
18.4.1	539 - Fixed Parts.....	87
18.4.2	921 – Routing Diagrams	87
18.4.3	922 - Routing Drawings.....	87
18.4.4	923 - Equipment.....	88
18.4.5	924 - Moveable Parts	89
18.4.6	925 - Assembly Drawing	91
18.4.7	927 - Final Assembly Line Specific Work.....	92
18.4.8	928 - Installation Drawing.....	93
18.4.9	929 - Definition Drawing.....	94
18.5	Release Process of Installation-Related “Objects”	95
19	Manufacturing Preparation	96
19.1	Treatment Concept	96
19.2	Sketch Phases	97
20	Testing the “Results”	98
20.1	Testing and Certification / Attestation Relevance	98
20.2	Ground Test	98
20.2.1	Ground Test Requirement (GTR).....	98
20.2.2	Ground Test Instruction (GTI)	99
20.3	Flight Test	100
20.3.1	Flight Test Request (FTR).....	100
20.3.2	Flight Test Modification	100
20.4	Special Conditions	101
20.4.1	Special Instruction (SI)	101
20.4.2	Permit to Fly	103

21	Correction Process of Changes	106
21.1	Introduction to Non-Conformity (NC).....	106
21.2	Changing the “Source” of NC (Design Query Notes (DQN)).....	107
21.3	Request for Design Rework (RDR)	109
21.4	Handling the “Results” of NC (Concessions)	110
21.5	Non-Compliance Process and Deviation to TDD92 / SIDP92.....	113
22	Delivery Preparation	115
22.1	Certification (MOD Closure)	115
22.2	Attestation-Related Documents	119
22.2.1	Constituent Assemblies - CA (EASA Form 1)	119
22.2.2	Aircraft Statement of Conformity (EASA Form 52).....	119
22.2.3	Release to Service (EASA Form 53).....	120
22.3	Handling of Requests during Handover	121
23	Configuration Conformity Management & Documentation	122
23.1	Definition of Outstanding Work	122
23.2	“Delta” Calculation.....	122
24	Aging Aircraft / In-Service.....	124
24.1	Service Bulletins (SB)	124
24.2	Technical Adaptations (TA).....	124
25	Appendix	126
25.1	Extract of ATA chapter (based on A350)	126
25.2	Reference Documents	127
25.3	Use of Electrics Technical Notes, Technical Reports and Technical Memos	131
25.4	Simplified Electric Process covering a “Change” with related Responsibilities, Documents & Levels.....	132
25.5	Requirements for Electrics Technical Signatories.....	133

26	Directories	134
26.1	Figures	134
26.2	Tables	136
26.3	Abbreviation and Acronyms	136

1 Introduction

1.1 Abstract / Overview of the Field of Application

The goal of this document is to give a brief overview of introducing a change into the “Electric Domain,”.

Therefore the “change” itself will be the subject of investigations and we will accompany the change all the way from the origin, the first idea, up to the end, the full embodiment of the change before delivery of the A/C at the Delivery Center.

The introduction of the “change” inside the organization and the A/C has to fulfill official requirements. These are, as a minimum, the EASA (European Aviation Safety Agency) / FAA (Federal Aviation Administration) requirements and the significant requirements of the Authority regarding the country of final A/C registration.

The requirements from the Authority are normally given on a high level of functional abstraction. For example, the CS (Certification Specification) paragraph 25.601 given by the EASA states:

“The aero plane may not have design features or details that experience has shown to be hazardous or unreliable. The suitability of each questionable design detail and part must be established by tests.”

The above paragraph for the CS means, in short, AIRBUS is not allowed to implement (electrical) systems, e.g. electrical cabling without tests.

This leads to the term “Airworthiness” [FLORIO2006; Chapter 2; Page 3 ff.] which means¹: “... the applicable regulations, aiming to define the minimum applicable safety (safety of the aircraft, of the transported passengers and of the territories flown over) level of the aircraft:

Designed and built according to applicable requirements

When operated within its intended environment and within its quantified and declared limitations

Maintained in accordance with procedures acceptable to the responsible Authority.”

The common procedure for handling a change through AIRBUS will be the focus of the following chapter.

1.2 Selected “Way of Working” in this Document

1.2.1 Approach/Intended Audience

In this document a System Theory/Black Box approach is used. The following chapters will use a high level of description/abstraction and describe main processes and process steps.

There will be no loops or iterations in the focus of this documentation, only the “normal flow,” the basic, will be the subject of discussion.

For this more or less normal flow it is important to note two entry points into the process flow.

¹ PURWINS, MICHAEL: Speech: “Product Integrity – JAA / EASA – MOD Closure,” October 2008.

One point is the start of the development of a new program. The other entry point into the process flow is the introduction of a change into the serial process after the end of the development. This is called the Fulfill Domain as opposed to the Development Domain (refer to AIRBUS Quality Manual (A01)).

While this sounds easy to understand, the difficulty is that these processes behave as follows:

The Development and Fulfill Domains work in parallel for one program,

The Fulfill processes of several programs work in parallel with impacts on changes which may possibly be relevant for several programs,

If a change is introduced to the Fulfill Domain, the investigations and impacts from this can lead to evaluations in an area related to the Development Domain.

Any detailed information beyond this point is outside this scope and further information will refer either to the Authorities (e.g. EASA), to AIRBUS documents or other parts within this document.

The processes described in this document are the “to-be” process based on the EASA and the experience from the “as-is” basis.

Many people carrying out different tasks at different levels are involved. The goal of this document is, on the one hand, to give a brief overview of the process to the holder of Airworthiness Signature, e.g. DAE (Designated Airworthiness Engineer), CVE (Compliance Verification Engineer) and to the holder of Technical Signature, for example the TAP (Technical Approver) or Technical Checker. On the other hand, every person, working or management level, should receive a good overview of the processes involved in the change of “aircraft-related objects” in the electric domain.

1.2.2 Software- /Documentation-Related Issues

The software tools are extremely program-dependent. Furthermore there is no common grade of realization regarding the requirements from the official documentation. On the other hand, the software is “only” the tool to implement the processes given in the procedural documentation stated in the AIRBUS Quality Manual (refer to A01).

Details regarding IT (Information Technology) Systems will not be featured in the scope of this guide; only a short overview of their characteristics and the related process steps will take place.

There will be no “tool training” as the focus here will be on the process part. Therefore the focus will be on the process side and not on the tool side.

Further on we will NOT deal with issues INSIDE the wires, i.e. we will not talk about Hardware (refer to DO-160) or Software² (refer to DO-178b³) issues. Also we don't care about signal types inside the wire and therefore we will stick almost entirely to OSI (Open Systems Interconnection) Layer 1, the Physical Layer.

This leads to the view that we are focusing on two elements:

² LADSTAETTER, GERNOT (HO Aircraft In-Service Security): Speech: “Security Impacts on Safety – Current Situation, Requirements and Initiatives,” presented by Product Security.

³ RIERSON, KLEANNA: “Using the Software Capability Maturity Model for Certification Projects,” FAA, Washington, D.C., 1998.

The documentation of the processes, the AIRBUS documentation system (refer to AIRBUS Quality Manual (A01))

The documentation of the contents of the processes itself (e.g. the Technical Repercussion Sheet (TRS) or the related Modification Approval Sheet (MAS)) which will be explained further on.

It is important to realize that the requirements given by the Authorities are mandatory for every change inside our organization and that this information is also specified in the related change document, the TRS (Technical Repercussion Sheet) that is considered a certification-relevant document.

This leads to the necessity for establishing a documentation system that must be synchronized with the Authorities. This task will mainly be the subject of the Design Assurance System (DAS) listed in the Design Organization Manual (DOM, refer to A20). The DOM is approved by the EASA as mentioned on the cover page of the DOM.

Therefore, based on Authority requirements, AIRBUS maintains a documentation system, which is also a mandatory requirement.

It is not sufficient to implement “only” the EASA requirements, we must also be aware that the AIRBUS documentation is mandatory.

The definition and description of the documentation system can be found in the “AIRBUS Quality Manual” (refer to A01) and “Manage Procedural Documentation” (refer to A5000) documents.

Consequently there are requirements for writing procedural documentations defined in “Requirements for Procedural Documentation” (refer to A1000) and in the document “Procedural Document Writing Rules & Recommendations” (refer to M2900).

1.2.3 What This Document Doesn't Cover

The document covers no detail data and overviews of the deeper involved areas in electric issues.

Specific procedures, possibly with special site related components as well as detailed examples are not addressed in this document.

All deeper information regarding the topics mentioned in this document should be retrieved mainly from:

MyDoc, the only official source of all (process) documents

ZAMIZ, the one and only long-term archive and source of all Design Data Sets, NC, DQN etc.

The “Faculties” providing deeper training upon special details

The different teams providing deeper knowledge regarding electrics

The community of all Design, Production and Maintenance Organization members of AIRBUS. Especially “In-Service” with the exhaustive experience retrieved from flying A/C.

This document is not updated on a “on-demand” or cyclic (e.g. weekly) basis. This means also that EVERY reference to documents given in this document MUST be checked for the latest issues.

1.2.4 How This Document Is Organized

This document is divided into 24 main chapters, organized by function and clustered in the following main parts introducing the field of application and the main item, the “Change”:

- Chapter 1: Introduction
- Chapter 2: Introducing a Change into Aeronautical Products
- Chapter 3: Configuration Management (CM)
- Chapter 4: Development Domain – The Current Development Process
- Chapter 5: Overview of Authority Documentation & Requirements
- Chapter 6: Organizational Requirements for the Company
- Chapter 7: Approval of Changes to the Type Design
- Chapter 8: Introduction to Responsibilities
- Chapter 9: Begin the Design with the End in Mind
- Chapter 10: System Design-Related Requirements
- Chapter 11: Type of Change Processes
- Chapter 12: Phase Concepts in the Different Change Processes
- Chapter 13: Origin of a Change
- Chapter 14: Change Items & Documents
- Chapter 15: System Design
- Chapter 16: Functional Definition
- Chapter 17: Harness Pre-Definition
- Chapter 18: Installation Definition
- Chapter 19: Manufacturing Preparation
- Chapter 20: Testing the “Results”
- Chapter 21: Correction Process of Changes
- Chapter 22: Delivery Preparation
- Chapter 23: Configuration Conformity Management & Documentation
- Chapter 24: Aging Aircraft / In-Service

1.2.5 Recommended Environment for the Reader

The field of application we are dealing with is highly complex and in constant motion. That means that one has to check ALL RELEVANT INFORMATION ON A SHORT-TERM BASIS to be up to date with the actual knowledge base and the state-of-the-art.

This leads to the necessity to have access at least to AIRBUS People, MyDoc and the EASA/FAA information.

It is urgently recommended to check MyDoc at least once a week for updated issues to be up to date with the documentation.

For training purposes, information on electrical elements can be found on the Glossary Server of the Structure Faculty (<http://myhosting1.airbus.corp/app053/public/>).

An introduction to EASA and FAA issues can be found in the book by the author Filippo De Florio titled *Airworthiness* (ISBN-10: 0-7506-6948-9).

It is urgently recommended that access to different i-shares be mandatory. The selection of these i-shares depends on the actual field of work; therefore, no further information can be given here.

1.3 Aircraft Construction at AIRBUS

AIRBUS Industry originally came into being in 1970 as a grouping of economic interests of the founding members Sud Aviation (France) and MBB (Germany). The Spanish CASA joined AIRBUS in 1971 and British Aerospace followed in 1979.

From the transnational roots of AIRBUS, it is also possible to trace the reasons for the work share we find in AIRBUS today; that means that the work-sharing taking place today has historical reasons.

The company's sites are distributed within Europe, and the main method for handling this scenario is the construction via the "section method." With this section method AIRBUS serves the market for cargo and passenger aircraft.

AIRBUS is a transnational company with its main sites located in France, Germany, the United Kingdom and Spain. There are a total of 12 sites including four in Germany, three in France, three in Spain and two in the United Kingdom.



Figure 1-1: Overview, Airbus Design Office Sites (as of September 2014)

2 Introducing a Change into Aeronautical Products

2.1 Considerations as to the Nature of a “Change”

Every variation of an existing AND/OR introduction of a new "item" into the AIRBUS system is, in terms of the authorities' regulations, a “change.”

That means that a “change” regarding the organization/(technical) processes/the product A/C itself must comply with the regulations as prescribed by the Authorities.

This is an extremely important sentence. Here we should come to realize that in the area we are working in every alternation of a situation is subject to be a “change” in the sense of the Authority.

For the sake of simplicity⁴ we will always consider the word “change” as a “change” in the sense of the Authority.

For example, the CS (Certification Specification) paragraph 25.869(4) defined by the EASA states:

“Insulation on electrical wire and electrical cable installed in any area of the aeroplane must be self-extinguishing when tested in accordance with the applicable portions of Part I, Appendix F.”

In other words, changing the insulation of an existing or introducing a new insulation, also possibly changing the supplier of the insulation or the fabrication processes of the insulation, is not necessarily a good idea unless the test requirements specified in Appendix F of the CS-25 have been fulfilled.

As a result for the electric and related parts at least in the proof of relevance regarding several CS paragraphs for applicability, e.g. beyond the CS-25.869(4), the following will apply:

CS 25.561 - Emergency Landing Conditions (General)

CS 25.869 - Fire protection systems

CS 25.1301 - Function and installation

CS 25.1309 - Equipment, systems and installations

CS 25.1353 - Electrical equipment and installations

CS 25.1360 - Precautions against injury

CS 25.1423 - Public address system

CS 25.1431 - Electronic equipment

CS 25.1701-33 EWIS (Electrical Wiring interconnections system)

It is important to note that every “variation” of the organization / (technical) processes / the product A/C itself is, according to the authorities, a “change.”

It is also important to realize that the “change” has one direction from the “entry point” in our organization to the delivery of the result of the change (possibly outside of AIRBUS).

⁴ Considerations regarding “human factor” are not in the scope of this documentation.

Every flow of messages, data or information in the opposite direction (e.g. changing a given Design Data Set in the design organization or determining a non-conformity in the production organization before delivery of the A/C to the customer that leads to a concession) are considered correction processes (refer to EN9100).

2.2 Authority & Quality Management Relations

It is mandatory that every task performed inside our organization will be in compliance at least with the EASA. Furthermore, the EN9100, Quality Management Systems, Requirements (based on ISO 9001:2000) and Quality Systems Model for quality assurance in design, development, production, installation and servicing (based on ISO 9001:1994) also apply.

This implies further the application of Configuration Management as specified in ISO10007 and related requirements (refer to Configuration Management (AP1007)).

3 Configuration Management (CM)

3.1 Embodiment of CM Inside AIRBUS

Product Safety is the number one priority for AIRBUS which leads to Quality and Airworthiness as the central topics inside AIRBUS (refer to "AIRBUS Code," A00, Part B, "Our Commitment to Product Safety").

Also within this scope is the fact that AIRBUS is responsible for its products for the entire product life cycle, which is assumed (as a theoretical assumption) to be 99 years. This is a long time during which at least "Continued Airworthiness" with all obligations, especially a coherent and complete documentation of the actual A/C must be assured.

CM comes into the game via several approaches. One of these approaches starts with IR-21A.4. This Implementing Rule states that the coordination between design and production must be such that proper support of continued airworthiness of the product can be assured.

The documents "Management of DOA-POA Arrangements and Direct Delivery Authorizations in AIRBUS" (refer to AP2585) and "Cooperation between Design Organization and Production Organization According to EASA Part 21 G" (refer to M2857) describe these topics in greater detail.

Also the satisfactory coordination between design and production must be in compliance with several paragraphs, one of those being IR-21A.165. This Implementing Rule describes the obligations of the holder of a production approval which involve the details set forth below:

"The holder of a production organization approval shall:

(a) Ensure that the production organization exposition furnished in accordance with 21A.143 and the documents to which it refers, are used as basic working documents within the organization.

(b) Maintain the production organization in conformity with the data and procedures approved for the production organization approval.

(c) 1. Determine that each completed aircraft conforms to the type design and is in condition for safe operation prior to submitting Statements of Conformity to the Competent Authority, or

2. Determine that other products, parts or appliances are complete and conform to the approved design data and are in condition for safe operation before issuing EASA Form 1 to certify airworthiness, and additionally in case of engines, determine according to data provided by the engine type-certificate holder that each completed engine is in compliance with the applicable emissions requirements as defined in 21A.18(b), current at the date of manufacture of the engine, to certify emissions compliance, or

3. Determine that other products, parts or appliances conform to the applicable data before issuing EASA Form 1 as a conformity certificate;

(d) Record all details of work carried out. ..."

For the approach taken here, the important text part is the reference to IR-21A.143 "Exposition" which states:

" ...

(a) The organization shall submit to the Competent Authority a production organization exposition providing the following information:

...

11. A description of the quality system and the procedures as required by 21A.139(b)(1).

..."

In turn IR-21A.139 states:

" ...

a) The production organization shall demonstrate that it has established and is able to maintain a quality system. The quality system shall be documented. This quality system shall be such as to enable the organization to ensure that each product, part or appliance produced by the organization or by its partners, or supplied from or subcontracted to outside parties, conforms to the applicable design data and is in condition for safe operation, and thus exercise the privileges set forth in 21A.163.

(b) The quality system shall contain:

1. As applicable within the scope of approval, control procedures for:

(i) Document issue, approval, or change.

(ii) Vendor and subcontractor assessment audit and control.

(iii) Verification that incoming products, parts, materials, and equipment, including items supplied new or used by buyers of products, are as specified in the applicable design data.

(iv) Identification and traceability.

(v) Manufacturing processes.

(vi) Inspection and testing, including production flight tests.

(vii) Calibration of tools, jigs, and test equipment.

(viii) Non-conforming item control.

(ix) Airworthiness coordination with the applicant for, or holder of, the design approval.

(x) Records completion and retention.

(xi) Personnel competence and qualification.

(xii) Issue of airworthiness release documents.

(xiii) Handling, storage and packing.

(xiv) Internal quality audits and resulting corrective actions.

(xv) Work within the terms of approval performed at any location other than the approved facilities.

(xvi) Work carried out after completion of production but prior to delivery, to maintain the aircraft in a condition for safe operation.

The control procedures need to include specific provisions for any critical parts.”

This requirement for a quality system leads further to EN9100(E) which states in chapter 4.3 “Configuration Management”:

“ ... The organization shall establish, document and maintain a configuration management process appropriate to the product.

NOTE: Guidance on configuration management is given in ISO 10007.”

This leads now directly to “Configuration Management” (AP1007) which is also directly linked to the ACMR, therefore to the DOM⁵ and back to the IR-21.4. Thus, the loop is closed.

A large number of references to CM are featured in the above mentioned paragraphs leading among others to the Sub-Processes of CM. The document “Configuration Baselines” (refer to AM2022) “... is written to support AP1007 to describe the concept and methodology of configuration baselines.”

In general CM⁶ is defined as:

“... A management process for establishing and maintaining consistency of a product's performance, functional and physical attributes with its requirements, design and operational information throughout its life.”

3.2 CM Disciplines

In order to fulfill the task required for CM as mentioned above five disciplines are defined inside the CM:

Identification; every object inside AIRBUS must be individually identified.

Change Control; this is the Discipline where at least the Change Process is located.

Status Accounting; the “book-keeping” of changes carried out, the CCMD (Configuration Conformity Management and Documentation) is the main sub-process in this discipline.

Audits; in general this covers any kind of test (e.g. GTI), attestation (e.g. EASA Form 1) and / or handling of “Non-conformity” and shall be performed in accordance with “Managing Quality Audits” (refer to AP2183).

CM Steering; understood in an initial approach as the “observation” (in a co-operational sense) of the above mentioned disciplines and, if deviation from the “must” is detected, the proactive supporting of measures to set “it” back on track.

An introduction to Steering is provided in the document “Steering Configuration” (refer to AP2579, Issue A).

⁵ Refer to “Design Organization Manual” (DOM, A20)

⁶ Definition of CM is taken from ANSI/EIA-649.

These disciplines are mapped against eight sub-processes defined for the CM. The CM Department is responsible for and in charge of the CM sub-processes.

3.2.1 “Helper Concept” Basic Approved ATA Breakdown Index (BAABI)

To identify all “areas” inside the A/C separately, several systems are used, one of which is the iSpec 2200.

The ATA has developed a system to identify the objects in reference to their system designation or function. An ATA reference is shown as a six-digit number as follows: ATAXX-XX-XX. This syntax follows the rules of meaning below:

The numbering of the first three digits follows the guidelines of the ATA and is mandatory.

The first two digits identify the main systems of an aircraft.

The third digit identifies subordinate systems or components.

The fourth digit stands for subordinate systems and caters to the peculiarities of individual aircraft programs.

Defining the fourth digit is one of the main aims of the BAABI at AIRBUS. Therefore, the BAABI features a complete reference list for the first four digits according to aircraft program.

The last two digits identify individual system components and can be taken from the ATA breakdown.

Additional information regarding ATA organizational is provided in chapter 5.2.5, Airlines for America (A4A).

ATA Defined part (to be followed by all the aircraft manufacturer) : XX-Xx-xx

System (ATA) (Chapter for Doc ATA)	21-00-00 Air Conditioning
Subsystem (ATA) (Section for Doc ATA)	21-20-00 Distribution 21-30-00 Pressurization 21-40-00 Heating
Domestic Part (to be defined by each aircraft manufacturer) Sub-subsystem (ATA)	21-21-00 Cabin Air Distribution
Subject (ATA) (Unit for Doc ATA) (Detailed items)	21-21-11 Blower 21-21-12 Detector Air Flow

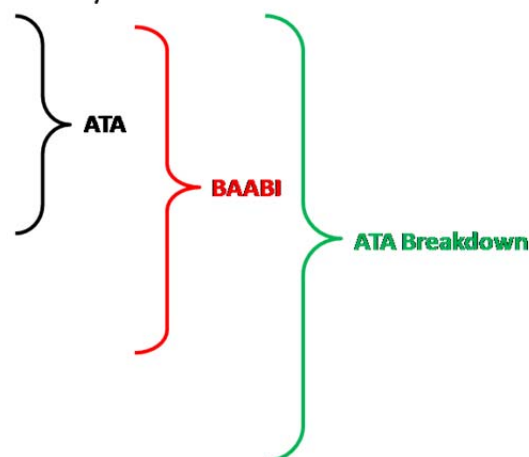


Figure 3-1: Overview, Structure of ATA Numbers

Further information on this topic, together with listings of the individual classifications, can be found on the ATA REFERENCE Intranet page in AIRBUS People.

The components designed, supplied and installed by the Electrics such as cable looms and equipment items are listed under ATA chapter 92/87.

The ATA chapters from the ATA-92 area most frequently applied are shown in the table below:

Sub-ATA Chapter	Short Description
92-00	Electrical and Electronic Common Installation
92-10	Routing Diagrams
92-20	Routing Drawings
92-30	Equipment Installation Drawings
92-40	Installation Drawings for Removable Supports
92-50	VU, VE, VX Components
92-60	Terminal Block Drawings
92-70	EDP Bundles Processing
92-80	Cable Loom Installation Drawings
92-90	Cable Loom Drawings

Table 3-1: Example for ATA Chapter 92 of A350 (refer to BAABI - A350)

3.3 Sub-Processes of CM

AP1007 "Configuration Management" defines 8 configuration management processes and their relation to the above mentioned disciplines (chapter 3.2). The following chapters are describing the basic principles of them.

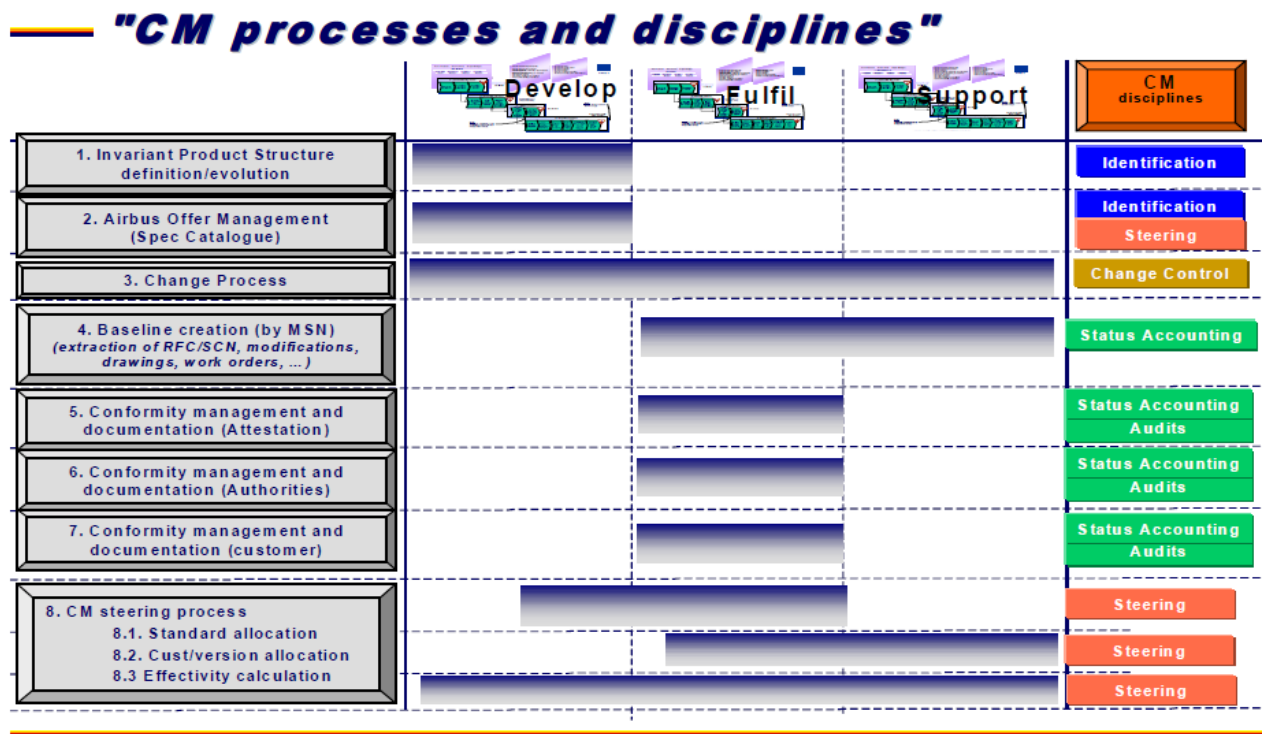


Figure 3-2: CM processes and disciplines allocated to the three Airbus domains (AP1007)

Next to Layers and Views, Product Structure is organized in a “Hierarchical Breakdown” containing 3 levels of different content (see Figure 3-4).

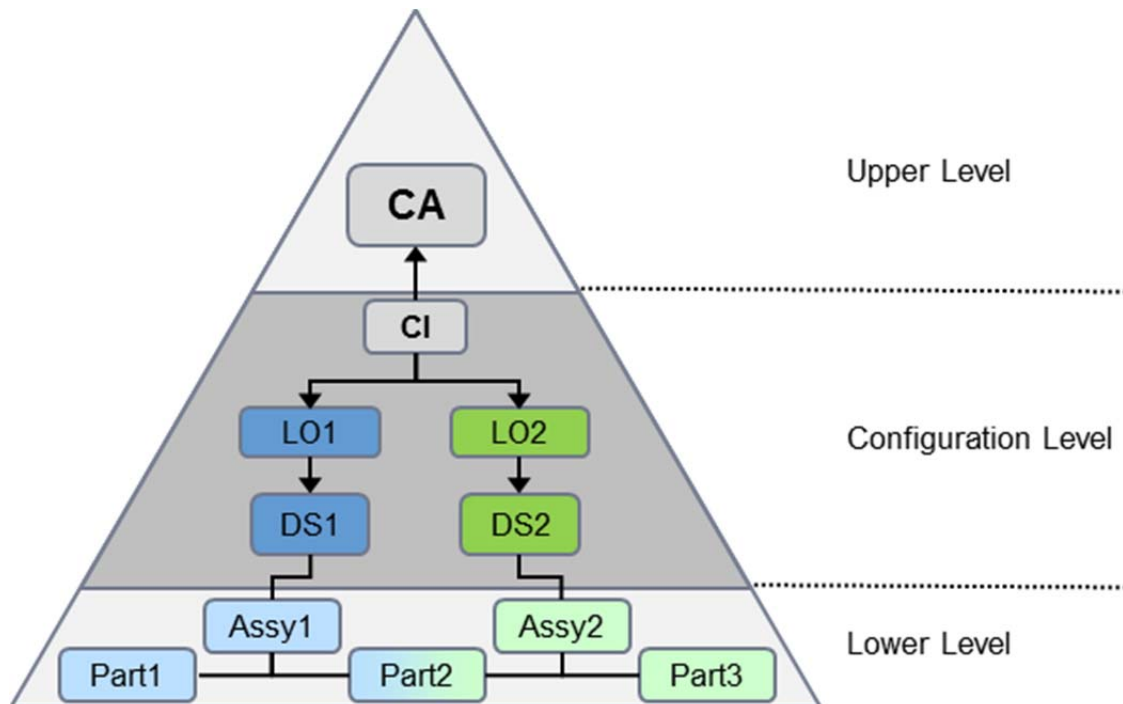


Figure 3-4: Three Levels of Product Structure

Upper Level: Contains Constituent Assemblies (CA) which are usually defined by more than one CI. Also other Cascade Components are to find in the upper level, refer to M2211.3 „Cascade Components - Concepts and Rules”.

While a **CA** represents a physical component of the Aircraft which is finally attested and documented in a CAIR, a Configuration Component (CC) represents another kind of upper Level node which is always linked to a **CA**. Under a **CC NSDW** for example, system describing documents such as PD or EPD are steered.

In contrary to a CC a CA has its own release document, the EASA Form 1 (refer to chapter 22.2.1).

No configuration takes place in upper level, all components are statically linked together because they constitute the aircraft and are necessary for each MSN (e.g. doors, sections, wing...).

Configuration Level: This level realizes the definition of different configurations per MSN. Using CI-LO-DS concept enables Airbus to implement customer specific changes and standard improvements in the design and production of aircrafts.

A Configuration Item (**CI**) represents a management node, which acts like a container for the information and is NOT a real object (e.g. L92C-75000-000).

With a CI, the various characteristics of a DS resulting from the different aircraft (A/C) configurations are linked together. In each case, only one DS is effective per MSN and CI. This clear linking is achieved by means of link objects, which contain the effectivity for the individual DS.

A Link Object (**LO**) links a DS with a CI. The LO assigns MSN effectivity to the DS.

A Design Solution (**DS**) constitutes the design implementation of the required technique, e.g. into Design Data Set. It therefore represents the actual product.

Lower Level: This level contains all different parts, assemblies and other items used by Design Solutions.

Each CI is “impacted” by one or more CIN’s. A CIN is a unique combination of MOD and MP. By means of MOD a technic describing MP becomes valid for a given range of MSN based on Standard or Customer Version.

The technique is described in a TRS which lists all impacted CIs on its sheet number 5.

Based on the Aircraft Allocation Table, which defines for each MSN standard, customer version and version rank, the specific CIN-Stack of a CI for a given MSN is calculated and answered by LO/DS.

Figure 3-5 shows how design solutions become effective for specific MSNs.

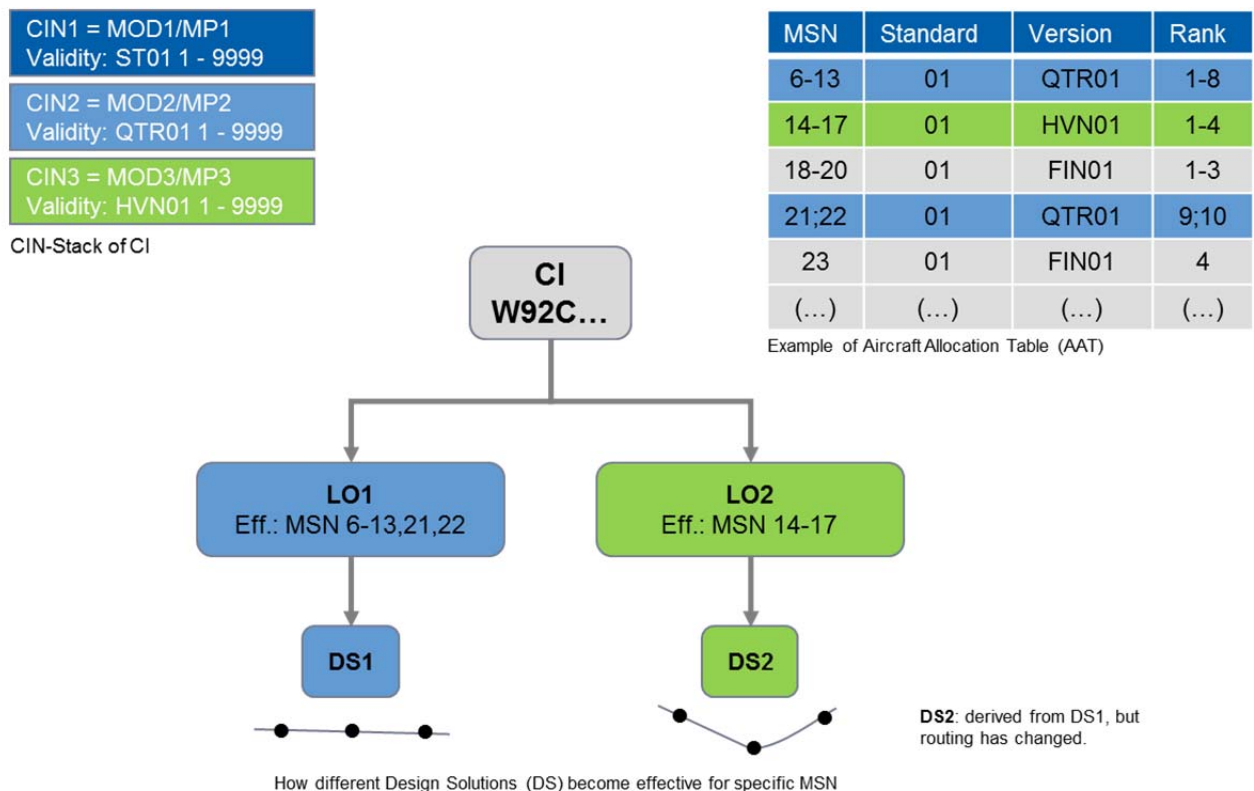


Figure 3-5: CI/LO/DS – transfer CIN-Validity to Effectivity (Effectivity for FIN01 not shown)

On Figure 3-5 we see 3 CINs impacting the CI, one “basic” standard CIN valid for all aircrafts based on standard 1 and two optional CINs – one for version QTR01 and the other valid for HVN01. By adding the optional MODs on the existing standard modification we have the need to answer two different CIN-stacks with design solutions. One stack consists of CIN1 and CIN2 and results in DS1, the other stack includes CIN1 and CIN3 and results in DS2. If we talk about CIN-stacks, we have on the one hand the complete stack of CINs impacting a CI and on the other hand we have the stack which is answered by a DS and where only CIN’s are involved which are impacting the same set of MSN.

Additive MOD system means, that once a design is released, all following changes are based on the initial design and implemented via pre/post descriptions.

3.3.2 P2 – AIRBUS Offer Management

In addition to Aircrafts based on a standard specification, Airbus offers options to the customers using Catalogues (Legacy) and Aircraft Description Document (ADD, A350XWB). These options are allocated to EPAC/TDU numbers (see also chapter 13.3.1).

The EPAC/TDU-concept is used in Airbus programs – legacy and new – to support and manage customization. But the way how the concept is used in design and configuration is different.

In A350 program, Airbus has implemented the Pre-Design approach (similar to A380's Custo-CoRe). Aim of Pre-Design is to offer solutions which design solutions already exist in a more or less maturity – so called Packages.

To perform Pre-Design in a configured 3D-environment, CIN's are created, having initially a package-validity and in case of ordered by a customer, getting the corresponding version validity. Figure 3-6 shows how the pre-designed solutions receive MSN applicability which means: "This solution could be used – if ordered – for the following MSNs: ...". Details on that topic are provided in AP5175 "Create Catalogue and Manage Standard Offer Data" and herein referenced methods.

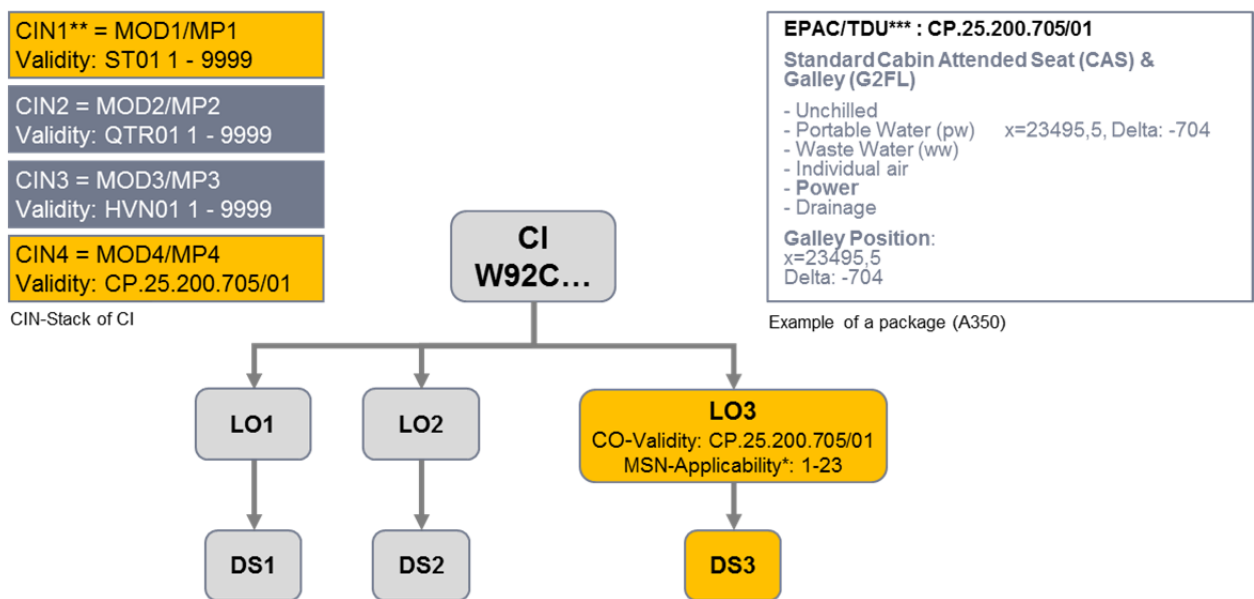


Figure 3-6: Package related CIN and corresponding LO

* Based on Standard-CIN-Stack, MSN Applicability is calculated in line with AAT on **Figure 3-5**

** Additional standard CINs may have impact on MSN-Applicability (e.g. in case a ST01-CIN, introduced with MSN 5, has impact on DS3. MSN 5 would not be listed in MSN-Applicability of LO3 anymore).

*** Package validity refers to an EPAC/TDU, which is nearer described in 13.3.1

More information on this topic is provided in the chapter describing the RFC and in the document "Manage Catalogues" (refer to A2843.2).

For program A350 the tool "APS" (Airbus PDM Services) provides additional product structure information like the above mentioned package validity and resulting MSN applicability. In the near

future, solutions based on Service Bulletins are also managed in APS and the tool provides in addition the chance to perform Pre-Studies in a configured DMU.

3.3.3 P3 – Change Process

With the implementation of the Change Process, Airbus fulfills the “Change Control” discipline required by ISO 10007. More details on Change Process are to find in chapter 12 and A2843.



Figure 3-7: Change Process Flowchart taken from A1080

The process in its simplified illustration is based on 4 phases with 3 decision gates (DG). During the first three phases “change investigation” takes place until the final decision is made on DG3: to implement the change by opening a MOD.

3.3.4 P4 – Baseline Creation (by MSN (Manufacturer Serial Number))

Please refer to the MSCN regarded in chapter 13.3.1.

3.3.5 P5 – P6 – P7 – Configuration Conformity Management and Documentation (CCMD)

The CCMD process ensures the conformity of the product which means that each single built MSN is fulfilling the expectations of the customer and the authorities. The process is categorized in three aspects:

Attestation (is every effective CIN fully embodied?)

Customer (are all RFC’s completely covered by modifications?)

Authorities (do all modifications have an airworthiness approval?)

The process monitors all change activities and reports outstanding work of all involved parties. Find details in chapter 23 and A2845.

3.3.6 P8 – CM Steering Process

The CM Steering Process covers mainly questions related to:

Standard allocation

Customer / Version allocation

Effectivity calculation

This topic is partially described in chapter 3.3.1 but is not in focus of this compendium.

4 Development Domain – The Current Development Process

For the programs SA/LR/A380/A400M the development process “Design New Aircraft” (DnA) was relevant (refer to AP2054). The process for existing programs to design a new electrical system is called the Design New Electric Process (DnE). For the recent electric development process refer to AP5997.



Figure 4-1: Overview, Simplified DnA Process

For the new programs (as of Dec. 2009) “Develop and Ramp-up Excellence (DARE)” was implemented (refer to A1350). DARE is the new development policy that becomes applicable to any new program development (e.g. A350XWB).

Both processes, DARE and DnA describe process steps for a completely new design of an aircraft type. For serial production, the Fulfill Domain, the incorporation of changes is described in the Fulfill Customer Order (FCO) process (refer to A2015). The FCO process itself has not changed significantly in A350 program, but package pre design and a robust standard aircraft not impacted by customization shall help to reduce the workload of head of version (HoV) design.

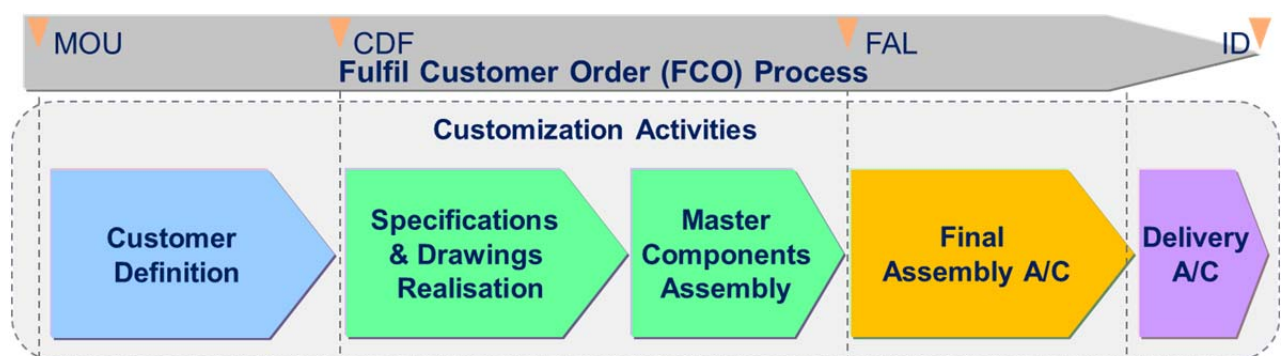


Figure 4-2: Overview, Simplified FCO Process

The Product Development Process starts from the early study phase and continues through the phases of design, procurement, production, certification, delivery, in-service operation and support.

In the document “Design Principles” (refer to AP2601) different principles used in engineering are provided. The “Developing Aircraft Definition Model” (refer to AP2600) covers the actions to compile a reference database to define the A/C.

DARE is an AIRBUS-wide change program (part of Develop Faster Module of Power8), launched in early 2006, with the goal of transforming the Aircraft Development Process in order to:

Reduce lead time: The objective is to have a six-year range development cycle from start of concept phase (M3/MG3) to Entry into Service (EIS),

Ensure robust production ramp-up: analysis has shown the clear competitive advantage of delivering new products at a higher rate.

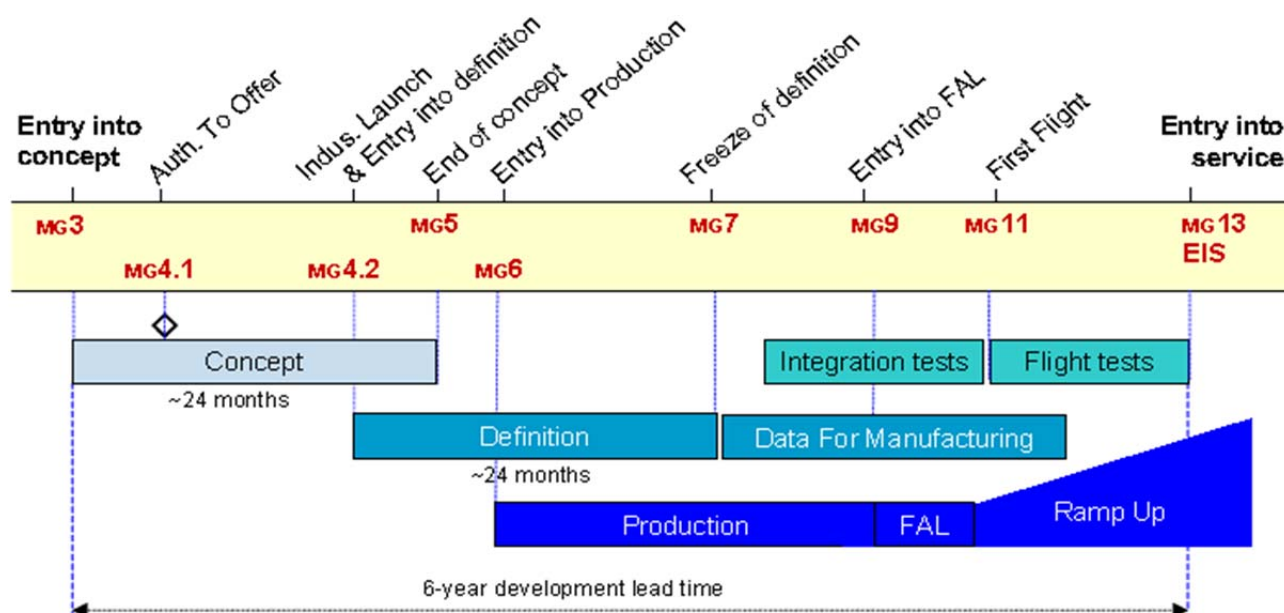


Figure 4-3: Overview, Simplified DARE Process (refer to A1350.0, Issue A, Chapter 3.1, “Overall Planning”)

The “Final Assembly Line & Major Component Assembly Line Industrial Process” (refer to AP2270) covers activities in the conception, definition and parts of the development phase⁷ (begin Final Assembly Line).

Ensuring on-time and quality deliveries clearly has a strong impact on the business case.

Deliver fully service-ready products: The target is to ensure very high maturity level at EIS (typically characterized by operational reliability figures like 99.5% for A350XWB).

Return of experience on recent program has highlighted the adverse impact of “Not mature” requirements, definitions or deliverables.

In particular, the product definition & design needs to be stabilized far earlier during the development process to limit the risk of late changes.

Late changes could be disastrous in a fast Ramp-up scenario due to their impacting many aircraft being manufactured prior to first EIS.

Beginning with a robust concept definition at the beginning of the process is a must.

It is absolutely necessary that all the market needs to be gathered, assessed and “frozen” (validated) in time. This is feasible given adequate interfacing with customers and adequate feasibility assessment.

⁷ The phases mentioned here are DnA / DnE phases.

The result is a highly front-loaded development process. Rigorous maturity management is required to tackle those root causes.

Thus as per DARE process, the aircraft development is paced and controlled through a set of Maturity Gates that are mandatory review milestones.

Generally, at each Maturity Gate during the process, the different aspects of the product and process performances are assessed in terms of quality and completeness.

The typical DARE development planning can be represented as follows, from Start of Concept phase until Entry into Service.

5 Overview of Authority Documentation & Requirements

5.1 European Aviation Safety Agency (EASA)

The EASA is the central organization of the European Union's strategy for aviation safety and supersedes the JAA⁸ (Joint Aviation Agency) on 28 November 2003 [FLORIO2006; Chapter 3.3; Page 10 ff.]. This leads to several advantages⁹:

Fits readily into the EU legal system

Fully compatible with EU aims and objectives

Has access to support from other EU institutions

Uniform rules and implementation all over Europe, therefore no more "Additional National Design Requirements" are possible

The EASA is established by the European Parliament and the Council of Ministers.

5.1.1 Document Structure & Naming Conventions

The EASA documentation follows rules similar to those of the AIRBUS documentation, but of course in a different way.

Initially, it is important to realize that the EASA is speaking primarily on behalf of the "EC," the European Commission Regulations.

The basic regulation relevant for AIRBUS is EC No. 216/2008, which describes the common rules in the field of civil aviation and establishes the EASA itself as the competent Airworthiness Authority in Europe.

The EASA issues different types of documents. The most important documents for an inexperienced reader are listed below with a short description.

5.1.2 Implementing Rules (IR) & Certification Specifications (CS)

IR is identified in relation to the basic regulation (EC) No. 216/2008. The IR states "for the airworthiness and environmental certification of aircraft and related products, parts and appliances, as well as for the certification of design and production organizations."¹⁰ The IR identified by the EC 1702/2003 is described in Annex Part 21 "Certification and Organizational-Related Issues."

Subpart J is relevant for the Design Organization Approval (DOA) and Subpart G is relevant for Production Organizational Approval (POA) (refer to IR-21A.135, "Issue of production organization approval").

⁸ Refer also to the article by: GIEMULLA, van SCHYNDEL, "Die Joint Aviation Requirements (JAR) und deren Überleitung in Vorschriften der EU und der Agentur für Flugsicherheit (EASA)," University of Berlin. This article is currently only available in German language.

⁹ PURWINS, MICHAEL: Speech: "Product Integrity – JAA / EASA – MOD Closure," October 2008.

¹⁰ See www.easa.eu; Citizens & General Information: Regulation Structure.

The CS describes the requirements for the certification of turbine-powered large airplanes. The CS-25 is relevant at least for the AIRBUS family Single Aisle, Long Range, A380, A400M and A350. The CS states requirements for the design of large aircraft.

The “Technical Design Directives (TDD)¹¹” and the “System Installation Design Principles (SIDP)”, for example, are documents within our organization that translate Authority documents such as the CS-25 into an AIRBUS-interpreted language for the designer. This constitutes a part of the “envelope” where the designer can act within his specific workspace.

5.1.3 Notice of Proposed Amendments (NPA) & Comment Response Document (CRD)

The NPA provides information from the EASA regarding proposed changes to the official regulations. Response given by different parties is presented to the public in the form of a Comment Response Document (CRD).

The document “Aviation Regulations Management: Screening, Information and Commenting” (refer to AP2663) provides further assistance to respond accordingly to a NPA from inside AIRBUS.

5.1.4 Acceptable Means of Compliance (AMC) & Guidance Materials (GM)

The AMC are intended to give guidance in finding methods to fulfill requirements required by Means of Compliance (MC). The AMC of the EASA finds its counterpart in the “Advisory Circular” (AC) issued by the FAA. The related documents formerly published by the JAA were called “Advisory Circular - Joint” (ACJs).

All of those documents provide a means “... but not the only means, by which a requirement can be met” [FLORIO2006; Chapter 4.4; Page 43].

A GM provides background information for selected areas to obtain a better understanding of the area of application for fulfilling a particular MC [FLORIO2006; Chapter 4.4; Page 45].

5.1.5 Airworthiness Directives (AD) & Certification Review Items (CRI)

The AD focuses on changes to the Type Design. Normally this is the case “... if the Authority reveals an unsafe condition in an aircraft ... that exists or has the potential to develop on similar types of aircraft...” [FLORIO2006; Chapter 9.2; Page 219].

This can also be a CRI, which has been meanwhile included in the normal workflow of the Authority. Normally a CRI requires changes to the Type Certification basis on an “on demand” basis. If practical experience shows that there is a need for a fast solution to safety findings [FLORIO2006; Chapter 2; Page 114] a CRI is issued by the Authority.

This procedure is in line with IR-21.16B “Special Conditions” and comes normally in conjunction with other rules. In the case of the CRI D-12¹² the ICAO Annex 8 (Amdt. 97) which belongs to the area of “International requirements to introduce Security in design requirements for large aeroplanes” introduces 8 separate requirements:

¹¹ The TDD will not be delivered to the supplier; instead the supplier will receive the SIDP. Cabin Module suppliers will receive a summarized TDD which is called EDR (“Electrical Design Requirements”)

¹² CRI D-12: A380 Certification Review Item; “Design for Security”; Issue 3, 30 JUNE 2003.

- Survivability of Systems
- Fire suppression
- Smoke and fumes protection
- Least risk bomb location and design
- Small arms and shrapnel penetration
- Interior design

The possible solutions in accordance with the requirements arising from the particular CRI are the object of discussions between AIRBUS the Authorities and other involved parties. These discussions will end with an agreement between the parties what exactly will be solved with what kind of processes or design methods / directives.

The resulting change is also visible in the official documents of the Authority. In the case of the CRI D-12, the part of the CS(JAR)-25 regarding "Security considerations" (refer to CS 25.795) was adjusted accordingly.

The result of the above mentioned working groups will be visible to the AIRBUS community as MODs and / or changes in the documentation (normally in MyDoc).

These changes triggered by the CRI D-12 introduced, among others, a change in the routing concept of the harness. In turn, this brought about changes to the relevant TDD (Technical Design Directive) regarding the Functional Design and SIDP (System Installation Design Principles) mainly for the Installation Design of the harness.

5.2 Related Authority Organizations

5.2.1 Federal Aviation Agency (FAA)

The FAA is the official organization of the United States of America with legislative power. The corresponding organization in Europe is the EASA. The FAA mainly issues the following documents for regulation purposes:

- **Federal Aviation Requirements (FAR)**
FAR are the regulations specified by the FAA and are similar to the IR / CS constructions given by the EASA. The divisions of the FAR given by the FAA are somewhat similar to the EASA regulations, but not entirely so. Furthermore the contents of the regulations are not entirely the same, and we have differences in the text, which are sometimes hard to interpret.
- **Special Federal Aviation Requirements (SFAR)**
A special FAR, called SFAR, is a regulation which is mandatory but not incorporated in the normal FAR divisions. As an example the SFAR 88 deals with issues related to the fuel tank system. This SFAR was, among other reasons, initiated by the accident on Trans World Airlines (TWA) flight 800 in 1996 (a video dealing with that accident is available within N24 news channel (television) / ESDC).

5.2.2 Civil Aviation Agency (CAA)

The CAA is the UK's specialist aviation regulator (www.caa.co.uk). The CAA is divided into four groups where the Safety Regulation Group is the most relevant group for our focus in this document. The Safety Regulation Group satisfies itself that aircraft are properly designed, manufactured, maintained and that all other criteria are met. For a first understanding of the CAA role and the relations between the EASA and other related organizations the Civil Aviation Publication (CAP) give further information. The CAP 747, "Mandatory Requirements for Airworthiness" for example is a good starting point for reading, because the CAP 747 is the means by which airworthiness requirements are made mandatory by the CAA.

5.2.3 Other Related Authorities

The relevant Authorities are determined by different requirements. Because AIRBUS is a European company the EASA always applies. Furthermore the FAA is in charge because more than 50% of the Stakeholders are of U.S. origin and nearly every customer will be using airports in the U.S.A.

The additional organizations in the area of the Authorities are:

National Transport Safety Board (NTSB) – U.S.A.

The NTSB (www.nts.gov) is an independent Federal agency charged with investigating civil transportation accidents occurring on U.S. territory. The NTSB is also responsible for civil aviation safety.

Transport Safety Board (TSB) – Canada

The TSB (www.tsb.gc.ca) is also an independent Federal agency similar to the NTSB and is in charge of investigating civil transportation accidents on Canadian territory.

Bundesamt für Flugunfalluntersuchung (BFU) – Germany

The BFU (www.bfu-web.de) is responsible for investigating civil aviation accidents in Germany.

Bureau Enquêtes Accidents (BEA) – France

The BEA (www.bea.aero) is responsible for investigating civil transportation accidents in France.

For future reference it has to keep in mind that issues related to military A/C within the European Community are handled by the European Defence Agency (www.eda.europa.eu). The EDA issues e.g. the EMAR-145 document which will be applicable for the A400m in the future.

5.2.4 International Civil Aviation Organization (ICAO)

The ICAO (www.icao.org) is an international organization whose members include (nearly) every country in the world working in the field of civil aviation.

The long history of the ICAO began in 1910 with the Paris Convention. In short, it is a special agency of the United Nations. Since the Chicago Convention of 1944 its purpose has been the promotion of future civil aviation development and safety.

The ICAO Annexes 1 to 18 help to clarify needs in civil aviation. The ICAO refers to these documents as a statute and designates them as ICAO Doc. 7300/9.

Two important annexes are:

ICAO Annex 8 - Airworthiness of Aircraft

ICAO Annex 13 - Aircraft Accident and Incident Investigations

5.2.5 Airlines for America (A4A)

The A4A was formally known as Air Transport Association of America (ATA) which was an umbrella organization of U.S. airlines. It was founded in Chicago in 1936 by 14 airlines and is, to this day, the most important union of airlines in the U.S.A.

The main task of the A4A is to represent the interests of the airlines before the U.S. Government with regard to regulations affecting air traffic.

The ATA established standards in many areas, which are adopted by airlines all around the world. One important standard, for example, was ATA100, which specifies the structuring for documentation on the individual systems of an aircraft.

This includes what we still call an ATA-Chapter like ATA 92 for electrics. We are able to do this because the ATA system has several chapters not assigned and these are free for the definition by aerospace companies. Nevertheless, it is important to keep in mind that it might be that other aerospace companies have no clue that ATA 87/92 is assigned to electrical installation.

Today, the formally known ATA100 Spec has been replaced by S1000D and herein, we find a Standard Numbering System (SNS) which is quite similar to the ATA 100 Spec.

The S1000D also defines technical documentation standards for other products than airplanes. The standard which was first applied for Boeings 787 can be found under "Air vehicle, engines and equipment SNS" (AVEE).

6 Organizational Requirements for the Company

6.1 Organization Approvals

AIRBUS has the ability to operate as a company dealing with aircraft as a result of:

DOA - Design Organization Approval (EASA Part 21, Sub-Part J); e.g. we may produce and change design data sets.

POA - Production Organization Approval (EASA Part 21, Sub-Part G); e.g. we may produce CA or integrate Sub-CA into "Upper CA." For details refer to "Obligations of the Holder," IR-21 (21A.165).

MOA – Maintenance Organization Approval (EASA Part 145, 147); e.g. we may perform repair¹³ work.

CAMO - Continuing Airworthiness Management Organization (EASA Part M Subpart G)

According to EASA Part 21 Airbus SAS is holder of the Design Organization Approval. The AIRBUS Organization Approval has been granted for:

Designing large airplanes, changes and repairs in accordance with the applicable type certification basis and environmental protection requirements;

Designing in accordance with the applicable type certification basis and environmental protection requirements supplemental type certificates to aircraft related to cabin interiors and associated systems, and structure;

Showing and verifying the compliance with the applicable type certification basis and environmental protection requirements; and

Demonstrating this compliance to the authorities.

AIRBUS is also holder of other Approvals, but we will focus here first on the DOA relevant for "Design work" to take place.

AIRBUS has demonstrated to the authorities that the Design Organization has the capability to:

- Design products or parts in accordance with the applicable certification specifications and environmental protection requirements;
- Show and verify the compliance with these certification specifications and environmental protection requirements, and demonstrate to EASA this compliance.

The planned and systematic actions necessary to provide adequate confidence are carried out by the following functions:

Design Function

Airworthiness Function

Independent Monitoring Function

The **Design Function** is defined as all the activities of the AIRBUS Design Organization:

¹³ Refer to IR-21, Subpart M, Repairs, 21A.431 ff.

To produce Type Design data relevant to an aircraft, and related products (aircraft engine or propellers), parts and appliances, in accordance with the applicable Certification Specifications and Environmental Protection requirements;

To produce change to Type Design data relevant to modifications, repairs and technical solutions for unintentional deviations from the approved design data occurring in production (concessions or non-conformances), in accordance with the applicable Certification Specifications and Environmental Protection requirements;

To provide continued design support to the in-service aircraft (preparation of Service Bulletins, etc.).

The **Airworthiness Function** is defined as all the activities performed by the AIRBUS Design Organization:

To show, verify and demonstrate to the EASA compliance of the Type Design data of an aircraft, and related products (aircraft engine or propellers), parts and appliances, with the applicable Certification Specifications and Environmental Protection requirements;

To show, verify and demonstrate to EASA compliance of design modifications, repairs and technical solutions for unintentional deviations from the approved design data occurring in production, with the applicable Certification Specifications and Environmental Protection requirements;

To ensure the continued airworthiness of in-service aircraft.

The **Independent Monitoring Function** is defined as all the activities performed by the AIRBUS Design Organization to independently monitor the compliance with, and adequacy of the documented procedures of the AIRBUS DAS through the DOM, ensuring its proper functioning, efficiency and compliance with Part 21 requirements.

6.2 Considerations as to Privileges Granted

It is important to keep in mind that the above mentioned approvals from the Authority are only granted for a limited time (e.g. two years) and that they can be revoked by the Authority at any time if the above organizations are not in compliance with the EASA Parts.

For example, in April 2001 the Authority was on the verge of revoking the POA privilege granted to AIRBUS because AIRBUS was unable to determine exactly the actual status of a given CA (Constituent Assembly).

The IR-21 ("Findings" 21A.158) describes the different cases if the Production Organization is no longer in compliance with the IR-21.

6.3 The Modification System & Related Top-Level Documents

The change inside AIRBUS is controlled and handled by a so-called MODIFICATION SYSTEM.

The detailed description is given in the Aircraft Configuration Management Rules (A1080).

In this context, a modification is an "object" used to handle a "change" inside AIRBUS. The modification itself can be classified in several ways. One classification of a MOD is the "Category" which gives mainly information about the "point of embodiment" of this MOD (refer to ACMR, Chapter "Category Definitions"). A MOD with category "A" for example is mainly seen as a MOD related to the Type Design and must be implemented before the first flight of the "first" aircraft of the program.

Another frequently used “type” of MOD is the so-called “Anti MOD” (which is NO category). The anti MOD is normally used to remove a MOD from a defined standard which will normally deliver a MOD “xy” to the customer but is “removed” from the delivered standard with an anti MOD.

The modification consists of Modification Proposals (MP), short descriptions of technical solutions.

The modifications are subject to the Modification Approval Sheet (MAS), meaning “Certification” under the responsibility of the Design Organization done finally by the DAE with the signature of the MAS.

The MP is the subject of attestation via the EASA Form 1 in relation to a particular CA under the responsibility of the Production Organization carried out by Certifying Staff / Quality.

6.4 Overview of the AIRBUS Document Structure

In order to ensure the approval as design, production and maintenance organization and to receive the certification under EN9100 and ISO14001, it is absolutely necessary that regulations are existent, stored, contributed and used within the company.

Placed above the pyramid of documents itself, there are the requirements to be fulfilled in the document-based management system - these are the “Regulations & External Standards.”

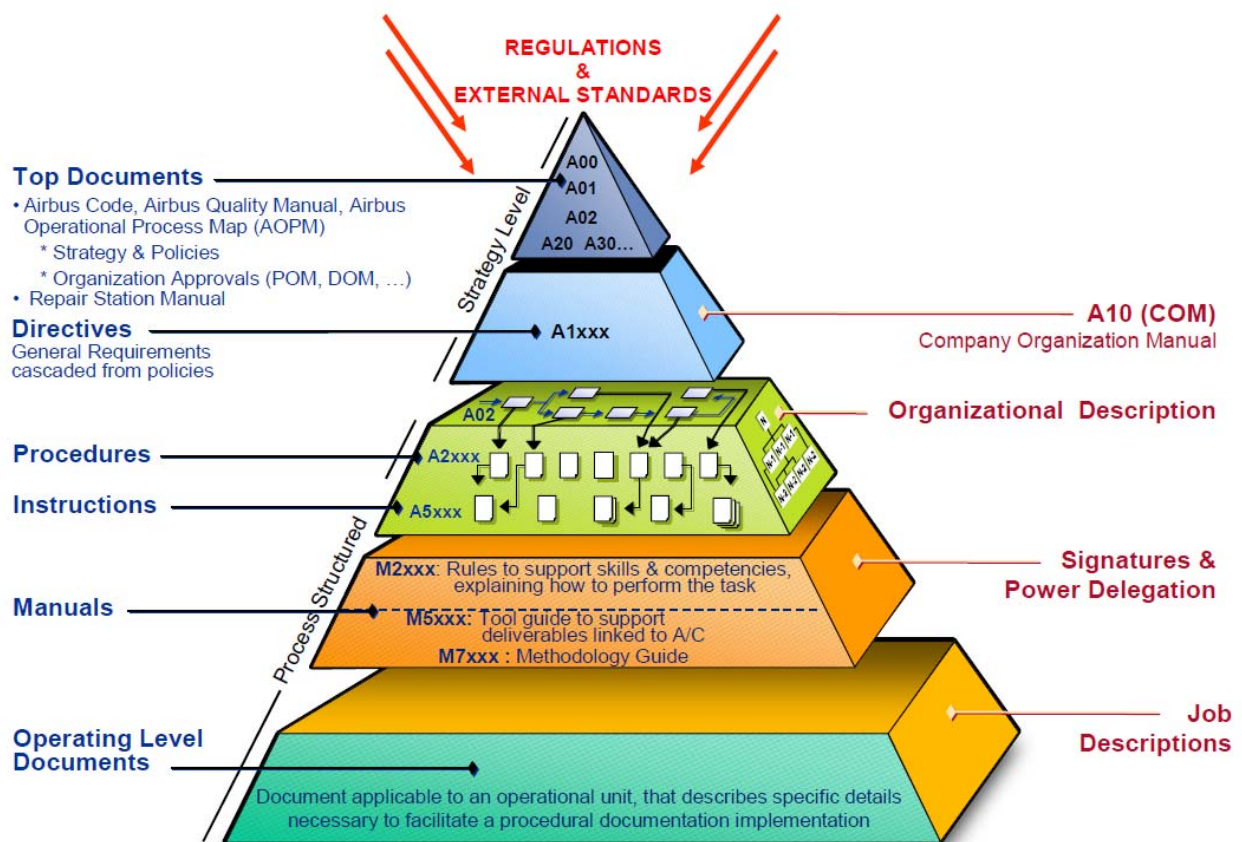


Figure 6-1: Overview, Airbus Documentation System (as of February 2010)

In order to meet the most varied requirements, the business must first comply with its own commitments that represent the guidelines of the highest level of the document hierarchy. Every document that exists below this level must undergo a consistency check against the content of the guideline and ensure that there are no contradictions. Requirements to documents are provided in "Requirements for Referential Documentation" (refer to A1000) and in the document "Procedural Document Writing Rules & Recommendations" (refer to M2900).

The highest document within AIRBUS is "**The AIRBUS Code**" (refer to A00), which describes the company's philosophy and objectives, its ethical foundation and general priorities.

After the philosophy the next stage consists of establishing and naming functionaries and domains, whose responsibilities include compliance with the rules and the guiding principles and ensuring the implementation of official ordinances.

Examples of the documents that describe these functionaries are the A20 Design Organization Manual (DOM), the A30 Production Organization Manual (POM), and the A40 Maintenance Organization Manual (MOM).

The next level comprises the so-called "**Directives**," specifying requirements to be implemented in certain operations within the business. For example, document A1001 describes the requirements for the safekeeping of documents and A1002 deals with the general requirements for project execution and management.

The requirement level is followed by the "**Procedures**" and "**Instructions**," describing the players and processes needed for the implementation of particular activities. Which decision follows which? Who makes what kind of input in the process (advice, decision, analysis, etc.), and when? For example, A2000 describes the general management for quality control and optimization; A5009 describes the various players and processes involved in creating a Technical Adaptation.

After the description of "procedures" and "instructions" **Directions** and detailed **Activity Descriptions** will follow. For example, M5112, whose eight modules explain the administration of the eTool for compiling PDs, WDs, SFMs, etc.

At the lowest level of the hierarchy are instructions that cannot be substantiated further. What should happen in the case of wiring looms that have no finalized attachment points after installation? They need to be rolled up and securely fastened, but how? The procedure is described in detail in the Germany site-specific document 80-T-40-3215¹⁴, a typical document at the lowest level.

All documentations **MUST CONFORM** with the "Lexinet" (refer to AP2080) which is a **MANDATORY** requirement from the "AIRBUS Quality Manual" (refer to A01, Chapter 5.2, "Documentation System").

Please keep in mind that AIRBUS maintains two (2) different structures of the documentation system. The former one works with the nomenclatures like AP, AM, ABD etc. while the new one is works with A, M and other items. Both of them are in use, but the system shown above is the master system.

¹⁴ Please refer in future to AIPS (Airbus Process Specification) / AIPI (Airbus Process Instructions)

6.5 Dossiers Types

Several Dossiers are used during the introduction of a change inside AIRBUS. The Dossiers are documents providing information regarding the status of several types of objects.

The Dossiers are mainly related to three different domains:

Design Organization

The Design Data Set (DDS) holds all relevant data, e.g. Bill of Material (BOM) and drawings in order to be able to transfer the data to the production organization.

A detailed description of the DDS is given in ABD0004 for running programs. The A350 will obtain an A1079 document that will supersede the relevant parts of the ABD0004 for the electric domain of the A350.

The DDS will be released mainly in the responsibility of the TAp and this is a legally relevant act. The Production Organization has the task of fulfilling the instructions given in the DDS one-to-one.

Production Organization (POM, A30)

The Production Organization defines the Industrial Dossier, which consists of the DDS, Production Dossier, and the Inspection Dossier. The DDS is supplied by the Design Organization. The Production Dossier contains all information for the physical realization of all items from the DDS.

The Inspections Dossier contains all relevant information to verify the conformity of the aircraft or its parts to the definition (supplied by the DDS).

Maintenance Organization

The Repair Dossier contains different data, e.g. inspection tasks (refer to MOM (A40)). This Repair Dossier is not in our focus and has therefore been omitted.

6.6 Certification vs. Attestation

The pair of terms "Certification" and "Attestation" are used in very different contexts, but basically they stand always for the same relation: A set of requirements has to be fulfilled by a person, an organization or a product (certification) and it has to be checked, whether the unique unit which wants to receive the so-defined certificate really fulfills the requirements (attestation).

Certification		Attestation	
Certificate	Requirement set	Unit to certify	How to show compliance to Requirement set
Engineering diploma	Learning plan of university	Student	Passed Tests, Diploma Thesis, List of attended lectures...
EN9100 Certificate	Norm EN9100	Organization	Written procedures, audit reports...
Type Certificate	CS 25, CS 36, DOA (...)	A/C Type	Certification Program, MC's
Certificate of Airworthiness	Set of approved Modifications	Aircraft	CAIR, AIR, Test Reports...

Table 6-1: Examples illustrating the difference between Certification and Attestation

For major changes, a Certification Plan is created by the Design Organization. This plan is the basis to get a modification approved by authorities. If it is approved, the built aircraft will be checked against all the content of the certification plan which is performed for all modifications effecting the single aircraft to receive its individual Certificate of Airworthiness (CofA).

This process is described more detailed in the modification closure process (refer to AP5204) and is under DOA responsibility (see also 22.1).

The embodiment of modifications is approved by Quality (Certifying Staff) on CA (Constituent Assembly) / CC (Configuration Component) or also on DA (Deliverable Assembly) basis by means of approval. DA is a term used inside a tool (e.g. PDMLink) and may or may not be a CA, however, if it is, it must be approved.

This is called the "Attestation", the Production Organization proves, with Quality/Certifying Staff in charge, that the production has produced this CA **exactly** as stated in the released documents (e.g. 928 design data set) from the Design Organization. Further information about the main documents are provided in chapter 22.2, "Attestation-Related Documents"

Special attention should be given to the fact that in some programs harnesses are defined as a CA. Therefore every harness needs the EASA Form 1 which is used for the CA Attestation.

If a MOD inside a CA is not CLOSED, it means we have no MAS and can only attest "NON-approved design data in the related EASA Form 1". A CA with "NON-approved design data" can move to another workstation but MUST have MAS approx. six weeks prior to delivery.

The top-level CA, the A/C itself, is approved with the signed and stamped EASA Form 52 by the Certifying Staff. The Approval is under the responsibility of the POA.

Here attention must be paid to the fact that every CA requires an EASA Form 1 AND a CAIR (CA Inspection Report, refer to AP2104). The CAIR is a prerequisite for the EASA Form 1 for that CA.

The prerequisite for the EASA Form 52 is the AIR (Aircraft Inspection Report, refer to AP5106) and the AIR requires the condensed data from all CAIR of all CA. So, if one CA with its CAIR fails, e.g. a harness is wrong in some way, no delivery of the A/C is possible.

One of the aims of the CAIR and AIR is to track the work left open during manufacturing. No outstanding work should be existent at time of delivery.

This leads to the related "Tests" to achieve needed answers to relevant questions. Further information regarding "Tests" (e.g. GTI) is provided in chapter 20 which is related to tests. Also the difference between Certification and Attestation should be kept in mind if talking about tests.

7 Approval of Changes to the Type Design

7.1 Type Certification (TC)

The TC is related to a Product and the associated production methods and the TC freezes all configurations. Hereby the Authority states that the design organization has shown compliance to all applicable requirements.

The Type Certification ends with the collective MAS for every MOD related to the TC.

In contrast to a TC the Supplemental Type Certification (STC) is used if any person alters a product by introducing a major change and is NOT the holder of the TC [FLORIO2006; Chapter 5.1.9; Page 85 ff.].

This is for example the case if Lufthansa-Technik is performing work in the Cabin Area and introducing a major change or changes to avionics. The holder of the STC must inform the holder of the TC about the performed work and the STC holder is responsible for the work performed.

The STC is described in Subpart E of JAR / FAR / EASA Part 21 and the organization applying for the STC shall demonstrate the ability to achieve a Design Organization Approval (DOA).

The TC is not a release to service or an authorization to operate the A/C; this is given by issuing the EASA Form 52. It should be taken into account that the A/C given by the so-called Type MODs is more or less a “minimalist” A/C.

To make things more “comfortable,” changes to the Type Design come in place and are the subject of the following paragraphs.

7.2 Classification of Changes

Every change to the A/C is a change to the Type Design and the method used for approval must be chosen. This leads to the question “Is this a MAJOR or a MINOR change?” (refer to IR-21, 21A.95 & 21A.97) which will be discussed also in the related GM.

7.3 Minor vs. Major Changes

“A Minor change is one that has no appreciable effect on the mass, balance, structural strength, reliability, operational characteristics, noise, fuel venting, exhaust emission or other characteristics affecting the airworthiness of the product.” More details are provided in “Approval of Changes to the Type Design” (refer to AP2243, Chapter 5, “Minor Changes Approval”). Every other change is automatically classified as “Major.”

Additional documents regarding 21A.95 & 21A.97 can be found in Annex 2.0 DAS Compliance Checklist (refer to “Design Organization Manual (A20)).

Major Changes are divided into Non-Significant and Significant Major Changes, whereas Significant Major Changes are based on actual requirements.

For Significant Major Changes the document “Type Certification Procedures” (refer to AP2184) applies.

8 Introduction to Responsibilities

8.1 Compliance with Part 21

On just 4 pages EASA's Part 21 Subpart J lists in 16 paragraphs the requirements for a design organization which wants to apply for a Design Organization Approval (DOA).

Airbus' basic "answer" to describe a design organization which fulfils all applicable requirements is written down in the Design Organization Manual (DOM, A20). Depending of what activities you want to perform in the framework of the approved design organization you need to show compliance to additional Subparts of Annex Part 21. You want to approve repair solutions on your own? Show compliance to Subpart M "Repairs"! You want to apply for a new type certification for a large aeroplane? Bring your organization in line with Subpart B "Type Certificates and Restricted Type Certificates"!

The roles, responsibilities and ways to delegate described hereunder are a mean to fulfill 21.A.239 and 21.A.245 of the above mentioned Subpart J. Paragraph 21.A.239 defines the Design Assurance System (DAS) which has to be described (procedures), implemented and monitored. Title of paragraph 21.A.245 is "Approval requirements" and points out that all staff members of the DAS have to have the necessary experience and that and are available in a sufficient number.

The procedures which are part of showing compliance to paragraph 21.A.239/21.A.245 and other applicable requirements of Part 21 impacting the design organization are to find in the Design Assurance System Compliance Checklist (DASCCL) which is stored on Airbus Peoples DOA eSite.

To ensure the fulfillment of the applicable airworthiness requirements (e.g. CS25) in all design activities, 3 ways of working have to be described:

How are the requirements integrated in the design activities and is compliance ensured?

How is "Independent Checking" applied to compliance data and who performs it?

How is ensured that an "Independent Monitoring" takes place, checking both, the compliance of the written procedures with applicable organizational requirements (Part 21) and the compliance of the "daily business" with the written procedures?

The following chapters explain the different delegation paths and related roles. The different paths are linked to the 3 main functions of the design organization. Each of these functions represents one of the three columns on which Airbus' design assurance system (DAS) is built:

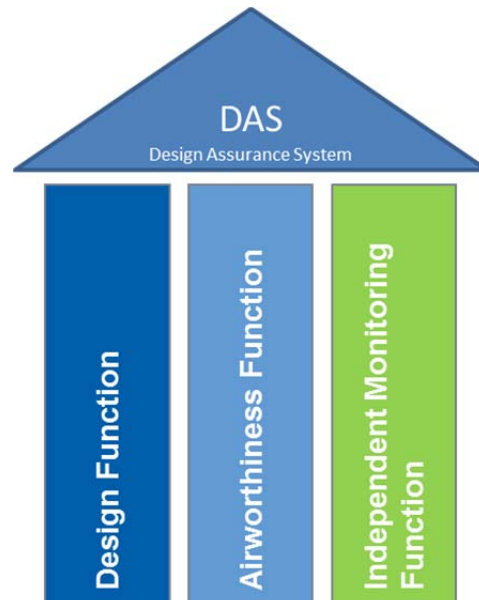


Figure 8-1: Design, Airworthiness and Independent Monitoring Function

Each delegation path starts at the Head of Design Organization (HdO) which is the holder of the overall authority. The term “Authority Delegation” is used, if the “Right to delegate” is forwarded to nominated personnel. “Design Authority” is the umbrella term for the three different kinds of authority - *technical, airworthiness and independent monitoring*.

Figure 8-2 shows how the authorities and responsibilities of these 3 ways are delegated top down in Airbus design organization.

AP1020 describes the delegation procedures in 4 modules with following topics:

- AP1020.0: Design Organization Authorities and Signatories (Main Module)
- AP1020.1: Design Organization Authority & Signatory Cascade
- AP1020.2: Supplier Requirements for the Delegation Management of Externalized Airbus Design Organization Signatories
- AP1020.3: Airbus Design Organization Signatories Roles, Responsibilities, Competence and Experience Requirements

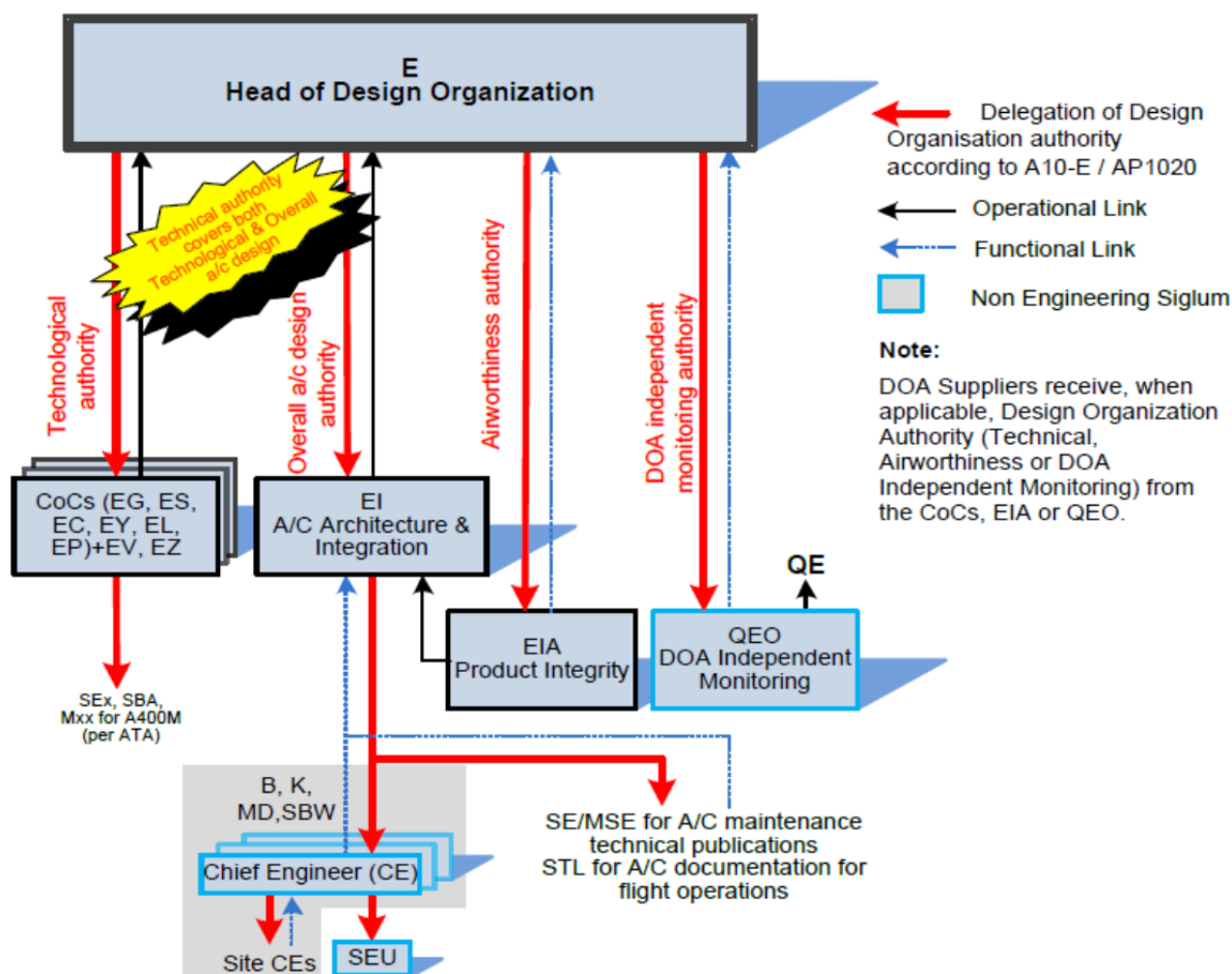


Figure 8-2: Way of delegation (source: DOM A20, Issue C Rev. 7)

8.2 Document Type Level

Design organization activities include the tasks that provide for (or directly contribute to providing for) the following categories of deliverables:

Level 1: Major Design Data contributing to the overall aircraft definition and performance, and related decisions.

Level 2: Data / Documents which influence or define the detailed design, design data evolutions, and design-related decisions.

Level 3: Other Data / Documents which support engineering activities, without any direct influence on the aircraft design.

An adapted ESD design Document Type Level List (DTLL) can be found in Airbus people searching for LS 1242078. In this list, you can find the level of documents and who (what role) is allowed to sign it.

8.3 Design Organization Signatories (DOS) Database

All relevant information concerning Design Organization Signatories awarded internally or by the supplier shall be retained by Airbus for its internal use within its information system or for submission to EASA upon their request. The DOS database is a IT solution with the task of showing on a transnational basis all internal and external Design Organization Signatories (refer to AP1020.1).

8.4 External delegation

Figure 8-2 shows the internal way of delegation. External companies can also achieve the right to nominate and delegate signatory rights to their own employees. First, they have to pass successfully an assessment (TDCA) before the company can receive the delegation by Airbus, and from that moment on, all delegations must be shown to Airbus. The delegated staff of external companies is also listed in DOS database and has exactly the same detailed restrictions (ATA, deliverable...) like Airbus internal signatories.

AP1020.2 describes how the externalization of signatory right is going on and which other documents are linked to it. A contractual document type level list (cDTLL) is created that contains a list of documents and the related signature which could be performed by the supplier linked to a specific workpackage. Another document is the design organization interface document (DOID) where it is explained, how the supplier's organization and way of work complies with the agreed procedures described in the Airbus Design Organization Manual.

Whilst a DOID is issued for "Design & Built Supplier" others may have an Airbus approval based on a Design Assurance Interface Document (DAID) both have to be kept up to date which is under the responsibility of nominated Design Assurance Coordinator (DAC) of the external company, the same function, which has to forward any new/extended delegation or other changes of the external design assurance system to Airbus.

Nevertheless all external delegations have the same result: The supplier approves design data under the Airbus DOA and not under the DOA of the suppliers company even if it has one.

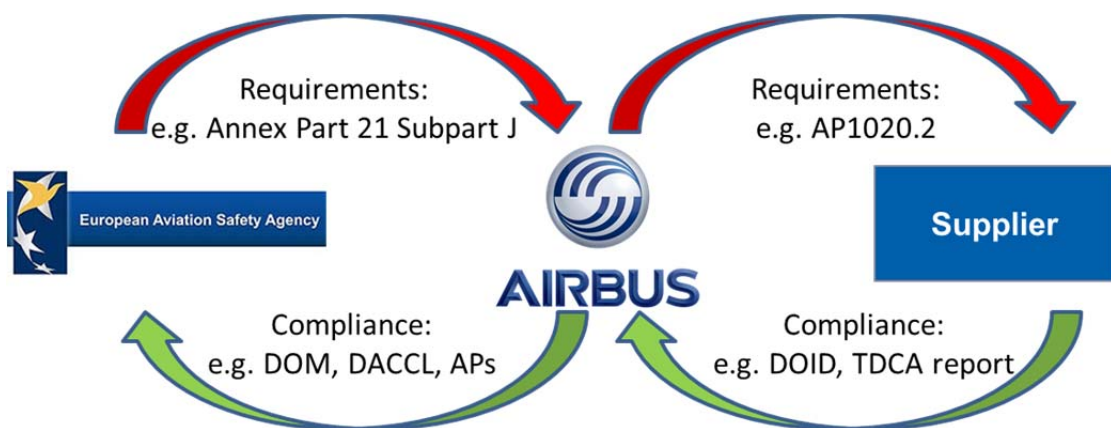


Figure 8-3: Design organizations requirements forwarded by Airbus

EASA wants Airbus to ensure the compliance with organizational (and airworthiness) requirements all over the extended enterprise. Airbus itself shows all the compliance to the Agency directly and in addition how the compliance on supplier's side is ensured, monitored and kept under surveillance.

In this context, the relation between Airbus and its suppliers is comparable with the relation of Airbus and the EASA (see Figure 8-3).

8.5 Design Function

The term “Design Function” covers all activities linked to the production of so called “technical documents”. It has to be ensured that during the production of those documents all applicable airworthiness and environmental protection requirements are met.

Linked activities to the production of technical documents are certainly the design itself, but also the proof of compliance with Airworthiness and Environmental Protection Requirements and the nomination and control of technical signatories.

8.5.1 Technical Authority (TA)

The Head of the Design Organization delegates the technical authority to the Heads of Center of Competence (CoC) and the Chief Engineers. These TA holders are allowed to delegate their authority to nominated individuals. It is possible to delegate technical signatory rights as well as design authority. That means, if we go deeper into the Design Organization, the usually combined right to sign/release documents and to delegate authority is split to different actors (refer to AP1020.1) – design authorities on the one hand and technical signatories on the other.

8.5.2 1st Independent Technical (Technological Expert) Signatories

First independent technical signature during the different release processes is the design checker.

It is a little bit confusing if you read in AP1020 that a Technical Approver is also called 1st independent technical signatory. This mixture of terms originates from the differences in how design deliverables are released (e.g. for stress sheet: compiled=authored, checked=approved and approved=authorized). For electrical definition dossier, we focus on the Technical Checker as 1st independent technical signatory.

The Technical Checker's activity generally takes place at Definition Dossier level per program and sub-ATA chapter. After the Compiler, the individual who put together the parts of the work, has done his work, the Technical Checker shall ensure that the technical content is “correct”.

The Technical Checker has to ensure that the work has been generated using appropriate methods, procedures and policies, and, if appropriate, perform any check calculations needed to verify accuracy. The Technical Checkers verify that the technical integrity of the content has been checked, and that necessary supporting evidence is available/has been retained.

In addition, they verify that the content takes proper account of, and where appropriate, meets the applicable performance, safety, quality and airworthiness requirements. Technical Checker are not allowed to sub-delegate their signature rights.

The SSCI confirms that the technical integration has been performed using the appropriate methods, procedures and policies and ensures that the technical content is validated against the appropriate documentation (TDD, SIDP...).

8.5.3 2nd Independent Technical (Technological Expert) Signatories

The Technical Approver's activity generally also takes place at Definition Dossier level per program and sub-ATA chapter.

The Technical Approver confirms by check & signature that the technical content has been generated using the appropriate methods, procedures and policies. The TAp ensures that the technical

integrity of the content has been verified, and that the necessary supporting evidence is available/retained.

In addition, the Technical Approver verifies that the content takes proper account of, and where appropriate, meets the applicable performance, safety, quality and airworthiness requirements. The TAp confirms that the engineering content, conclusions and recommendations have been verified when necessary. The TAp also endorses the responsibility on the technical content. Technical Approvers are not allowed to sub-delegate their signature rights.

Related to the technical content of the design deliverable, checker and approver have to perform the same kind of check, but the approver checks in addition if all applicable technical and procedural documents are delivered to designer and checker and if the design is consistent to the TRS.

8.5.4 Specialists

Depending on technical content and type of deliverable, different kinds of specialists can be involved. During the release of electrical schematic diagrams a system specialist and a system reliability check specialist could be involved. "Stress" specialists declaration confirms that correct dimensions and considerations of effective measures were used to assure the required lifetime of any component. For external equipment installation you may need confirmation of an aerodynamic specialist but what is common to all specialist signatures is that they are in the most cases not automatically involved in the release. So, in case of doubt or facing a technology on the deliverable where you are not skilled enough in to make a decision: ask the specialists!

8.6 Airworthiness Function

8.6.1 Compliance Verification Engineer (CVE)

The CVE represents the technical, airworthiness and certification competence for his respective division at AIRBUS. The CVE takes responsibility for all documents requiring airworthiness release or approval, such as ICS, certification plan, permit to fly, etc. This signature means that the "independent checking function for the showing of compliance" has been performed by a person not creating the compliance data.

The department they are working for shall nominate the CVE. The Airworthiness Technical Directorate shall perform the final appointment of the CVE. Within AIRBUS, regardless of the design area and across programs and ATA chapters, the CVE mainly assumes the airworthiness responsibility. His activity takes place predominantly at MOD level. However, in the event of technical obscurities the Technical Approver escalates these issues to the responsible CVE. Verification of the technical correctness and accuracy of supporting data can be performed either by the CVE or the technical authority delegation principles according to AP1020. As a consequence, signature for "independent checking of the showing of compliance" must then be performed by a person not creating the compliance data.

In order to perform MOD closure the CVE has to ensure the continuous, complete and on-schedule implementation of the definition and documentation from RFC up through MOD closure. Within this process, he has to ensure that quality standards, guidelines and instructions are respected. CVE names must be recorded on the signatory list of the DOM.

8.6.2 Designated Certification Specialist (DCS)

The Designated Certification Specialist (DCS) is the Airworthiness specialist nominated for a given discipline (ATA or sub-ATA level or for Approved Manuals) to carry out the certification work.

Their main activities are within the Type Certification Process where they participate in certification meetings at panel level with EASA specialists.

8.6.3 Designated Airworthiness Engineer (DAE)

For each program a DAE is nominated ATA overlapping to manage and coordinate the airworthiness activities within his scope of responsibility. Airworthiness release is undertaken by the DAE whose signature on the appropriate documents authorizes their release to the authority (e.g. certification documents, major MAS, etc.). The DAE monitors and signs every released document of CVE on airworthiness level (independent checking function). In addition, the DAE manages the interface between AIRBUS and responsible authorities by signing respective documents (e.g. minor MOD approval). DAE cooperates with the CVE regarding review of certification requirements.

For type certification (TC), supplemental type certification (STC) and major changes, airworthiness approval is granted by EASA.

8.7 Independent Monitoring Function

Independent monitoring also delegates approvers and authorizers but for deliverables others than design. An example of a document released by independent monitoring function is the Design Organization Interface Document (DOID) with which Airbus demonstrates that suppliers also fulfill the organizational (including procedural) requirements of Part 21. Design Assurance Nominees (DANs) are supporting the DOA independent monitoring department in case of any findings (e.g. wrong application of procedures) having a DOA root cause. The independent monitoring department is also the owner and approver of the DOM.

8.8 Chief Engineer (CE)

The Program Chief Engineers have a wide range of design responsibility and in addition, they have received a delegation of power which is not in scope of this compendium (ref. A10).

As the holder of the overall design authority of the program they are linked to, they sign for example on certification documents cover sheet and by that, they authorize them – e.g. Aircraft Certification Summary for a new Type or significant changes (AP2184).

By having the overall responsibility, chief engineer is not a “Design Function” only, the role also covers airworthiness and independent monitoring aspects and is based on a multi-ATA delegation as shown in Figure 8-2 “overall a/c design authority”.

To learn more about the outstanding role of the chief engineers inside Airbus, please refer to AP1020.

9 Begin the Design with the End in Mind

9.1 Continued Airworthiness

One of the most important things to keep in mind while building an A/C is the product life cycle. The organization responsible for the building of aircraft and putting the A/C into circulation is responsible for this product during the entire life cycle of the program which is assumed to be 99 years.

The ICAO states in Doc No. 9760 (Airworthiness Manual Vol. I & II Ed 1, Volume I — Organization and Procedures, Volume II — Design Certification and Continuing Airworthiness) that continued Airworthiness "... is the process ensuring that, at any time in its life, an aircraft complies with the technical conditions fixed to the issue of the Certificate of Airworthiness and is in a condition for safe operation."

The ICAO recommends that "... contracting states are required to have a system that ensures aircraft are in a condition for safe operation."

European regulations were based on that recommendation and EC No. 1056/2008 (27 October 2008) amending Regulation (EC) No. 2042/2003 on the continuing airworthiness of aircraft and aeronautical products, parts and appliances, and on the approval of organizations and personnel involved in these tasks enter into force as Annex Part – M of this regulation. The EC 1056/2008 must be in compliance with EC 1702/2003 Annex Part – 21.

This leads to the application of the 21A.3 ff. and the AMC 21 & GM 21. More details are stated in the DOM (refer to A20), chapter "DAS Compliance List with EASA Part 21 Requirements."

The relevant paragraph states that the DOM formulates the task in chapter 3.3.5 "Continued Airworthiness" and establishes an AIRBUS specific system for:

Collection, Investigation and Analysis of Data

Reporting to the Authority (for occurrence reporting refer to AMC 20-8 in AMC 20)

Investigation of Reportable Occurrences.

The details for this are specified primarily in "Continued Airworthiness Procedure" (refer to AP2020) where this procedure is referred to as module 3 of the AIRBUS Product Safety Process (AP2750) and refers to procedures (ENGO/MANO/ISO) covering the collection of events within AIRBUS as well as among suppliers (refer to AP2750).

ISO (In-Service Occurrence):

Is the AIRBUS internal procedure specifying the guidelines to be followed by the customer service representatives to sort events having occurred in-service to be presented as occurrences to the Screening Panel.

The ISO process is currently documented in AP2750 "In-Service Occurrences (ISO)".

Refer also to A5411.1 "Operators Information Transmission (OIT)", A5412.1 "Alert Operators Transmission (AOT)" and "Airbus Safety Criteria and Guidance for Issuing Product Safety Process (PSP) Inputs" (M1750).

ENGO (ENGINEERING Occurrences):

Is the AIRBUS internal procedure giving guidelines to be followed by the Engineering (COC/COE including

extended Enterprise and suppliers) representatives to report events known at designer or specialists level and to be presented as occurrences to the Screening Panel.

MANO (MANufacturing Occurrences):

Is the AIRBUS internal procedure giving guidelines to be followed by the Manufacturing/Quality representatives to report events known before delivery, and to be presented as occurrences to the Screening Panel.

Please note: As per the DOM, the CAE nominates a member of the CAE team to manage the Continued Airworthiness day-to-day work. This member is designated as Certification Airworthiness Manager (CAM) or Head of Continued Airworthiness and acts on behalf of the CAE for Continued Airworthiness activities. Therefore it shall be understood that each time the CAE is referred to, the CAM may replace the CAE for these tasks.

9.2 Electrical Wiring Interconnection System (EWIS)

Introduced by the EASA with NPA 2007-01 the EWIS is now integrated in the official rules of the EASA in Appendix H of the CS-25.

“... The Regulation includes system requirements, servicing information and maintenance instructions, including the frequency and extent of inspections necessary to provide for the continued and approved airworthiness limitations section in the ‘Instructions of Continued Airworthiness’ (ICA).
...”

EWIS is first covered by the addition of a new Subpart H to FAR-25 and also to find in current CS-25 issued by EASA. You can find the EWIS related authority requirements following the number 25.17xx.

Within EWIS there are two main clusters during the process:

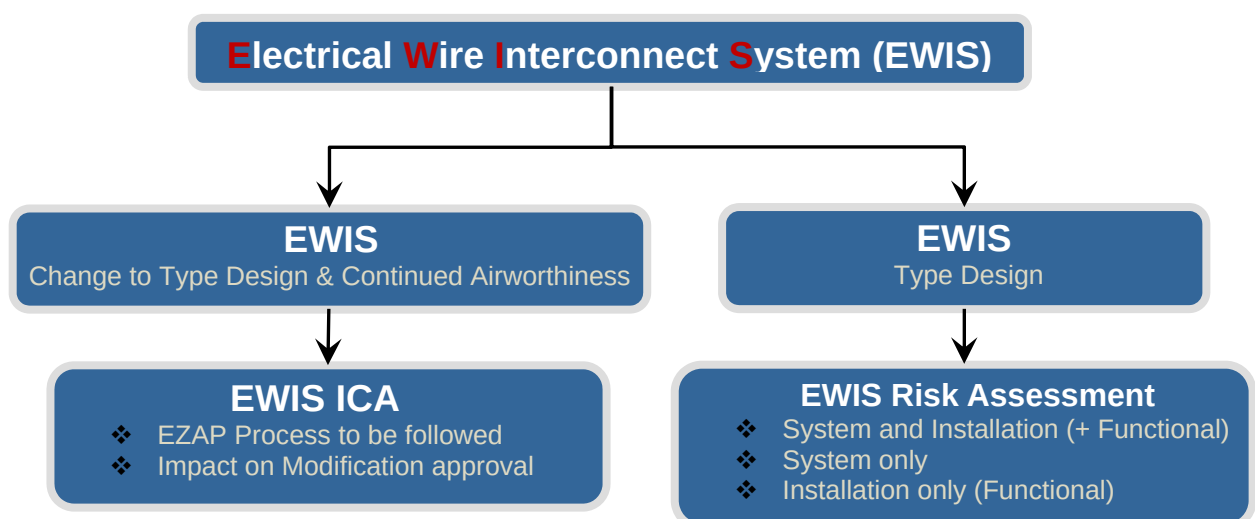


Figure 9-1: EWIS for Type Certification and changes to type design

9.2.1 EWIS (ICA) Change to Type Design & Continued Airworthiness

These requirements have to be fulfilled by the design of any large aircraft. In addition, Part 26 “Continued Airworthiness and Safety Improvements for Transport Category Airplanes,” includes EWIS requirements (26.11) concerning the continued airworthiness by the order to develop Instructions of Continued Airworthiness (ICA).

Since 10th December 2009 each modification needs to be assessed with respect to its impact on EWIS ICA. Part 26 is part of the applicable requirements to any design change to an aircraft. If the determination of impact/non-impact is not available, compliance with Part 26 cannot be shown, and the modification cannot be approved.

These topics are also part of the “Enhanced Airworthiness Program for Aging Systems” (EAPAS) which has been in effect since the late 1980s to reduce safety concerns related to “wire issues.” Most of the related data to this topic arises from accidents and incidents.

You see that EWIS requirements are not a topic of new developed A/C like A350 only, they also impact the flying fleet and all design changes of any program.

This leads to the philosophy “protect and clean as you go” which is discussed in detail in document AC 25-27A published by the FAA in 2010.

The AC 25-27A stresses the importance of inspecting EWIS in the Design and Production Organization as well in the Maintenance Organization which lead among others to enhanced zonal analysis procedures (EZAP).

This leads in turn to the fact that occurrences during an EZAP in the Maintenance Organization can have significant major impacts on the certification base of an aircraft even if EWIS was not relevant for the type certification of this aircraft in question.

Different to A350, EWIS was not part of the certification base for legacy programs (incl. A380) and has been introduced to them by mean of CRI and special conditions. In legacy programs, an EZAP has to be performed, identifying special conditions in the areas, where EWIS components are installed, e.g. the probable accumulation of combustibles.



Figure 9-2: Risk of smoke and fire shall be additionally prevented by EZAP results

9.2.2 EWIS Type Certification

For A350, EWIS impacts the ATA92 certification activities the way that all the electrical installation will be **certified like a system**.

That does not mean that there are new requirements existing (certainly there are, but this is usually the case if you develop a new A/C on an actual certification base), it means that you need to show compliance for all the electrical installation as one.

This also covers the functional aspect which was in the past under responsibility of system definition. But ATA92 needs to show that the “electrical wiring interconnection system” works probably.

During the design this makes new processes necessary. To secure the requirements on EWIS Level a “Risk Assessment” has to be implemented to ensure the independencies between function and installation. For the A350 a ATA92 DCS was nominated.

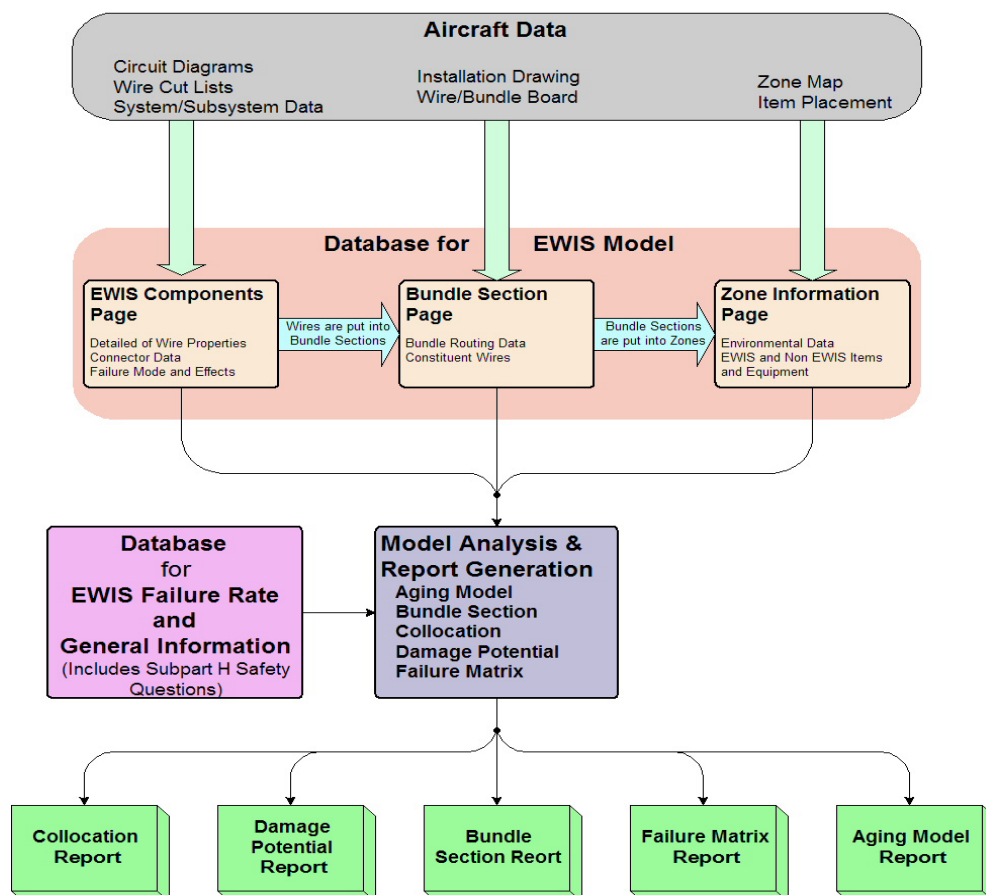


Figure 9-3: EWIS Risk Assessment

10 System Design-Related Requirements

10.1 “Helper Concept” Functional Item Number (FIN)

In relation to the ATA, each significant component of an aircraft which has a specific function is assigned a number which clearly defines both the function and the installation location of the component. This number is called the Functional Item Number (FIN, refer to AM2212 and ABD0036).

The FIN is one element to support the requirement “Identification” coming from Configuration Management (CM) discipline, which states that every “object” must be uniquely identified.

Typical examples of components that are assigned a FIN are equipment items, connectors, manifolds or grounding / bonding points. In order to guarantee a clear definition of the installation location of a component, a FIN can only be used once per aircraft.

One has to keep in mind that a piece of Software within an A/C is also identified with a FIN and that it's urgently important during (re-)assembly to ensure that the Software FIN in question is loaded to the correct Hardware FIN.

A FIN can be assigned to several parts if, for example, the components relate to an equipment item. Each part has a part number (PN).

Virtual Circuit	Meaning
VA	Avalanche Diode Module
VB	Wire Bundle/Harness
VC	Connectors
VD	Diode Modules
VE ¹⁵	Panel (Commercial)
VF	Thyristor Modules
VG	Ground Terminal Modules
VH	End Line Resistor
VI	Virtual Interface / Connector Plate/Mount
VM	Integrated Control Panels
VN	Ground Points
VP	Pressure Seal
VR	Resistor Module
VS	Splice
VT	Terminal Block
VU ¹⁵	Panel
VX	Printed Circuit Boards
VZ	Spare Wire

Table 10-1: Overview of virtual circuit / Dummy codes for a circuit

¹⁵ It is no longer possible to draw a clear distinction between VU and VE, as both components are increasingly merged into a single type of equipment. On the A380 program, apart from a few exceptions (e.g. 5100 VU = Rear eBay), VE are mainly used.

The FINs specifically used for the electrical domain are made up of a four-digit sequential number with special circuit identification. These circuits are called “virtual circuits,” which defines the meaning of a function within a circuit.

The “V” inside the FIN means “Virtual” which in this context means that several ATA chapters use this “virtual” component (e.g. VC – Virtual Connector).

A VB (virtual bundle) is virtual, because it is not said, that the bundle has also one single physical reference. For system definition, it is one bundle, but on its way to become a producible and installable physical cable collection, it could be broken in sections due to workshare and/or manufacturing reasons.

10.2 Architect's Point of View

One of the main points the architect must keep in mind is the function in relation to the requirement and the interactions and interfaces between the different Systems (ATA) and their unique identification (FIN).

The development of a new program begins with questions such as, “What principles from the running programs seem to work with the new one?”

For example, the question “Is it a good idea or not to use the hot air from the turbines, bleed air, and what about the weights?”. If the answer to this question is “yes,” this topic will be part of the Top-Level System Requirements (TLSR). The TLSR estimate the systems and basic concepts in the new program.

The next main steps are:

fixing the first digit of the ATA,

defining the essential or non-essential routes,

defining the line-name for a connection between one object and another object.

The identification of objects will be realized with different systems. The unique identification of an object item in the A/C is done with the Functional Item Number (FIN) Concept (refer to AM2212 for more details).

The system designer will now start to produce Block Diagrams (BD) and these BDs will show the connection between FINs and the line-name.

The architect is now in charge of clarifying the consistency of this BD and the relation to the TLSR.

In case of A350 and its Electrical Structure Network (ESN) the return path, voltage drop and ground voltage fluctuation has also to be controlled and checked by the architects. A current injection and length assumption table (CILAT) filled by system designers provides all necessary information. The electrical architect controls and negotiates discrepancies of these assumptions and the specific design work (for details see technical report “V2477RP1104240”).

10.3 Requirement-Based Engineering (RBE)

As stated in the “Requirements-Based Engineering Policy” (AP1034) the RBE is an integral part of the domains Development, Fulfill, Support and Enable (refer to “AIRBUS Operational Process Map (AOPM)” (A02)). The RBE application in use in this case at AIRBUS is DOORS (Dynamic Object Oriented Requirements System).

The aim of the process is to ensure assurance that requirements are valid, to provide feedback on the development process and to ensure customer satisfaction.

The Validation & Verification process intersects with the Requirement Based Engineering Process (defined in AP1034). The focus will be A350XWB Requirements Cascade (with regard to ATA 92 Electric). The V&V plan describes the V&V process managed by the Electrical architecture team in interaction with the other teams (Fuselage, Wing, RFE, Systems, ...).

The Electrical V&V process is fully in line with the AP1034 and the A350XWB V&V plan. As there is no deviation this document can be referred to for more information. To change such data, all requirement changes in the TLElecRD have to be managed via the RCN process (refer to AP2078.3).

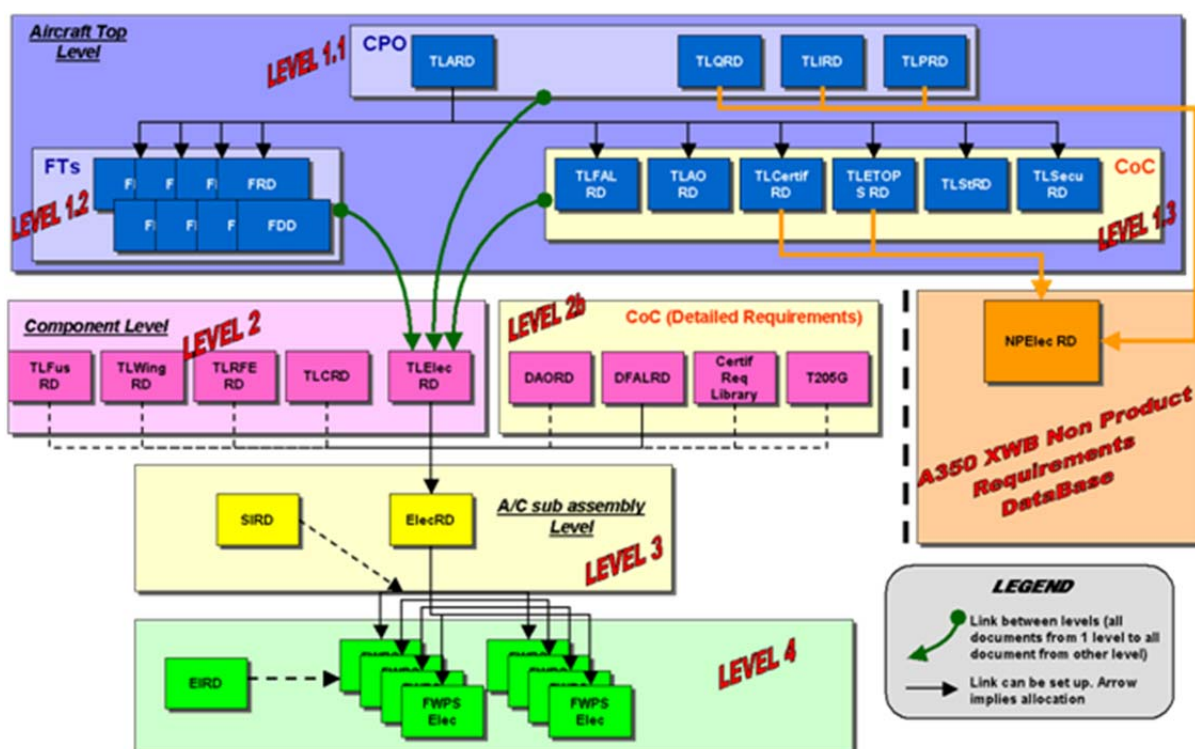


Figure 10-1: Overview, RBE, Requirement Cascade

10.4 Validation & Verification (V&V)

V&V is finally introduced in the EN9100 and described in the scope of RBE in AP1034.

Validation means that there must be proof that the design output meets its requirements for the design, meaning e.g. the EASA Part-21 and CS-25. This is in the words of the Authority a Certification, a MAS. In other words “is this design in accordance with Authority (e.g. EASA) requirements?”

The EN9100:2003(E) states in chapter 7.3.5, “Design and Development verification”:

“... to ensure that the design and development outputs have met the design and development input requirements. ...”

Verification means proof that the product meets the requirements specified in the Definition Dossier, in other words, “Is this product exactly what the Design Organization requires by the DD?” This is in the words of the Authority an “Attestation” (e.g. EASA Form 1).

The EN9100:2003(E) states in chapter 7.3.6, “Design and Development Validation”:

“... to ensure that the resulting product is capable of meeting the requirements for the specified application or intended use, where known. Whenever practicable, validation shall be completed prior to the delivery or implementation of the product. ...”

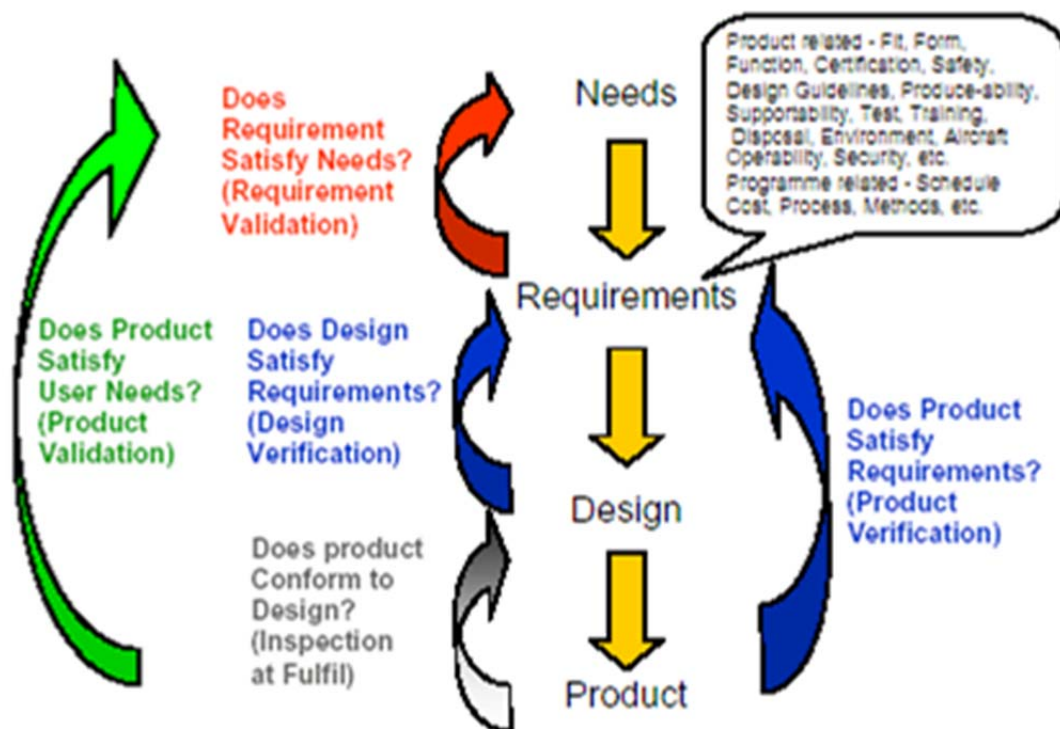


Figure 10-2: Overview, V&V Flow

The most important steps can be formulated as:

- Check that the requirements internally to the supplier (e.g. Airbus) meet the customer (e.g. Airline) needs (Requirement Validation)
- Check that the design meets the requirements (Design Verification)
- Check that the product meets the design
- Check that the product meets the requirements (Product Verification)
- Check that the final product delivered meets the customer needs (Product Validation)

For A350 V&V electrics, document V92PL0802812 describes the steps to finally proof compliance of the design. It is important to know that only critical requirements of SIDP20 are additionally checked, all other requirements are proofed under Technical Approvers responsibility by signing a checklist provided by the designers which contains:

List of verified requirements

For each group of requirements, an example on a dedicated location showing the compliance to the requirements. The compliance will be proved based on a DMU snapshot.

Reference of the deviations opened following the verification.

By signing the list, the technical approver ensures that the requirements are fulfilled in related areas and not only on the shown examples.

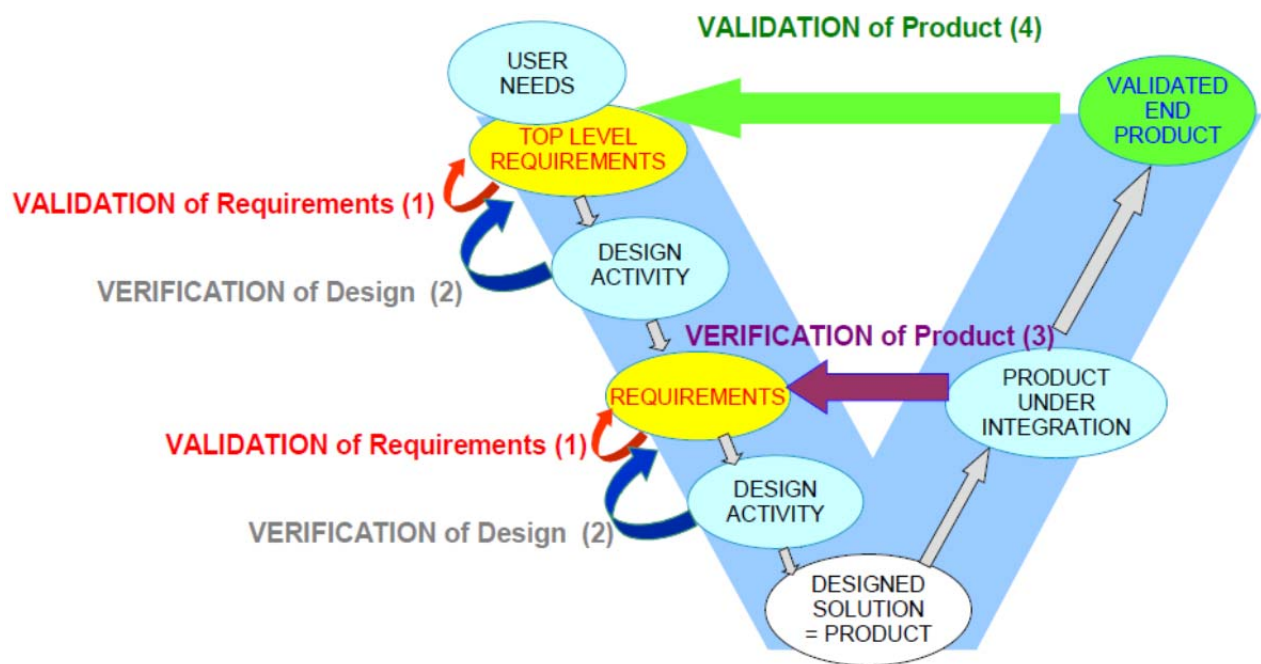


Figure 10-3: Overview, Simplified V & V Flow (refer to AP2296)

10.5 Top-Level Requirements Dossier (TLxxRD)

Top-Level Requirements Dossiers are a key element to Requirement-Based Engineering (RBE) which in turn is essential for AIRBUS as a key contributor to the discipline of Systems Engineering (SE) understood in accordance with ISO 15288 (Systems and Software Engineering — System Life Cycle Processes).

RBE includes Functional Analysis (FA), Requirements Engineering (RE), Configuration Management (CM) and Verification and Validation (V&V).

Starting with the user needs, Top Level Aircraft Requirements are formulated (e.g. number of passengers, weight, payload, range, etc.). These requirements are broken down into requirements for Major Components of the A/C, e.g.:

- Fuselage
- Wing
- Empennage
- Systems

The decomposition of the requirements goes further until the manufacturing level is reached. Further details are provided in the document “Requirements-Based Engineering Policy” (refer to AP1034) which references the following additional documents:

- Develop New Aircraft Business Process Definition (refer to AP2054)
- Aircraft Project Management Policy (refer to A1002)
- Aircraft Functions Definition Process (refer to AP2161)
- Requirements-Based Engineering Method (refer to AM2150)
- Validation & Verification Process at A/C Level (refer to AP2296)
- Functional Analysis Method (refer to AM2389)

Special attention should be given to AM2150, as the implementation method used for the Tool “DOORS” is described in this document and especially with regard to A350 to the TLElecRD.

The purpose of the Top Level Electrics Requirements Document (TLElecRD) is to define the technical requirements applicable to the development of the A350XWB PAX all variants regarding Electrical Installation and Architecture.

According to previous consideration we put in place during the “solution process” full verification of:

- xxRD requirements
- SIDP 20 & 92
- ERHCD
- ESN SIRD

and finally to ensure the design verification of these “specific” requirements today included in the TLElecRD.

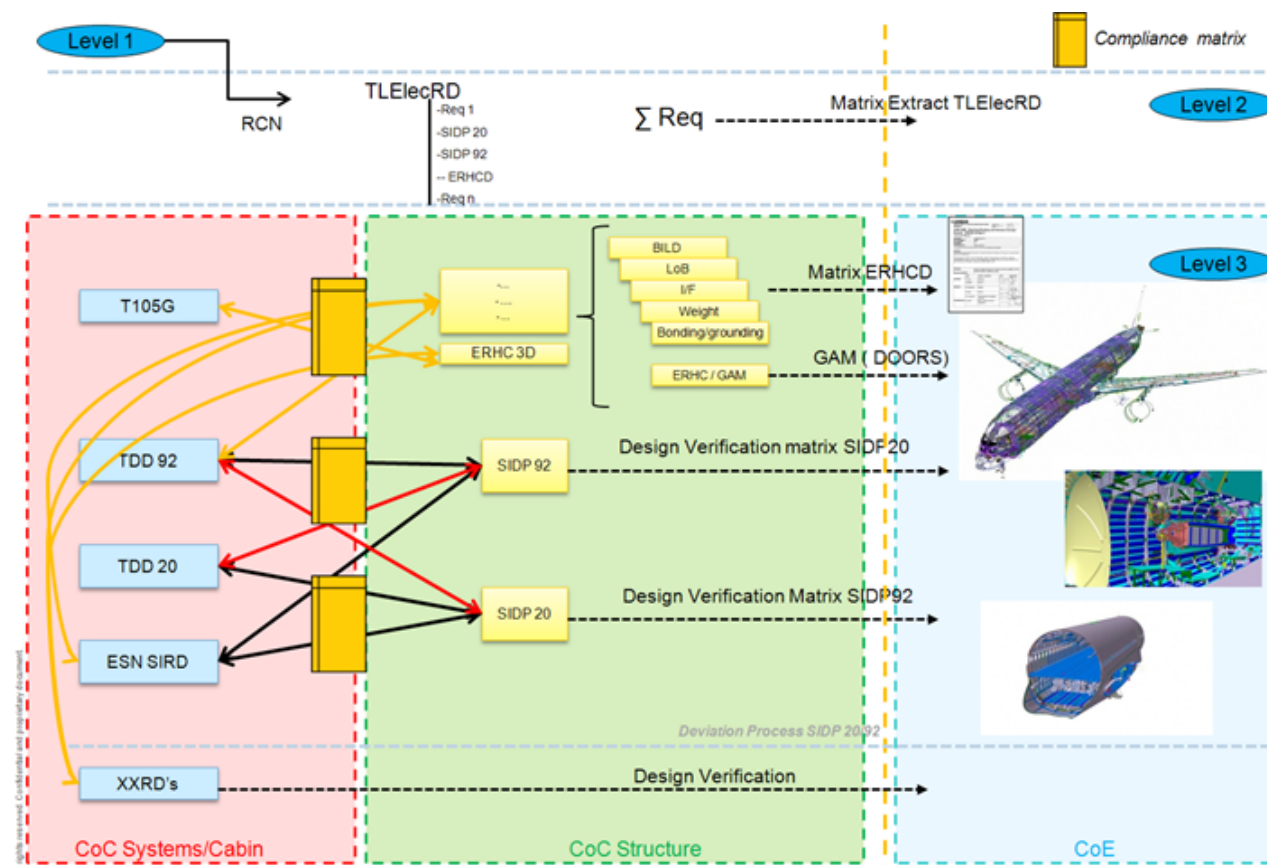


Figure 10-4: Overview, TLElecRD – Position in the TLxxRD Process

10.6 System Definition & System Requirement Dossiers

The different System “xxx” Dossiers and related documents are introduced in “Electrical Definition and Installation Process” (AP5997.2). More details on system design are to find in chapter 15.

- System Installation Requirement Dossier (SIRD)
- Equipment Installation Requirement Dossier (EIRD)
- System Wiring Requirement Dossier (SWRD)
- Bay Installation Requirement Dossier (BIRD)
- Bay Installation Layout Dossier (BILD)

11 Type of Change Processes

11.1 Requirement Change Note (RCN) Process

This process is applicable to the A350 and future programs and is one part for implementing requirement-based engineering. The description of the process is featured in "Requirements Change Note Process" (AP2078.3) and is located in the Development Process.

The Development Process is described in the "Develop and Ramp-Up Excellence - DARE" (A1350) and RCN is located between the Maturity Gates (MG) 3 and MG 4.1.

11.2 Change Note Process

The Change Note (CN) is located before the transition in the Full Change Process, which starts after Milestone M7 (for the running programs) or MG5 for the A350 and future programs.

The Change Note process starts after MG 4.1 and is similar to the Full Change Process but without the Modification System.

The description of the process for A350 is featured in "Change Note Process for A350XWB and Future Programs" (AP2078.2) and in "A380 Change Process for Concept and Definition Phases" (AP2078.1).

11.3 Full Change Process

After MG5 for DARE (Develop and Ramp-Up Excellence) and/or M7 for DnA/DnE (Develop new Aircraft / Develop new Electric) the Change Process is carried out by means of Modifications.

The Modification System as given in the A1080 (ACMR) will be activated during the Transition Phase.

This is also the transition from the SAM (Maturity C) into the SFM/GRM.

The documents will change from the Change Note (CN) to the Change Items and the Change Documents.

The Change Items, the Change Request (CR) and the Modification Proposal (MP) are described in "CR/MP Concept and Rules" (M20091).

The Change Documents, the Technical Repercussion Sheet (TRS), the Technical Dossier (TD) and the Modification Approval Sheet (MAS) are described in "Change Process – Technical Change Documents (TRS, TD, MAS)" (AP5130).

The analysis of the Industrial Repercussions is described in greater detail in "Change Process – Industrial Repercussions Elaboration" (AP5157).

Whatever the reason for the change, the two basic types of change can be differentiated as follows:

- Changes originated by the Customer, known as "Product Customization". This is normally generated by means of a Customer Query (CQ) and/or a Request for Change (RFC).
- Changes originated inside AIRBUS, known as "Product Changes". This could be generated by a Request for Work (RFW) (or this can be also the consequence of a RFC).

The diagram illustrates the project lifecycle, organized into five main phases represented by blue curved shapes at the top:

- Feasibility Studies & Definition**
- Design / Design Specification(outsourcing)**
- AC-Delivery**
- In-Service**
- Aging Aircraft**

The process flow is depicted as a series of interconnected boxes and documents, with arrows indicating the sequence of activities:

- Initial Documents:** No Signature, DAE, CVE.
- Feasibility Phase:** RFC/RFW/CQ.. leads to CR, which leads to MP/TRS, then MP opening, and finally BD.
- Design Phase:** BD leads to SAM A/B/C, which leads to GRM. GRM leads to TRS / Sub-TRS, MOD opening, and Full-TRS / TD. Full-TRS / TD leads to RDR. RDR leads to Design / Design Specification(outsourcing).
- AC-Delivery Phase:** Full-TRS / TD leads to MCx, which leads to MOD Closure. MOD Closure leads to DDP, which leads to Drawings. Drawings leads to DQN, which leads to NC. NC leads to Concession. Concession leads to ICS, which leads to MAS after TC. MAS after TC leads to TA.
- In-Service Phase:** ICS leads to xSB.
- Aging Aircraft Phase:** TA leads to In-Service.

Additional milestones and documents shown include:

- Checkpoints:** RTE 1/2, Checker, RTE 3, TAp.
- Other Documents:** WD, Referential, PD/EPD, FD, and SAM A/B/C.

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12 Phase Concepts in the Different Change Processes

12.1 Overview

The Change Process is divided in different Phases which will be used with different content in the Change Note (AP2078) and Full Change Process (A2843). A first short description is given in the following chapters.

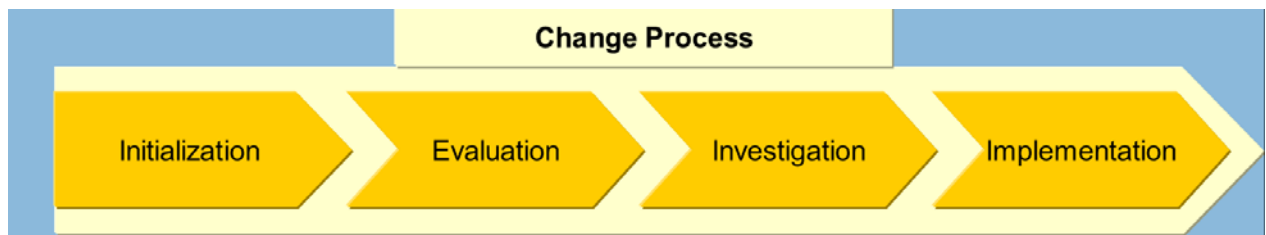


Figure 12-1: Overview, Change Process Phases

12.2 Initialization

The initialization phase formalizes the request given by one or several triggers (e.g. CQ, RFC, RFW, etc.).

The aim is to identify and formulate clearly in a first step the following items:

- Is the request consistent?
- Is the request complete?
- Is the request easily understood by any actor?
- Are the necessary process steps for resolving the request clear?
- Are the responsibilities in the process clear?

12.3 Evaluation

The Evaluation Phase attempts to give answers for the possible technical solutions.

An evaluation of costs and possible risks is also within the scope of the evaluation phase.

12.4 Investigation

The Investigation Phase provides details on the technical solutions and information useful for MOD Opening. All actors involved estimate their technical, industrial and costing positions.

The Modification Proposals (MP), i.e. short descriptions of the technical solutions, are normally created during this phase.

12.5 Implementation

MOD Opening is performed with the data from the MP and the detailed solutions.

The detailed solutions are provided by the TRS and the TRS is bound to the MP.

The industrial data for the embodiment point (ERS/PRS) and a proposal for effectivity are provided.

The site performing the role of the MOD Leader as well as all other parties involved will be defined.

12.6 Conclusions Regarding Change Note/Full Change Process

The principle steps and the main questions in the Change Note and the Full Change Process are quite similar. The main difference is in the documents involved, primarily the Modification System used in the Full Change Process. The Change Note is more flexible, which could be seen as a good idea in the early stage of the Development of the A/C.

13 Origin of a Change

The origin of a change can arise from a vast number of reasons. Examples of such Changes could be:

- Product improvement
- Cost reduction
- Production improvement
- New product development
- Correction of “defects” (which would be called for example “Non-conformities”)
- New official requirements (by the Authorities)

All of those origins will be divided into two parts (Product Change, i.e. implemented by AIRBUS, or Product Customization, i.e. tailored to the customer’s needs or wishes).

13.1 Nature of Changes in the ATA 92/87 Domain

The nature of this type of change is such, that normally ATA 92/87 itself is not always the origin of a change. Of course this depends on the program, but “other” ATA chapters introduce also changes with an impact on electric components.

It is important to remember that ATA 92/87 has to be informed by the change owner in order to state their impact. Not being informed in time could lead to extensive delay of ATA 92/87 design deliverables.

Changes may occur from inside or outside AIRBUS.

13.2 Changes Introduced by the Authorities

Changes by the Authorities are normally introduced by means of an AD or CRI. This leads normally to an internal AIRBUS Request for Work (RFW) which after approval by the Authority follows the normal flow of a change.

13.3 Customer-Initiated Changes

13.3.1 Request for Change (RFC)

A RFC (Request for Change) can be every kind of “need” formulated by the customer and translated by AIRBUS, e.g. the Customer requests a special IFE (In-Flight Entertainment) System. An RFC contains a good and clear description of the intended changes needed to fulfill the customer request for change.

The process to create a RFC normally begins during talks with the customer. This in turn leads to “customer wishes” which can be a “known subject” or a “new subject.”

In the case of a “known subject” the EPAC (Entity per Aircraft Change)/TDU (Technical Description Unit) will be selected from catalogs (e.g. the SCG (System Configuration Guide)). The EPAC is the “Function” and the TDU is the technical solution.

This will lead to the creation of a RFC, if several EPAC/TDU are collected; a GRFC (Global RFC) is created and passed to Customization Screening.

If the subject is new, the Contract Manager and the Cabin Engineer will collect the customer request and will formulate a concrete and complete “need.” The output is a specific request with sufficient information to proceed in the process.

EPAC/TDU management and Chief Engineering will allocate and/or create an EPAC/TDU and assign this with a TDU description, a selling policy, the fleet and the expected A/C to the created RFC. This formalized RFC will be passed on to Customization Screening.

The document “Create Catalog and Manage Standard Offer Data” (refer to AP5175) describe the process to create and update A/C catalogs.

In every case customer screening will probably propose studies, a leader and a target date.

All results arising from this process, MSCN (Manufacturer Specification Change Notices), SCN (Specification Change Notices), RFC or other customization documents must be well documented.

A MSCN (Manufacturer Specification Change Notices) is a proposed contractual change to the basic aircraft introduced by AIRBUS. This describes an evolution of the standard and will be part of the contractual definition if the customer accepts a MSCN.

The SCN details the “translated” customer wish into “AIRBUS language.” The SCN also specifies applicability, price and other details and is therefore the AIRBUS answer to a RFC. If both parties sign a SCN this SNC becomes part of the contractual definition.

Additional information is given in “Manage Fleet” (refer to M20196).

13.3.2 Customer Query (CQ)

CQ (Customer Query), e.g. the customer detects violation against the contractual framework during the production of the A/C before delivery.

13.4 Changes Initiated within the Organization / Request for Work (RFW) Series

Examples of product changes include:

- RFW (Request for Work) 1000 (engineering OR customer service), e.g. new information from aging A/C implies changes also for that customer.
- RFW 5000 (Engineering Department)
- RFA (Flight Test Department)

14 Change Items & Documents

14.1 Change Items

Change Items are objects formalizing the flow of the intended Change through its related process. Two Change Items have been distinguished, the Change Request (CR) and the Modification Proposal (MP).

Therefore the Change Items are not actually “real Documents,” but constitute a “folder” and / or a process that is identified and characterized by a distinct number (of course the CR and the MP can be printed out at this point).

The document “CR/MP Concept and Rules” (refer to M20091) provides a detailed description of these items.

14.1.1 Change Request

The Change Request¹⁶ represents the technical and invariant part of the Request (provided for example by incoming RFC, CQ, RFW or other triggers) and is the invariant part of the request. The Context represents variant parts, line-requested applicability and other data, like the Origin Document Reference. A Change Request can group several associated Context parts arising from documents of differing origin. The document “CR/MP Concept and Rules” describes this in greater detail (refer to M20091).

14.1.2 Modification Proposal

An MP is the short description of a technical solution. A MP will be bound to one or more Modifications (MOD).

To avoid any confusion, it is important to know that the A380 had defined three types of MP. This definition is given in the Harness Agreement letter, published in the “TRS Handbook” (formerly known as “TRS Cookbook.” The three types of MP used in the A380 program are:

MP1: Elements which must be installed on section level and do not depend on the customer cabin layout (this is true mostly of provisions),

MP2: Elements which must be installed on section level possibly depending on the customer cabin layout, and with the benefit of reducing the workload of FAL1,

MP3: Remaining elements that must be installed in FAL1.

This system is only valid for the A380 and the type of the MP is NOT directly visible in any tools nor specified by any kind of documents inside the official documentation system (MyDoc).

¹⁶ Refer to “ICC Training – Viewers,” prepared by EDC – Configuration Management.

14.2 Change Documents

14.2.1 Repercussion Sheets

14.2.1.1 Technical Repercussions (TRS)

A TRS contains the (more or less) detailed technical description of the change. A TRS is always bound to an MP (which in turn is bound to one or several MODs (Modifications)). Different types of TRS are defined which “exist” in different phases of the Change Process.

An Evaluation TRS gives a rough description of consolidated repercussions from leader and involved organizations. The Evaluation TRS exists normally between Decision Gate (DG) 1 and DG 2 within the Evaluation Phase. The Evaluation TRS is normally required for complex CR and for support in creating a SCN (to answer a RFC)

A TRS for MOD Opening should contain the complete description of the technical solution for the change and this will be the technical basis for the MOD decision.

The Full TRS contains the complete and final solution provided with this TRS and finalizes the repercussions for Attestation. The Full TRS will also be the basis for TD and MAS which is relevant for the Certification of this change.

The document “Change Process - Technical Change Documents (TRS, TD, MAS)” (refer to AP5130) provides more details as to the different TRS.

14.2.1.2 Production & Point of Embodiment (PRS/ERS)

The central term is the Point of Embodiment (POE), which defines a set of A/C where the change will be applied and this will become the validity of the change. The decision as to the POE is taken in DG3 only and is driven in the case of a product customization by the related customer.

The main data for taking a decision about the POE is contained in industrial repercussions which will be expressed via a “Production Repercussion Sheet” (PRS). This PRS will feed data into the decision process and formulate a proposal for the POE.

With this position the PRS process is a major set of data for the POE and will be used in Evaluation and Investigation phases for production only.

“... The Embodiment Repercussion Sheet (ERS) expresses the POE proposal for DG3 resulting from all constraints: industrial, FCO Managers, Customer Service, Engineering Program.” More details regarding PRS/ERS are provided in the document “Change Process – Industrial Repercussion Elaboration” (refer to AP5157).

14.2.1.3 Costs & Weights (CRS / WRS)

The Cost Repercussion Sheet (CRS) contains information as to the upcoming costs of the change and will normally be evaluated during the Investigation Phase. The CRS data is for internal use only. Generally, it is not transferred to the customer, but the CRS results will be a point of departure for talks with the customer.

The Weight Repercussion Sheets (WRS) evaluate weight impacts coming from different objects (e.g. wire length) and are required for filling out TRS Sheet 6 data.

14.2.2 Technical Dossier (TD)

The TD contains all technical impacts of a MOD and is the prerequisite for creating a MAS. The TD contains all relevant information from all Full TRS contained in one MOD (bound to the MPs).

Under special conditions (e.g. the MOD contains only one MP) the creation of the TD may be omitted and the MAS can be created directly. Further details are provided in the document "Change Process - Technical Change Documents (TRS, TD, MAS)" (refer to AP5130).

The TD contains the sheets 0 to 4 only because the information on sheets 5 and 6 are not relevant for the preparation of the MAS (refer to AP5130).

14.2.3 Modification Approval Sheet (MAS)

The Modification Approval Sheet is an internal airworthiness document that serves as an internal certification summary and Mod approval evidence used internally at the end the EASA approval of the modification.

"It is issued after the compliance demonstration is complete (meaning that all compliance documents, irrespective of category, are available and complete) and the final compliance statement is made to EASA through the MCCA. This document does not contain in itself demonstration of compliance. Therefore, no CVE signature is needed." (refer AP2243)

The MAS confirms and documents the completeness and accuracy of the documents describing a Modification (MOD) which can consist of several Modification Proposals (MP).

The MAS contains the sheets 1 to 3 of the TD from which it was created. The information is exactly the same as in the corresponding TD.

The MAS contains the information relevant to acquiring certification of the MOD and the Design Organization is responsible for signing the document on behalf of the DAE.

15 System Design

15.1 (Non-)Specific Design Work (NSDW / SDW)

“... Non-Specific Design Work (NSDW) describes systems and their interconnection. It is aircraft oriented and concerns every team where the system will be defined / installed.

Specific Design Work (SDW) defines the physical wires of a circuit, including manufacturing and installation information. It concern one team.”

15.2 Electric Schematic Diagrams (ESD) System Design

All ESDs shall be created with only one tool called SEE XP (SEE Electrical Expert, refer to AM5014). Even if other tools are still in use, they shall be replaced by SEE XP in the future (like CCD in A-D). The two ESDs, BD and FD are part of System Design, while the PD, EPD and WD are part of functional definition. All of these documents are also important for preparing and providing Customer Services with the definition of the systems installed in the A/C.

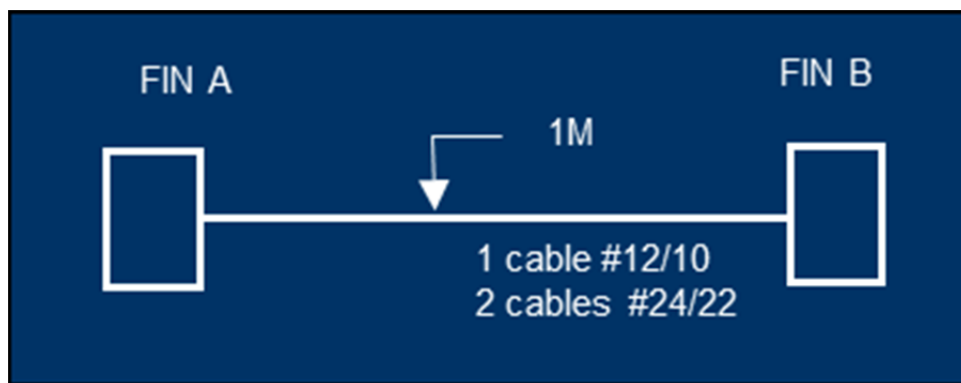


Figure 15-1: Block Diagram (BD) illustration example

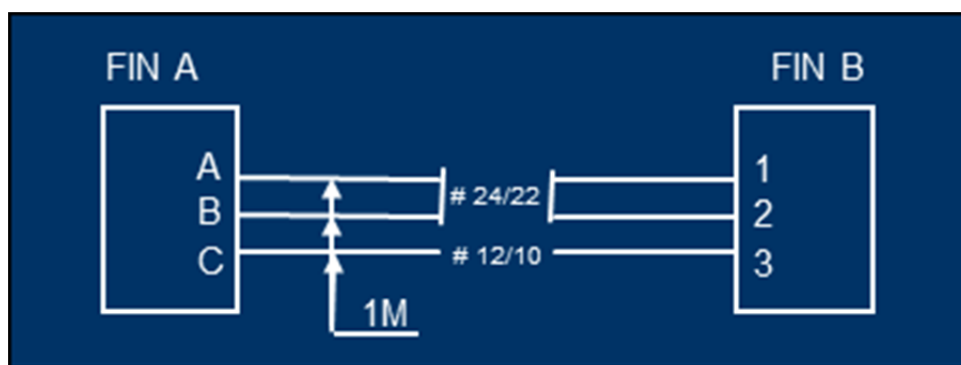


Figure 15-2: Functional Diagram (FD) illustration example

Content of BDs is a system architecture overview and used for pre-sizing activities. The FD is used for system definition, pre-sizing activities, EPD design and test purpose. While a BD only contains “connection lines” between FINs, the FD already contains “logical wires”.

15.3 Block Diagrams (BD)

A Block Diagram (BD) is an element of the “Electrical Schematic Diagram (ESD) – Family.” The BD is related to the System Design and therefore Non-Specific Design Work.

A short description is given in the document “Create Specific Electrical Functional Design” (refer to AP5997.1). The position of the BD in the process flow of the ESD is also provided in that document.

The BD is basically a System Wiring Requirement Dossier (SWRD) with Maturity A. Details on the BD can be found in “Specification of System Wiring Requirements” (refer to ABD0200.1.1.4).

Starting after MG3 the following information will be incorporated into the BD preparation:

- TLSysRD (Top Level System Requirement Dossier)
- TLCabRD (Top Level Cabin Requirement Dossier)
- TLElecRD (Top Level Electric Requirement Dossier)
- SRD (System Requirement Dossier)
- CRD (Cabin Requirement Dossier)
- TDD (Technical Design Directive) from previous programs

This leads to the definition of the system architecture and allows the identification of equipment and the interaction & links with other systems.

In a first step during BD preparation the routes will be identified, the first digit of the Circuit will be fixed and the routes will be divided into Essential or Non-Essential routes.

Further on the Equipment and the FINs will be allocated and the second digit of the Circuit will be fixed.

The BD contains Connection Lines between several “objects” identified by a Functional Item Number (FIN). The connection (logical link) itself is identified with a line-name and consists of first Pre-Sizing route information. The line-name is a unique identification provided by system design. Additionally the number of wires will be fixed.

In short, a Block Diagram (BD) is system level (mono- or multi-circuits)-oriented and shows mainly connection lines.

When the Maturity of the BD evolves, the BD becomes Maturity B and is called a Functional Diagram (FD).

As all other ESD, BDs are stored in ESDCR (Electrical Schematic Diagram Common Repository) storage tool.

15.4 Functional Diagrams (FD)

Based on the data provided with the BD and further functional wiring requirements and feedback on the system and its interfaces FDs are generated which illustrate the different functions for each sub-system. The FD will be generated with respect to DARE for MG 4.2.

In contrast to the BD the FD mainly adds connection lines. Further information regarding FD is provided in “Specification of System Wiring Requirements” (refer to ABD0200.1.1.4).

A FD may detail a system from the nose of the A/C to the tail and only WD details the system for a given section. That is one of the reasons why the FDs must be split into several WDs and EPDs.

The FD splitting is the responsibility of Customer Service. More information regarding the functional process is provided in the document “Create Specific Electrical Functional Design” (refer to AP5997.1).

The splitting defines for all WD and EPD:

- The title
- The reference (901, 902, 002, 102...).
- The schematic format (F5, F6...)

In short, a Functional Diagram (FD) provides mainly information about logical lines and is mono-circuit-oriented.

Customer Services receives all the first issues of the Functional Diagram (FD) to be split in one or several Wiring Diagrams (WD) and EPDs depending on the Site responsibility & FD complexity.

Customer Services realizes a functional splitting of the FD and transmits the proposal to the circuit responsible for the design of the EPD. Splitting is done only once, on first release of a FD.

15.5 Equipment-Related Issues

Equipment-related issues are covered for the Release Process in the document “Release Process: Equipment” (refer to AP2763.2) which provides further assistance on this topic.

General requirements for suppliers of equipment designated for use in AIRBUS aircraft is provided in the document “Equipment - Design – General Requirements for Suppliers” (refer to ABD0100).

The requirements mentioned in the ABD0100 are applicable to equipment design and industrialization in parallel with the Purchaser Technical Specification (PTS).

These general requirements, the specific PTS and the applicable documents (e.g. standards DO160, DO178, etc.), all defined in the PTS, represent all the requirements applicable for Equipment design.

The ABD0100 is made up of four parts:

- Part 0 - General Requirements
- Part 1 - Product (equipment) Requirements,
- Part 2 - Design Process Requirements,
- Part 3 - Documentation Requirements.

The term Design Process is understood within the ABD0100 as all documents cover the design and industrialization up to the qualification for certification of the item of equipment (Declaration of Design and Performance (DDP)). If System design is also part of the work, then the relevant sections of the document “Requirements and Guidelines for the System Designer” (refer to ABD0200) will also apply.

16 Functional Definition

16.1.1 Overall Workflow

Functional definition takes place in the “two-dimensional space.” Therefore, every Design Data Set in this area has only two dimensions. The hyponym for these drawings is ESD - Electric Schematic Diagram consisting in the functional design of:

- PD - Principle Diagrams; mono-circuit, FD and section break, logical wires
- EPD - Extended Principle Diagrams; mono-circuit, FD and section break, industrial elements, physical wires and logical wires
- WD - Wiring Diagrams; mono-circuit, FD and section break, industrial elements, physical wires.

Electrical Schematic Diagrams are part of the Electrical Design Data Set. The WD is created from EPD or PD depending on the program and the decision of the circuit leader. EPD and PD reside on the same level. As reference, the global view of the electric process for the different programs is shown in document AP5997.1. A380 specific procedures can be taken from AP5997.1.2. A400M specific information is given in AP5997.1.1. The relevant instructions during definition and development phase are given in document AP5997.2. Electrical Installation Architecture and Integration specific for A350 is addressed in document AP5997.5.

It is important to keep in mind that different programs use different subsets of the ESD. The A350, for example, does not use PD but EPD, while A380 and A400M use a mix of PD/EPD.

The main difference between PD and WD is that PD shows logical cables while WD shows physical cables and all necessary industrial elements for manufacturing.

The main difference between PD and EPD is that in EPD the leader of the circuit details his part with physical cable information. All others are still working with WD, which is linked to the EPD.

The relevant diagrams are split into several parts depending, for example, on their complexity.

Customer service is in charge of the splitting. This information is urgently needed for the AWL (Air-craft Wiring List) provided by customer service. A description of all involved functions can be found in the document AP5997.1.

For the description and definition of the documents see AP5997.1. Further information regarding ESD, especially (E)PD/WD, is listed in ABD0004.3. It has to keep in mind that the A1079 is the relevant document for A350, but it is possible that the A1079 refers back to dedicated parts of the ABD0004 modules.

The ESD are created with the IT tools like SEE XP (refer to AM5014), legacy programs also use PowerPoint, CCD and others.

The release process of the ESD documents PD, EPD, WD for A380 is listed in AP2763.3.1, and for A400M, A350XWB and future programs the document AP5763.3.2 applies.

The BD and FD are not currently involved in the release process and are not taken into account by the AP5763.3.

The data of the functional domain is handled by CIRCE-C.

The storage of the ESDs is carried out with the IT tool ESDCR or helper tools such as KIWI to provide a connection between e.g. SEE XP and ESDCR. The functionalities of the ESDCR tool itself are described in document AM5014.15.

It is important to keep in mind that CIRCE is a database and does not contain “real” objects such as “Design Data Sets”.

CIRCE-C has a pointer to the ESDs (in this case to ESDCR) and other relevant data is stored in the CIRCE-C database.

This is also true of other CIRCE Modules. For example, the pointer to the 929 Design Data Sets is stored inside CIRCE, but the real Design Data Sets come from GILDA (for A380) or from PRIMES (for A400M and A350).

The numbering of the ESD follows the rules defined in “Electrical Drawing Set – Definition” (refer to ABD0004.3, chapter 9.2 “Lists and Drawings” (one circuit)) and in “Rules with Specific/Restricted Use Conditions” (refer to ABD0004.5, chapter 4.6 “Sequence Number within a Type of Document”) and depends on the program.

For example the following ESD Number

HL L 213 9 1 004 001 A

shows in detail:

HL	- Circuit Letter
L	- Aircraft Project Code ¹⁷ , this is A380
213	- First THREE digits of the ATA Chapter
9	- Site Code for AIRBUS Germany
1	- FOURTH digit of the ATA Chapter, resulting now in 2131, Air Conditioning / Pressure Control & Monitoring
004	- Sequence number within a type of document, in this case a PD
001	- Solution i.e. Drawing Configuration
A	- Issue

The main input for the work of the designer in this process is defined in the Functional Sub-TRS.

If we are in the Change Note process before M7/MG 5.x the information comes from SYSTEMS, mainly by means of SWRD, SIRD and EIRD.

The principle of the Functional Sub-TRS was first described in the TRS Handbook (refer to WD0601263).

16.2 Electric Schematic Diagrams (ESD) Functional Design

The PD, EPD and WD are part of functional definition. All of these documents are also important for preparing and providing Customer Services with the definition of the systems installed in the A/C to realize the Aircraft Wiring Manual (AWM).

¹⁷ To avoid confusion, the Project Code is not the same as the Project Key (e.g. 56 for A300 / A310, 80 for A330 / A340, 81 for A318 / A319 / A320 / A321 and 85 for A380) which is defined in the site Germany-specific document TOH1-180, Page 9.

16.2.1 Principle Diagrams (PD)

PD is no longer in common use and is replaced for A350 and future programs with EPD only. The A380 and the running programs SA/LR are still using the PD.

The PD describes the connection and function of electrical equipment and components for an electrical system. PD for complex systems will be divided into several sub-PDs in cooperation with Customer Service.

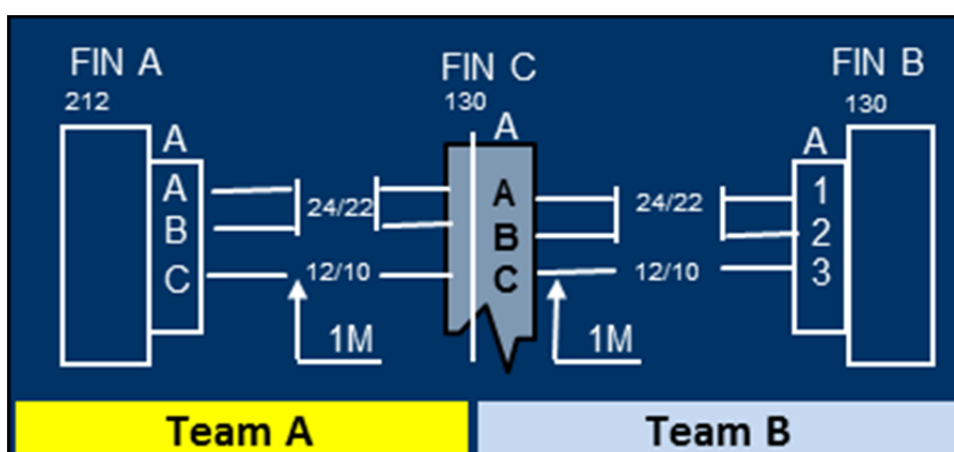


Figure 16-1: Principle Diagram (PD) illustration example

One or more MPs can be incorporated into one PD which describes a design solution (DS) for changes originated from a CR. The main difference between a PD and a FD is that a PD is a document for one system related to one MOD.

Data which has to be shown in a PD is defined in “Electrical Drawing Set – Definition” (refer to ABD0004.3). The CVE/TAP are responsible for the relevant system and must ensure that all MPs are incorporated in the PD.

16.2.2 Extended Principle Diagrams (EPD)

EPD are in common use starting with A350 and PDs are no longer present in A350 and, of course, the existence of the EPD is not really new.

It was always possible for the leader of a circuit to produce EPD instead of PD by adding the data normally provided with a WD into a PD.

After this step, adding WD data for the circuit leader parts, into the PD, the additional data “extends” the PD and this leads to an EPD, an “EXTENDED Principle Diagram.

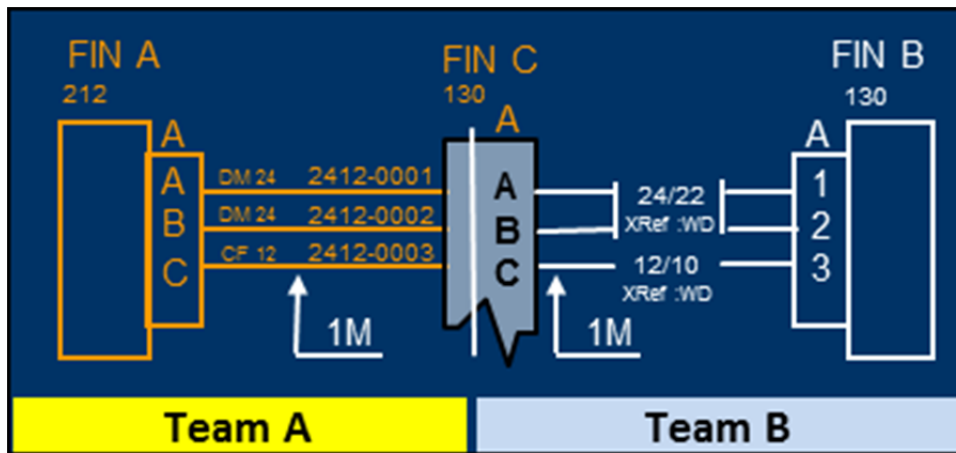


Figure 16-2: Extended Principle Diagram (EPD), logical wires of PD where wire numbers and other information are added to

All other involved parties (not the circuit leader) provide the WD information as usual in a WD with a cross reference to the linked EPD. The EDP provides also data to retrieve the related WD via cross links to the related WDs.

16.2.3 Wiring Diagram (WD)

A WD shows greater details on a specified section of an EPD (or a PD). The WD describes physical cables on the A/C.

The information given in a WD is used for two main “customers”:

- 929 Creator, as a basis for setting up the corresponding loom variants
- Customer Service, for producing, among others, the Aircraft Wiring List (AWL).

The preparation of the WD (FIN positioning in the schematics, Pin allocation, etc.) is done within AIRBUS with the goal of preparing the work and sending it to the sub-contractor who is actually designing the WDs. This transfer of the WD to the sub-contractor is supported by coordination actions (e.g. Fiche de liaison technique (FLT)).

WD preparation also details where a FIN should be located on a WD (upper right side of the sheet, lower left side...). The Circuit Responsible OR “Sub-Assembly Responsible” (in A-D) is in charge of WD preparation.

The WD is used to provide manufacturing with a correct definition in order to manufacture harnesses. This is done through entering WD data inside CIRCE-C.

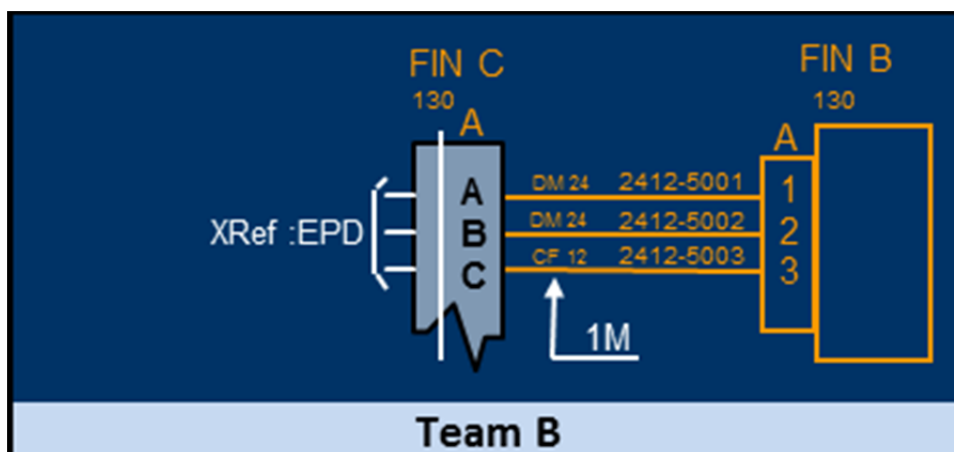


Figure 16-3: Wiring Diagram (WD) example showing missing information of the EPD (Team B)

16.3 Release of ESD (PD, EPD, WD)

16.3.1 Engineering Change Note (ECN)

The ECN is the document which releases the Design Data Set (DDS) (or parts of it) for the Design Organization (DO) to the Production Organization (PO). The PO has the task of implementing on a one-to-one basis the solution specified in the DDS in the A/C. The ESD are part of the DDS, so they are released via the ECN.

16.3.2 Release Process of ESD (PD, EPD, WD)

As Electrical Schematic Diagrams (ESDs) are Design Data Sets (refer to ABD0004 part 3 and A1079), they have to be released, details of the "Release a Design Data Set" (refer to AP5763) are defined in accordance with AP1020.

The workflow runs through different stages

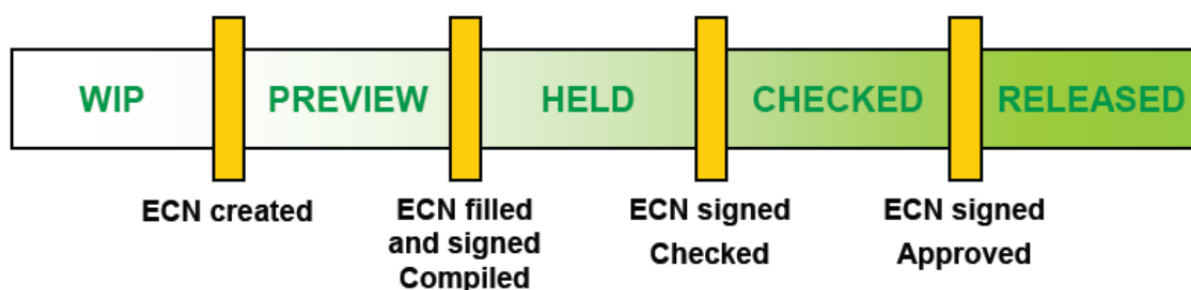


Figure 16-4: ESD workflow

The document "Release Process for Electrical Schematic Diagram" (refer to AP2763.3.1) describes the concepts for A380 and Electrical Schematic Diagram Release Process for A400M, A350 and Future Programs (refer to AP5763.3.2) accordingly.

Even though release process is currently not similar for all programs we have to keep in mind that all technical documents with a design impact have to be released according to AP1020 signature rules. Therefore the release of ESD has to be performed by authorized individuals (Technical Checker Technical Approver) only. Please note that release of ESD implies also an approved system input (e.g. FD) according to AP1020.

17 Harness Pre-Definition

The Harness Pre-definition (HPD) is first part of the electrical definition process. The aim of Harness Pre-definition is to lead efficiently the functional and installation design work. The Harness Pre-Definition permits rational concurrent engineering between the Harness Installation and the Functional Definition by sharing and ensuring FIN properties coherency.

The aim of the Harness Pre-Definition is to synchronize and align the Functional and the Installation Design. This is supported by the Topology, Electrical Routing Harness Concept Dossier (ERHCD) and the Referential / Referential in Configuration (RiC).

17.1 Topologies

Topologies give a brief overview of the main branches of the harness in a section. The application of topologies is limited to A380 and A400M. The A350 and future programs use the Electrical Routing Harness Concept Dossier (ERHCD). The document "Harness Pre-Definition (refer to AM5161.5) states, "... Topologies provide an overview of the harness morphology¹⁸ into a section. They contain main links between equipment and/or VU/VE by route or harness". A Topology is build-up in Maturity grades and is mainly used for:

- creating Harness-oriented data
- main branch pre-sizing
- decision support for architecture
- help for installation designer, circuit manager and referential administrator.

The Topology (in different Maturity grades) also supports the creation/setup of the Space Allocation Model (SAM) in Maturity A and consists normally of a set of 2D Drawings. The position in these process steps can be retrieved from "Design Structures & System Installation" (refer to A2240)."

For additional information the documents "Specify System & Equipment – Introduction" (refer to ABD0200.1.1), "Product Structures for Electrical Harness Modeling – CATIA V5" (refer to AM5011.11), "Functional Definition Process" (refer to AP5997.1) and "Electrical Definition and Installation Process" (refer to AP5997.2).

17.2 Electrical Routing Harness Concept Dossier (ERHCD)

Starting with the idea of a "Topology" the ERHCD goes far beyond this point with the goal in mind of ensuring the correct implementation of functional and manufacturing requirements. This concept is applicable to A350 and future programs. The creation of the ERHCD is part of the architecture.

The ERHCD evolve with time and use the concept of Maturities¹⁹ fitting mainly the needs of the SAM design. To achieve this goal the main inputs to the ERHCD are:

¹⁸ Morphology can be understood as the geometrical orientation of the harness.

¹⁹ For a description / definition of Maturities the document "Design Structures & System Installation" (refer to A2240, Issue: C, Chapter 1 "Process Golden Rules") provides further assistance.

- TDD 92
- SIRD, SWRD
- Manufacturing requirements

The main deliverables of the ERHCD are:

- Weight and diameter estimations
- 3D layouts of main electrical routes
- Interface plate definition
- Harness breakdown (List of bundles per area)

which will be used mainly by the SAM. The coordination between SAM and ERHCD is described in the document “Digital Mock-Up Rules & Life Cycle” (refer to AP2266).

Further details are provided in the document “Electrical Definition and Installation Process” (refer to AP5997.2) regarding goals of the ERHCD the Maturity Table is shown. The role in the SAM Process is provided also in AP5997.2.

17.3 Referential Concept / Referential in Configuration (RiC)

The Referential is an integral part of the Harness Pre-Definition. The location of the Harness Pre-Definition, between Functional Definition and Electrical Installation, in the Electrical Process is given in the document “Electrical Definition and Installation Process” (refer to AP5997.2). The connection between the Referential and CIRCE-C as a tool for the Functional Definition is provided in the document “CIRCE-C” (refer to AM5061.1).

The position inside CIRCE I as tool for the Harness Installation is provided in the document “CIRCE-I – Manages Electrical Installation Data” (refer to AM5061.2).

The aim of the Referential in this case is to²⁰:

- Link FIN to VB
- Integrate manufacturing responsibilities for FIN
- Support concurrent engineering between functional and installation teams
- Formalize agreement on FIN definition
- Replace CIRCE-I missing data (during CIRCE-G Treatments).

The Referential is realized with the Referential Tool as a central database. The data structure inside the Referential Tool is divided into three parts, Key Data, MP and properties.

A first example of a part of a Referential is provided in the document “CIRCE-I – Manages Electrical Installation Data” (refer to AM5061.2) with more details given in the document “Harness Pre-Definition” (refer to AM5061.5).

²⁰ Refer to “Electric Process Overview for A380, A400M and A350 “, prepared by EDSBL / EDSKD, Structure Faculty, 2009.

Updates to the central database of the Referential Tool are managed via the Referential in Configuration (RiC) process. There are four requests possible:

- Functional impact
- Installation impact
- Impact due to a Design Query Note (DQN)
- Impact due to a discrepancy raised in a CIRCE-G Treatment

Praxis has shown that roughly 80 % of these updates are originated from functional definition.

The RiC defines several roles in this process:

- Functional Designer
- Circuit Manager
- Referential Responsible
- Installation Designer

Information in deeper details is provided in "CIRCE Treatment Monitoring & Electrical Database Administration" (refer to AM5061.11), "Harness Pre-Definition Rules" (refer to AM5061.5.1) and "Harness Pre-Definition" (refer to AM5061.5)

18 Installation Definition

Design Data Sets illustrate the relevant parts of the product. These drawings constitute at least the real parts, the DS of the product structure defined in AP2641.

Depending on the program different drawings are used to describe the Product "Harness," their installation, manufacturing and relation to other elements (e.g. structure).

Based on the program related BAABI drawing-numbers could have the following meaning, e.g.

- 924 - Installation Drawings of Removable Supports
- 928 - Cable Loom Installation Drawings
- 929 - Cable Loom Drawings

These Design Data Sets constitute the remaining mechanical drawings, BOM and ECN the Definition Dossier. For 921, 922 and 923 please refer to chapter 18.4.2 ff.

With the given effectivity this is equivalent to the Design Solution (DS) and is the document describing the real product.

The flow of the drawings inside the process is controlled and handled by means of FLT ("Fiche de Liaison Technique") used as a coordination memo (see internal document L92PP0700442). The main input for the work of the designer in this process comes from the Installation Sub-TRS. In A380 and A350, the FLT management is supported by the tool Imade.

At least for A350 program the master reference for the design is the Digital Mock-Up (DMU), especially for ATA92 using Full3D Data Sets instead of drawings.

DMU owner is the SSCI (System Structure Cabin Integration) manager who is responsible for completeness and consistence regarding the cDMU (configured DMU) Quality Assurance Process (M2832.x).

Within this responsibility the SSCI manager has to ensure consistency/compliance with geometrical impacts of SIDP.

The DMU is also the source of the requests to structure ATA 539 for the brackets. The Brackets/Interface Requests for ATA539 follow the process described in AP5156 supported by the tool PDMLink SSCI IPM.

It is important to keep in mind that the transition from the SAM to the DMU is located at M7 / MG 5.x and this is called GRM (Geometrical Reference Mock-Up).

The DMU is also the basis for the flattening process to feed the form-/jig-board with the necessary data for manufacturing the harness.

It is important to note that "clean" installations of the harness (928) cannot be obtained if the DMU is not in correct condition, because the 539 requests for the brackets come from the DMU and the 924 for the moveable brackets and the 929 for the manufacturing.

Simply stated, if the DMU is not in order, the harness cannot be either.

CIRCE-I deals with the issues in the Installation domain and the data in the DMU is the same as that inside CIRCE-I, the so called "skill model."

The "skill model" in CIRCE-I is abstract data and not used for visualization.

The data in CIRCE-I is loaded from CATIA V4/V5 drawings using IT tools such as “Prop-Tool.”

In contrast to the DMU and CIRCE-I skill model the data in the referential is also the same as in the DMU (and therefore CIRCE-I) but the referential has no data regarding the length.

For the A380 the “Dressing” takes place just after the picture drawing step. The task is to add all bullets referring to GILDA BOM on 928 and 929, also to add installation principles.

In A350, the Full3D approach requires all necessary information for installation and harness definition stored in a DataSet which replaces the classic drawing.

Basic rules for creation, configuration and release of the definition dossier are given in ABD0004 which is superseded for A350 program by A1079 – nevertheless some ABD0004 modules are still valid for A350.

18.1 Space Allocation (SAM)

The Space Allocation Model (SAM) is set up on three levels of Maturity. The SAM is an A/C DMU and not a Non – A/C DMU as described in document “cDMU as Master” (refer to A1069) and used as a model for definition.

The document “Space Allocation Design Process” (refer to AP5617) states, “... the Space Allocation Mock-Up is a digital mock-up made of simplified 3D solid models in CAD format (Computer Aided Design), used mainly during concept and definition phases (note: in A350 just concept phase) and matured into final definition models during the development phase (note: in A350 definition phase) (constituting the Geometrical Reference Mock-Up).” The SAM guarantees hereby requirements feasibility.

If the SAM reaches Maturity C the transition to the Geometrical Reference Model (GRM) is performed. This will be in the DnA Process Milestone M7 and in the DARE after MG 5.2 / MG 6.

The 3D model is all the same all the time, it only gains maturity. More and more details are added to the model, starting with SAM A (first routing, no brackets), SAM B (interface definition with related ATA) to SAM C with nearly all relevant data (Connectors, Length, etc.). For more details refer also to AP5617.

The Electrical Product Structure in the lower level for different SAM Maturities is described in the document “Electrical Installation Definition Product Structure, According to the End-to-End Process Variants” (refer to AM2211.7).

18.2 System Frontier Model (SFM) / Geometrical Reference Model (GRM)

The SFM and the GRM are one and the same. SFM is the term used in Site Germany and GRM the term in Site France for the Design Mock-Up²¹. In the A350 program GRM is used only.

The point in time for the transition from SAM C to GRM is described in document “DARE – Electrics Development Policy” (refer to A1350.3 Issue A, Chapter 1 “Electrics Activities and Deliverables”). Further details on this process step are provided in “Design Structures & System Installation” (refer to A2240)

²¹ Refer to “Electric Process Overview for A380, A400M and A350”, prepared by EDSBL / EDSBK, Structure Faculty, 2009.

The document "Validation and Publication of 3D cDMU Data" (refer to AP5766) describes the Design Review Activities of different roles with regard to the GRM (and other elements not already mentioned).

18.3 Configured Digital Mock-Up (cDMU)

The idea behind cDMU as master is (more or less) clear. Only one cDMU shall be used and all business areas shall adopt the DMU as the master product referential.

The cDMU consists of the 3D Geometry, the "Meta Data," means the product structure and related attributes (names of parts, version, etc.) for each configuration, meaning at least one cDMU for every specific delivered A/C. Furthermore, that means that the "c" stands for embedded Configuration Management decisions as described in document "Configuration Management" (AP1007) and related documents. Additionally this cDMU is NOT a static but a dynamic view of the Product Life Cycle Management (PLM) / Product Data Management (PDM).

The term "master" in this context means the primary source of information, which, in turn, means:

- Geometry is always created as 3D cDMU and triggered by "changes" leading to a CR
- The 3D cDMU is used as the primary reference, 2D is only a projection of the 3D space, meaning that EVERY geometrical component MUST be present in the DMU.

Additional information regarding cDMU is provided in the document "cDMU as Master" (refer to A1069), "Definition Dossier - Management Rules" (refer to ABD0004) and "Digital Mock-Up General Process" (refer to AP2266).

Handling "trouble" inside the (c)DMU or performing reviews is described in "DMU Trouble and Review Management" (refer to AM2037.4).

The document "Delivery of Electrical 3D Models in Accordance with CIRCE-G Treatments" (refer to AM5061.3.1) describes the method of the delivery of 3D models for harness manufacturing. The document also covers the method for exchanging data between team managements of the harness morphology.

The document "Validation and Publication of 3D cDMU Data" (refer to AP5766) describes the process that ensures the cDMU and Design Quality Assurance during Definition (DnA)/Concept (DARE)/Development (DnA)/Definition (DARE) Phase.

The document "cDMU Quality Assurance: Process, Roles & Responsibilities" (refer to M2832.1) describes the process rules to ensure the quality of the design using cDMU through mandatory formal blocking Design Quality Checks (QC):

- DQC1: cDMU Data Quality Check;
- DQC2: cDMU Design Quality Check;
- DQC3: Data Check of Drawing Set;
- In addition: FCO DMU Milestones and Quality Gates

18.4 ATA92 related definition dossier elements

The ATA breakdown for the different programs is not unique, also not in the specific interpretation between sites (as mentioned in the previous chapter). An overview of the commonly used relation

of sub-ATA (e.g. 923, 924, etc.) and the intended Design Data Set content is given in the document "Electrical Definition and Installation Process" (refer to AP5997.2).

18.4.1 539 - Fixed Parts

Validated by Structure 539 this DDS contains the definition of fixed brackets. Further information regarding this field of application is provided in the document "Manage Interfaces Between System, Cabin & Structure from Request to Attestation of Installation in Aircraft" (refer to AP5156).

18.4.2 921 – Routing Diagrams

The basic idea behind the usage of ATA92-10 drawings is organizing the ATA92-90 drawings in a cascade (92-10 is the summit (Top-Drawing), 92-90 the Bill of Material).

This summit was used to provide a cluster for the harness delivery for installation. Harnesses, which were mainly located in one installation area, were placed under the same ATA92-10 drawing.

The system of clustering drawings was also used by ATA92-30 and ATA92-80, but there it was not called ATA92-10, they use their own ATA to cascade.

QUAN- TITY	ITEM REF.	GRID REF.	OBJECTNUMBER VENDOR	DESCRIPTION	FUNCT- ID WEIGHT /KG	MM M2	REMARKS MOPE GRID-REF	KS IDENT-Nº LOG CLC
1	5061		L92C-65061-000 D8518	CI-CABLE-HARN. 5061UB				KK50603995
1	5062		L92C-65062-000 D8518	CI-CABLE-HARN. 5062UB				KK50603997
1	5063		L92C-65063-000 D8518	CI-CABLE-HARN. 5063UB				KK50604000
1	5131		L92C-65131-000 D8518	CI-CABLE-HARN. 5131UB				KK50604005
1	5141		L92C-65141-000 D8518	CI-CABLE-HARN. 5141UB				KK50604007
1	5180		L92C-65180-000 D8518	CI-CABLE-HARN. 5180UB				KK50604008
1	5189		L92C-65189-000 D8518	CI-CABLE-HARN. 5189UB				KK50604009
1	5209		L92C-65209-000 D8518	CI-CABLE-HARN. 5209UB				KK50604013
1	5211		L92C-65211-000 D8518	CI-CABLE-HARN. 5211UB				KK50604014
1	5212		L92C-65212-000 D8518	CI-CABLE-HARN. 5212UB				KK50604015
1	5217		L92C-65217-000 D8518	CI-CABLE-HARN. 5217UB				KK50604018
1	5292		L92C-65292-000 D8518	CI-CABLE-HARN. 5292UB				KK50604019

AIRBUS DEUTSCHLAND GMBH D.O.Orig. : D8518	B I L L O F M A T E R I A L A I R B U S	TITLE: S12 8 B LD CEILING LE PART NO: L921 70022 010	FORM INDEX AA	ISSUE: AA 28.11.06 SHEET: B01 FOLLOWING: END	IDENT-Nº 50155500
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Figure 18-1: Example, ATA-921

18.4.3 922 - Routing Drawings

ATA92-20 drawings are mainly used in Airbus France for route installation containing similar information as ATA92-80 drawings.

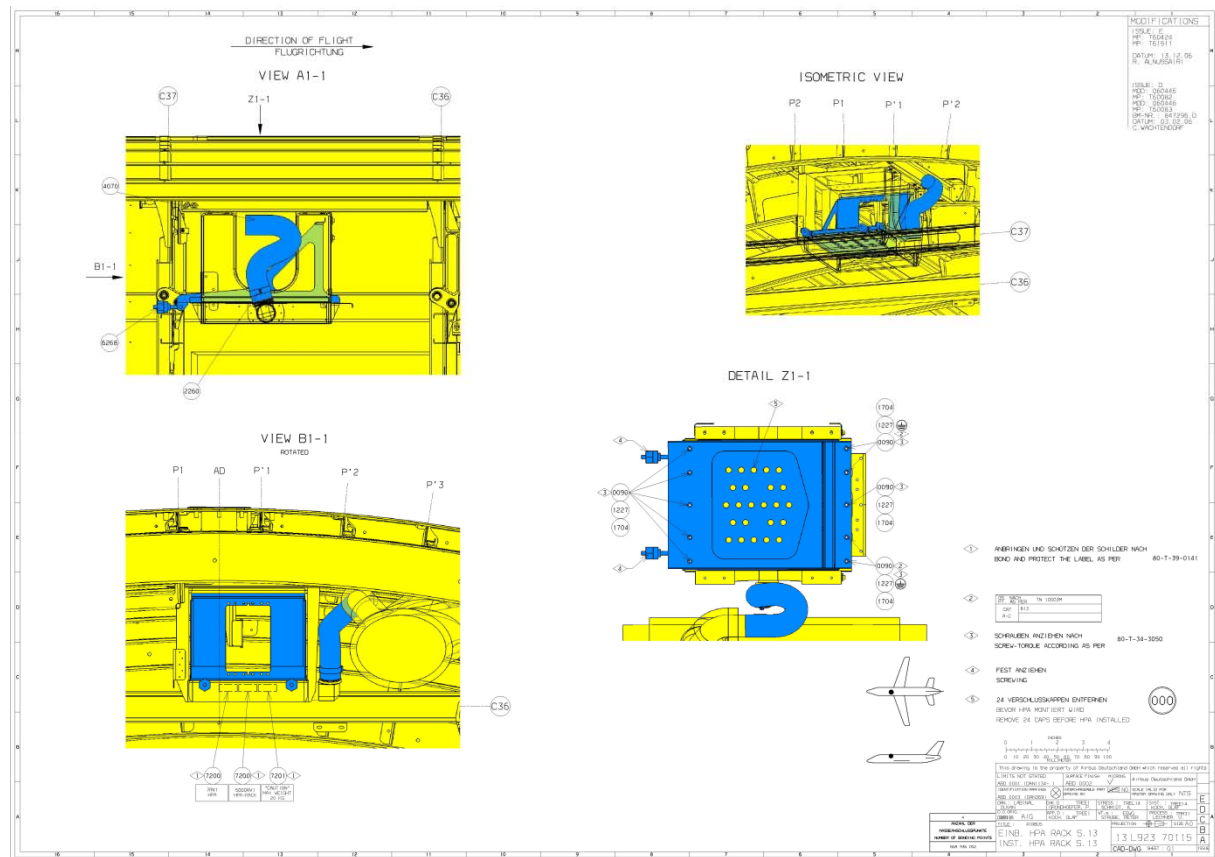


Figure 18-3: Example, ATA-923

18.4.5 924 - Moveable Parts

A 924 drawing contains mechanical and part drawings. For example the drawing for an Overhead Equipment ramp or/and ESN Raceways Installation Drawings.

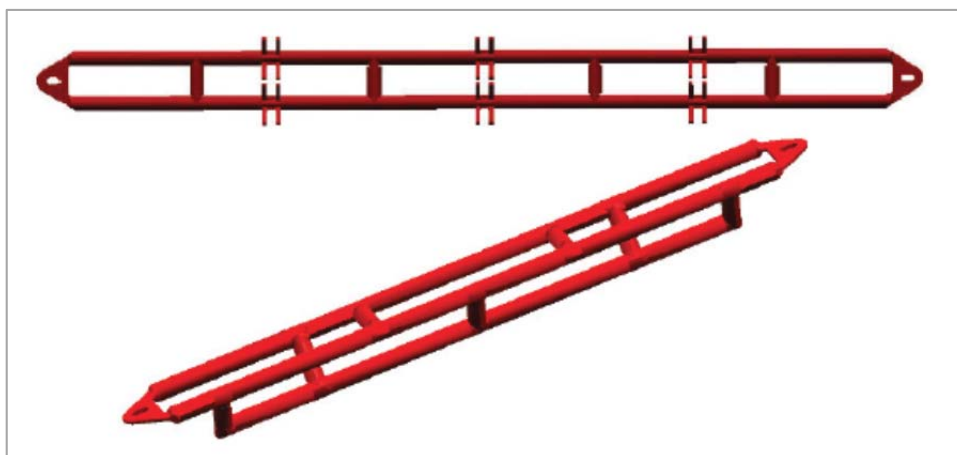


Figure 18-4: Example of Parts, ATA-924

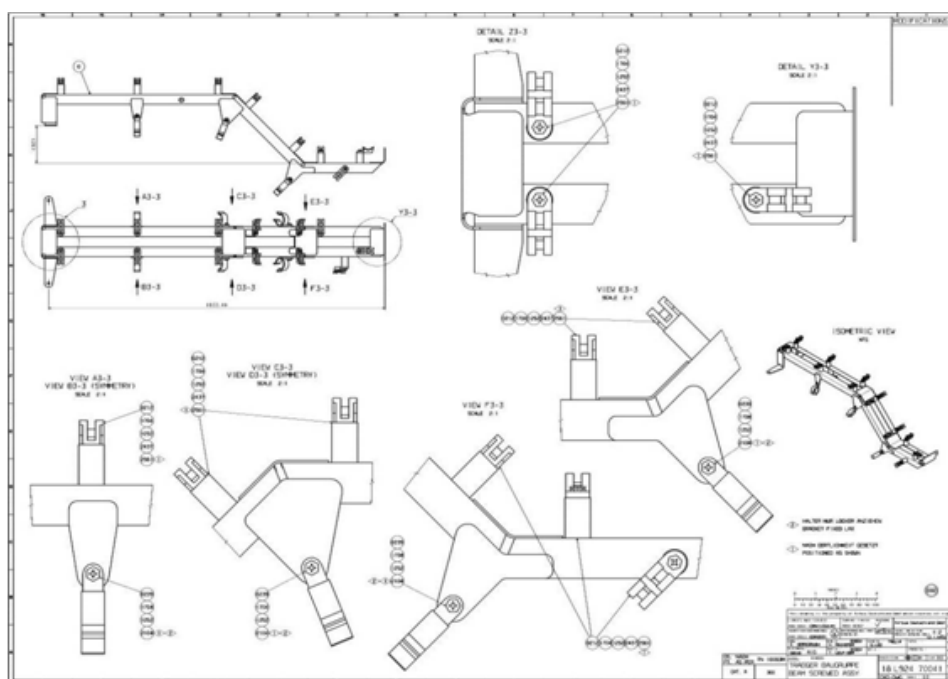


Figure 18-5: Example drawing, ATA-924

18.4.6 925 - Assembly Drawing

A 925 drawing is usually used for VU/VE and shows details for specific wiring inside the VU.

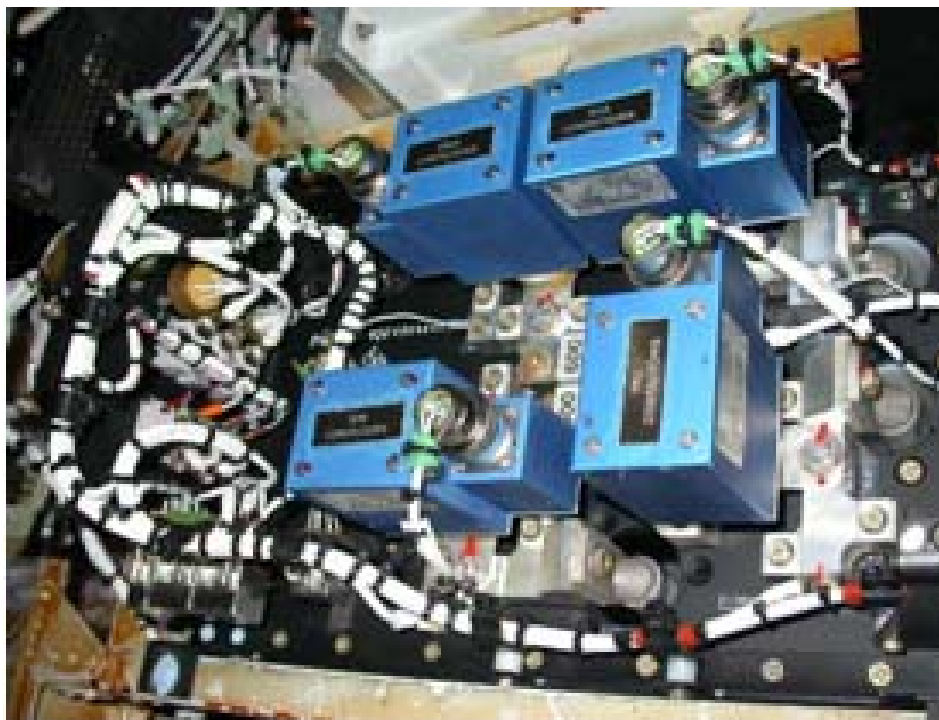


Figure 18-6: Example of parts, ATA-925

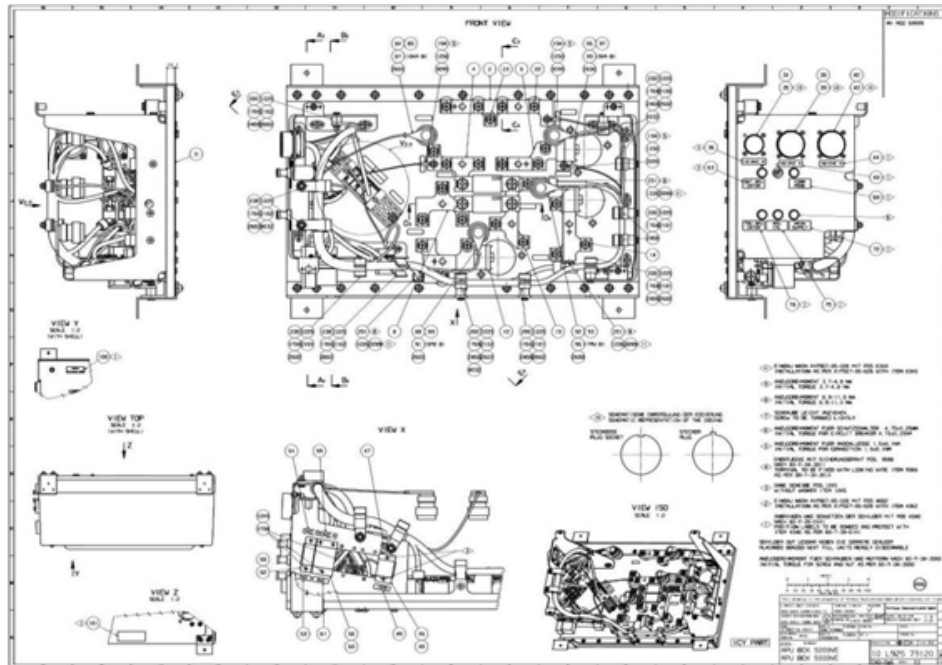


Figure 18-7: Example drawing, ATA-925

18.4.7 927 - Final Assembly Line Specific Work

L927 Installation Connection Drawings are used at FAL to locate the different physical connections to be performed on A/C. They can be linked to a L928 Installation Drawing if the connection is implemented during the same installation operation (refer to M2277.1.2).

L927 drawings are also called Z- or C-Connection and normally do not provide any BOM, only 2D drawings and the connection list information from CIRCE. Several types of connections are defined²²:

A null Connection (empty field) is described as follows:

- The FIN is delivered connected to the Deliverable Assembly
- No operation has to be done at FAL

A C-Connection is described as follows:

- The wires have been cut at good length with End Items in workshop
- The connection has to be done at FAL

A Z-Connection is described as follows:

- The wires have to be cut at good length at FAL
- The End Items have to be installed at FAL
- The connection has to be done at FAL

²² Refer to "Electric Process Overview for A380, A400M and A350", prepared by EDSBL / EDSKD, Structure Faculty, 2009.

Generally, it is considered a good idea to avoid this kind of connection if possible.

18.4.8 928 - Installation Drawing

The 928 drawing is used to install those items produced as a result of 929 and the Mobile Brackets (924). The 928 will show how the harness should be installed in the A/C. The 928 BOM will contain mechanical items such as mobile brackets, etc. A User Guide from EDSBL focusing on the eNDP (A380) is also available.

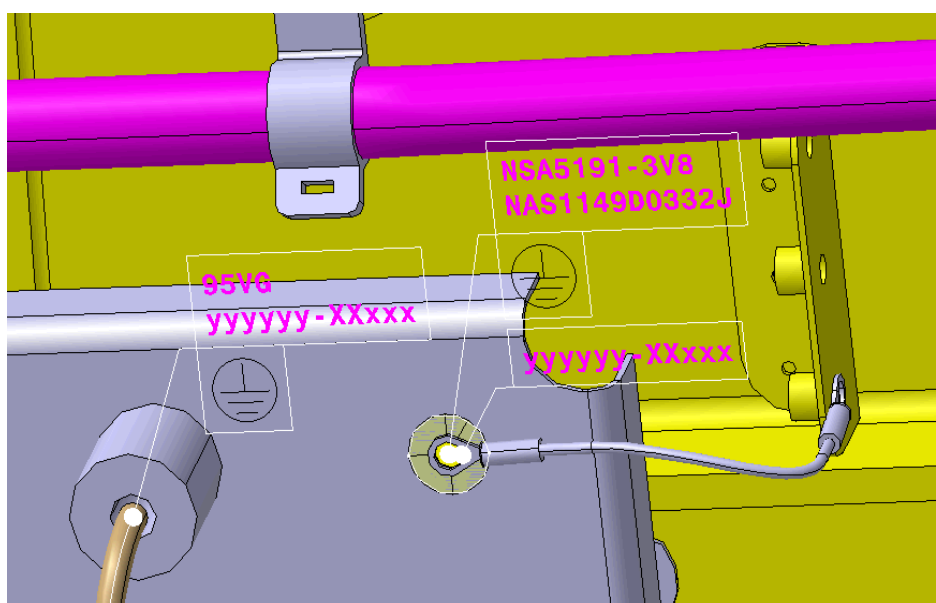


Figure 18-8: ATA-928 Example Full3D

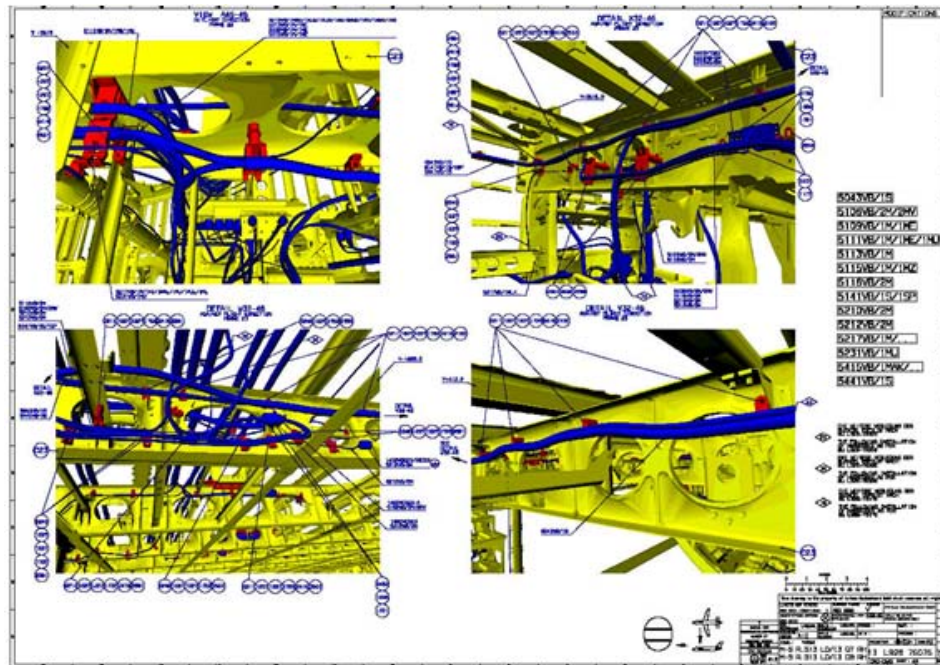


Figure 18-9: ATA-928 Example picture drawing)

An A350 928 Full3D DataSet for installation also contains the RG20 code which replaces addition ground tests for bonding points. RG20 is called a PCM code standing for Process, Class and Method. Process describes the manufacturing process how to install physically the bonding connection, (bonding) class is linked to maximum resistive which also gives the target value for the (test) method results.

Full3D DataSets have to fulfill the same archiving requirements like all other design data. This is solved by using an Adobe step format in which the data are archived in ZAMIZ.

18.4.9 929 - Definition Drawing

The 929 drawing is used to manufacture the harness in conjunction with the 929 BOM. The 929 BOM lists all items which are produced in line with the harness. A detailed description of 929 is provided in the document "Harness Definition Drawing Set" (refer to M2277.2.1). This document provides a detailed description of the "Harness Definition Dossier" content.

In A350 the 929 drawings are also replaced by Full3D DataSets (see example below).

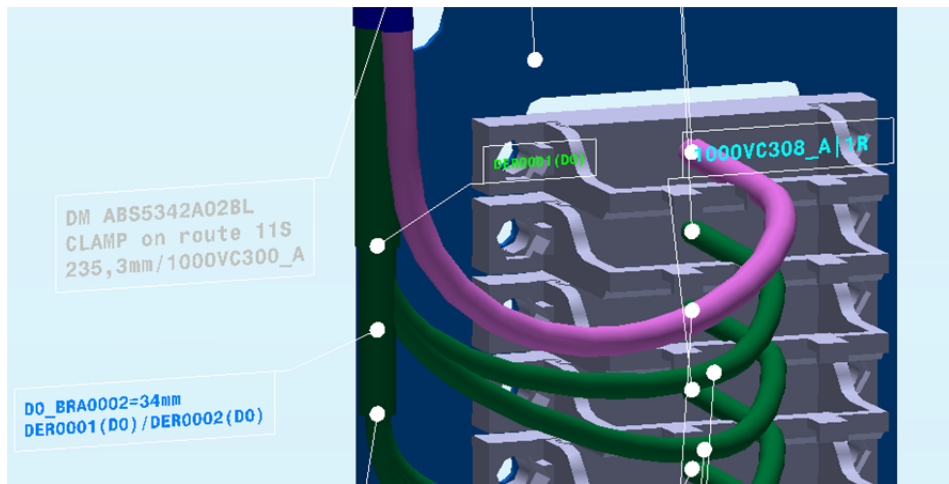


Figure 18-10: ATA-929 Example – Bundle with annotations

18.5 Release Process of Installation-Related “Objects”

As usual the release of “objects” from the Design Organization to the Production Organization is done by means of the Engineering Change Note (ECN).

In the case of installation objects the ECN refers to the cDMU as master and the related Design Data Set consisting of:

- 3D Model(s)
- 2D Drawing(s)
- Bill Of Material(s)
- ECN

A more detailed description will be provided in the document “Drawing Set Definition - Common Rules” (refer to ABD0004 Part 1/A1079 for A350).

The Release Process is described in detail in the document “Release a Design Data Set” (refer to AP5763.0).

19 Manufacturing Preparation

19.1 Treatment Concept

The data from the functional domain and the installation domain merges during CIRCE-G treatments. The treatment takes place every three weeks for selected MSNs of one “instance” (i.e. . The treatment creates the “drum beat” in the whole system.

The main term in this context is the DA - Deliverable Assembly. It is important to keep in mind that the term DA is CIRCE-specific and that a DA can be a CA.

The treatment should lead to a gradually increasing quality of data in the DMU. That means that with every treatment more and more data will come from the DMU itself. In turn, an increasing amount of referential data will not be referenced during the treatments. This leads to the goal that we wish to have 100% DMU data in use and the quality of increasing complete DMU-validated data.

The main workflow / steps during Treatment Cycles are:

- Effectivity Calculation with data provided by the AAT (Aircraft Allocation Table) and the GLM (General List of Modification)
- Wire to DA Calculation (in a first step a FIN to DA allocation will be performed)
- Length and “other” calculations
- End-fitting allocation

The results of the “real” treatments are stored in one of three databases, the BDO, the database of the design. The other two databases are:

- BDF - database of the production organization
- BDH - history database for old treatment data.

Anomalies are analyzed and the remaining deviations are treated as discrepancies. A discrepancy means that we have data inside CIRCE-C not covered by data in CIRCE-I or the corresponding backup data in the referential is also missing.

The results of the treatment are the parts of the Design Data Set (DDS)

- Drawings (e.g. ESD, 929, 928, etc.)
- Bill of materials (BOM)
- Engineering change note (ECN) used for the release of the DDS

It is important to point out that the signatory of the ECN is a legally binding signature by the TAP to release the DDS from the DO to the PO.

This data is transferred to the manufacturing units via the connection to CIRCE-P.

CIRCE-P deals with the production organization and reads the manufacturing schedule and copies data from the BDO to the BDF database if a DA is frozen.

It is important to mention that the production organization has no information about the DA in charge until these DA are “frozen.” The contents of the Definition Dossier are transferred via so-called transfer files to the manufacturing units.

19.2 Sketch Phases

This Treatment behavior changes after “Power On” to an additional treatment on an occasional (possibly daily) basis called “Sketch Phase” and takes place in addition to the normal treatment.

In addition, the information flow between design and production is done by retrofit process sheets, which will be replaced after “power on” with sketches.

Retrofit process sheets describe the change in terms of a full PRE-/POST-description. Sketches describe only the difference.

The document “Create and Manage Sketches in CIRCE” (refer to M5061.9) provides further information on handling these sketches inside CIRCE.

20 Testing the “Results”

20.1 Testing and Certification / Attestation Relevance

To avoid confusion, the test of conditions has to be divided into two different groups. One of those tests is required for the compliance in the framework of fulfilling the Means of Compliance for Certification. This means that those results are needed for the Modification Approval Sheet (MAS) in the framework of MOD – Closure described in greater detail in “Change Process – Modification Closure” (refer to AP5204).

This will normally be carried out just once, BEFORE the serial production, and called: “MOD – Closure (MAS achieved)” (Certification of this MOD is completed). All this happens for a new A/C development in the DEVELOPMENT domain.

Then in the FULFILL domain, the serial production, EVERY MSN will be tested before Attestation with the EASA Form 52 at least.

If a change is detected which would lead to a CR, leading in turn to a MOD classified as MAJOR, the process will return to the DEVELOPMENT domain, the procedure will start over again and tests will be performed, possibly several times but for only ONE event, the MOD – Closure (MAS).

In addition we must keep in mind that we (actually) realize that “concurrent engineering,” means the DEVELOPMENT and FULFILL domains are working in parallel.

Further information regarding Certification and Attestation was already given in one of the previous chapters (refer to chapter 6.6).

20.2 Ground Test

20.2.1 Ground Test Requirement (GTR)

GTR describes how tests should be performed to fulfill the Means of Compliance (MC). In this case the GTR is used to show compliance with MC5 – Ground Tests.

Additional information regarding Means of Compliance is provided in “Type Certification Procedures” (refer to AP2184), Appendix 1, IR-21A.33 “Investigation and Tests” and related AMC and GM.

GTR are in the responsibility of the Design Organization and are created once per program. In comparison to GTR, GTI are in the responsibility of the production organization and are created once per MSN.

The document “Guidance on Validation and Verification” (refer to AM2388.2.3) provides additional information regarding GTR.

GTR are divided into several types, which are specified in ABD0200.2.9. The type of GTR and other information are provided in the GTR numbering, e.g.:

GTR 8-2515-001-0001-A

GTR – Type - 8

ATA Chapter - 25

ATA Sub-Chapter - 15

Number- 001

Variant - 0001

Issue - A

The GTR Types are as follows:

Attestation	8	Ground Test in A/C (Functional Test) Requirement for a whole system to achieve MOD - Closure / MAS. These are tests in the framework of Type Certification (TC) and are documented as MC5 (refer to the relevant ACMR for the specific program)
	7	Test in Constituent Assembly (CA)
Certification	6	Interchangeability
	5	Test for system development, no GTI will be generated
	4	Test for system qualification, no GTI will be generated

Table 20-1: Overview of GTR types

Every GTR has a similar document structure with the following chapters:

- Introduction
- Aircraft conditions BEFORE test
- Test description including the necessary test equipment
- Special conditions for development A/C
- Test close-up and Aircraft conditions AFTER test
- Appendix

GTR, which are not in status “Approved“, shall be reported as Delta 1 per MSN and CA.

20.2.2 Ground Test Instruction (GTI)

The Production Organization is responsible for the proper translation of the GTR into reasonable GTI. The types of GTI are site-specific and defined in AIQI 2.24.034.00.

If the GTI leads during function tests in the MSN to errors, a GCO (Ground Test Requirement Change Order) is issued to change the corresponding GTR in the Design Organization (DO).

If in the sense of CCMD the GTI are not part of Delta 1, they shall be reported as Delta 2.

20.3 Flight Test

Prior to delivery the Production Aircraft Test Manual (PATM) has to be compiled. The PATM consists of a Complete PATM and a Flight PATM. The Flight PATM contains only the Flight Test parts, while the Complete PATM contains the Ground Test Part and the Flight Test Part. The department in charge for all Flight Tests is EV.

Flight Clearance in conformity with the document "Flight Clearance" (refer to AP5215) is mandatory. Also all other requirements from the "Design Organization Manual (DOM)" (refer to A20) will apply.

20.3.1 Flight Test Request (FTR)

To ensure at least the airworthiness of the development of new airborne systems or the incorporation of a MOD (classified as major (or possibly as minor) change) tests are required. This Process is located in the "AIRBUS Operational Process Map (AOPM)" (refer to A01). Those tests will be tests of components in Flight or also on ground requiring special environmental conditions.

The goal is to ensure:

- that theoretical results of research are reasonable
- approval of the status of a system for Certification (MAS)
- integration in an aircraft as a formal modification.

All these tests must be subject to an official request, the Flight Test Request (FTR) approved and authorized by the Chief Engineer (CE). For Certification Flight Tests the document "Type Certification Procedures" (refer to AP2184) will apply.

- A complete and signed FTR is mandatory to perform tests in the following cases:
- if the aircraft will fly in order to achieve the test results
- if the aircraft will stay on the ground, under Design Organization Approval (DOA) status, under Production Organization Approval (POA) status, but the test requires the active involvement of a Flight Test specialist on board [Flight Test Engineer (FTE), Test Flight Engineer (TFE) or EV Ground Specialist].

Otherwise a GTR is necessary and sufficient. Further details regarding the FTR are provided in the document "Flight Test Request Process" (refer to AP2695).

20.3.2 Flight Test Modification

To perform a (special) ground test or Flight Test a modification is mandatory. This MOD will be a so-called "Z MOD". This MOD is outside the normal modification system and the Flight Test department is in charge of this MOD.

For details regarding this topic the document "Aircraft Configuration Management Rules" (A1080) provides further assistance in the chapter "Definition of Modification".

In the case of development A/C, which must be transferred between Production and Design Organization the document "Transfer of Development Aircraft and Items Between Design and Production Organizations" (refer to AP2146) will apply.

Two different types of modification are related:

- Flight Test Installation and Removal

- Flight Test Installation Provisions and Refurbishing

Further details regarding this topic are provided in the document “Aerostructure Work Packages - Technical General Requirements” (refer to AP2190.2).

With regard to “Digital Mock-Up Rules & Life Cycle” (refer to AP2266) Flight test components are not shown in the DMU.

The process to validate each of the flight test parameters is described in the document “Flight Test Installation Validation Process” (refer to AP5801).

The flight test installations must be present at MG6 as required in “DARE – Prerequisites and Maturity Criteria for MG6 Review” (refer to M2350.6).

The methods and rules describing the creation of an “Electric Definition Dossier” are also mandatory for the flight test team (ATA89) provided by the document “Method and Rules for Creating Electrical Definition Dossier for A400M and A350 Using CATIA, PRIMES and CIRCE” (refer to M2277.2).

20.4 Special Conditions

20.4.1 Special Instruction (SI)

Special Instruction (SI) is used to define experimental design solutions and temporary configurations before delivery to an airline. The document “Special Instruction Management” (refer to AP2761) provides a detailed description of this Process.

An SI is used to temporarily change the A/C configuration in order to perform tests (experimental modifications). The SI can be applied for both development and series production A/C.

During test phases the A/C configuration is indicated by a set of applied MOD/MP and SI.

The A/C **cannot** be delivered to the customer with SI applied.

If at the end of the test phase the change is to be permanent SI must be refurbished and the whole configuration must be managed through the full Change Process via MOD/MP.

The applicability of an SI of type “E,” “B,” and “C” is limited to one MSN.

Other types of SI can be applicable to more than one MSN.

Any temporary modification that should require agreement from the AIRBUS Airworthiness Department should be submitted using the full Change Process.

All SI must be stored in a common repository with the impacted CA.

SI shall not be used for advanced application of a modification except with specific authorization from program.

A SI shall not be used to cover a permanent deviation to the A/C definition or specification. In such a case a concession shall be raised according to A2406.

The following table shows the different types of Special Instructions:

SI Type	Example
E	Used to install equipment Install on A/C development equipment (CMS 50...89) Install on A/C instrumented equipment (equipment with sensor, for example) for which the instrumentation can be carried out by AIRBUS or by the supplier Install already certified serial equipment on a development A/C (to test it for a different function or location, for example) Uninstall temporarily an equipment part of the target definition: this process is covered by the installation of equipment with P/N "NOT-FITTED" These SI install the same equipment reference on one or several locations (one CMS & several impacted FIN per SI)
W	Used to make a modification to the wiring definition Make modifications to the wiring standard definition of the A/C due to test constraints Install specific wiring for test purposes
S	Used to make modifications to the structure of the A/C Make modifications to the structure standard definition of the A/C due to test constraints Install specific structure parts for test purposes
T	SI that does not change the A/C configuration Cover SI clustering several SI applied at the same time to achieve a common purpose
C	Used to request a change on a Constituent Assembly that is under the responsibility of a supplier These SI make it possible to track a change request between the organization that requires the change and the organization that is responsible for the impacted Con-

	stituent Assembly
B	Used to install on A/C a consistent set of equipment Install several kinds of equipment (several CMS & several impacted FIN per SI): these SI are an extension of the SI of type E
G	Used to implement specific test configurations that do not fall into the previous categories

Table 20-2: Different Types of Special Instructions

20.4.2 Permit to Fly

There are two different kinds of Permits to Fly. A detailed description is provided in “Issurance of Permit to Fly for Development A/C” (refer to AP5070) and “Approval of Flight Conditions Document for Issuance of Permit to Fly for In-Service A/C” (refer to AP5067).

More details are provided in the Design Organization Manual (refer to A20) and 21A.707, 21A709 and related paragraphs provided in the EC No 1194/2009 dated 30 November 2009 amending Regulation (EC) No 1702/2003 laying down implementing rules for the airworthiness and environmental certification of aircraft and related products, parts and appliances as well as for certification of design and production organizations.

The applicable EASA Form is Form 37, Application for Approval of Flight Conditions for a Permit to Fly. Form 37 also includes EASA Form 18b; FLIGHT CONDITIONS FOR A PERMIT TO FLY – APPROVAL FORM.

A “Permit to Fly for development aircraft” is the legal document allowing an aircraft to fly in the absence of a valid Individual Certificate of Airworthiness.

As per Part 21A.701, Permits to Fly shall be issued for aircraft that have not yet formally demonstrated compliance with airworthiness requirements, but are capable of performing a safe flight under defined flight conditions.

As per Part 21A.708, Flight conditions are defined by:

- the configuration(s) for which the Permit to Fly is requested,
- any restriction necessary for safe operation of the aircraft including:
- the restrictions put on itineraries or airspace, or both, required for the flight(s),
- the restrictions put on the flight crew to fly the aircraft,
- the restrictions regarding carriage of persons other than flight crew,
- the operating limitations, specific procedures or technical conditions to be met,
- the specific flight test program (if applicable),
- the specific continuing airworthiness arrangements including maintenance instructions and regime under which they will be performed,

- the substantiation that the aircraft is capable of safe flight under the conditions or restrictions of subparagraph (b),
- the method used for the control of the aircraft configuration in order to remain within the established conditions.

Flights are authorized for the following purpose:

- development,
- compliance demonstration with regulations or certification specifications,
- design organization or production organization crew training,
- support of commercial campaigns, including customer's crew training.

AIRBUS as holder of Permits to Fly shall ensure that all the conditions and restrictions associated with the Permits to Fly are satisfied and maintained.

The Permit to Fly can be surrendered or revoked in the event that the associated detailed conditions above are not satisfactorily maintained.

A Permit to Fly can also be issued for in-service aircraft.

AIRBUS, as Type Certificate & Design Organization Approval holder, is entitled under Part 21 Subpart J to approve the conditions under which a "Permit to fly for an AIRBUS In-Service Aircraft" can be issued. This approved data is referred to as a Flight Conditions Document (FCD).

Upon request from an operator to fly an Aircraft that has sustained system and / or structure damages, AIRBUS is entitled to issue a Repair Approval Sheet (RAS) and / or a Technical Adaptation (TA) if the compliance to relevant Certification Basis can be demonstrated.

If the compliance demonstration to the Certification Basis is no longer possible AIRBUS is entitled to support the operator's application for a Permit to Fly by providing approved data defining the required flight conditions to perform a safe flight.

Such Flight Conditions Document is approved and released by AIRBUS under its DOA to support the operator's application for a Permit to Fly (PtF) to its National Aviation Authority (NAA).

For in-service aircraft that no longer hold an individual Certificate of Airworthiness and for which the continued airworthiness is ensured under the AIRBUS Continuing Airworthiness Management Organization approval (CAMO), Airbus may need to put in a request a Permit to Fly from the relevant National Airworthiness Authority, even if the aircraft is fully compliant with its Certification Basis. In that case AIRBUS must justify that:

- the aircraft definition is approved,
- the aircraft is compliant with the applicable Airworthiness Directives,
- the aircraft is maintained in accordance with the applicable maintenance instructions.

If the demonstration of compliance with the Certification Basis is no longer possible, safe flight means:

- for the structure, that the aircraft is still able to sustain the load limits, possibly associated to particular flight conditions for this specific flight (such as restriction on cabin pressure, weight, Mach, speed, altitude...),

- for the systems, that the overall probability of a catastrophic failure condition shall not exceed 2×10^{-6} per Flight Hour (cut-off level given in the GM 21A.3B(d)(4) "Defect correction - sufficiency of proposed corrective action").
Similarly, the probability of a hazardous failure condition shall not exceed 2×10^{-4} per Flight Hour. All alleviating factors must be considered first (operational limitations, specific crew procedures, specific maintenance actions...).

21 Correction Process of Changes

21.1 Introduction to Non-Conformity (NC)

NC is an event occurring in Production Organization which disturbs the normal work flow. It is documented in a Non-Conformity Notification. The "Control of non-conforming items" (refer to A2406.1) states that an NC "... is a non-fulfillment of a specified requirement (refer to ISO 8402, ISO 9000:2000), e.g. the Definition Dossier."

The source of this process requirement is first specified in 21A.139 Quality System and in EN9100(E), chapter 8.3 "Control of Non-Conforming Product."

Therefore Non-Conformities (NC) are used by Manufacturing / Manufacturing Quality to declare failures that appear during operations scheduling, manufacturing or design.

The document "Managing NonConforming Items by NonConformity-Notification (NCN)" (A2406.1) describes how the NC is recorded in a NonConformity-Notification and how the lifecycle of an NC looks like.

Non-Conformity is classified as significant or insignificant. Only the Design Organization may determine what an insignificant non-conformity is. The parameters to declare a NC as insignificant are collected and approved by the DO in the Concession Support Decision Manual (CSDM – AM2205).

An insignificant non-conformity is a non-conformity that fulfils the requirements specified by the Design Organization in the final status (after rework/repair) and has no effect on any of the above conditions.

The Quality Organization, or authorized persons within the AIRBUS integrated MAP (Mise Au Point) or Material Review Board shall assess an insignificant non-conformity in accordance with criteria defined by the AIRBUS Design Office.

A significant non-conformity results in a concession or DQN and is a non-conformity which after the final status (i.e. rework/repair) may have an effect on one or more of the following, as appropriate (refer to A2406.1):

- airworthiness,
- strength (static and fatigue),
- installation,
- functioning,
- processes,
- interchangeability,
- quality,
- documentation,
- in-service implications,
- cumulative effect due to other nonconformities.

21.2 Changing the “Source” of NC (Design Query Notes (DQN))

A DQN is a process to introduce changes into the Definition Dossier after the occurrence of NC in the Production Organization. The DQN normally involves several ATA and responsibilities.

As soon as design documents cannot be implemented by manufacturing within the aircraft they must be changed. Therefore the design office must issue a Design Query Note (DQN).

The process is described in “Manage Design Non Conformance” (refer to A2838).

The DQN provides validated and industrialized engineering solution against non-conformity identified by the manufacturing of a released design.

The workflow starts with a Non-Conformity (NC) request from Manufacturing as soon as a non-conformance has been identified.

A DQN can only originate from Manufacturing Non-Conformance.

The DQN technical solution is applicable at least for the MSN concerned (= MSN for which the DQN request was raised), and for other MSN based on the effectivity of the DS listed as impacted.

The DQN process shall be in line with the “DMU as Master” policy. This ensures that the technical solution described in the DQN and applied on the A/C is consistent with the DMU at all times.

As a DQN has a direct impact on aircraft design, the DQN approval has to be performed by a Technical Approver according to AP1020 signature rules.

Manufacturing shall use (for a limited timeframe) the technical solution provided by the DQN.

This technical solution must systematically be used to update the Configured Definition Dossier (including DMU).

The elements of the Definition Dossier requiring updating to solve the non-conformance are considered as impacted by the DQN.

Impacted elements (Design Solution and associated Design Data Set) must be especially listed in the DQN.

For example, the Non-Conformity 12063603 describes a clash of bundle 8045VB with a waste water pipe in section 13 of A380.

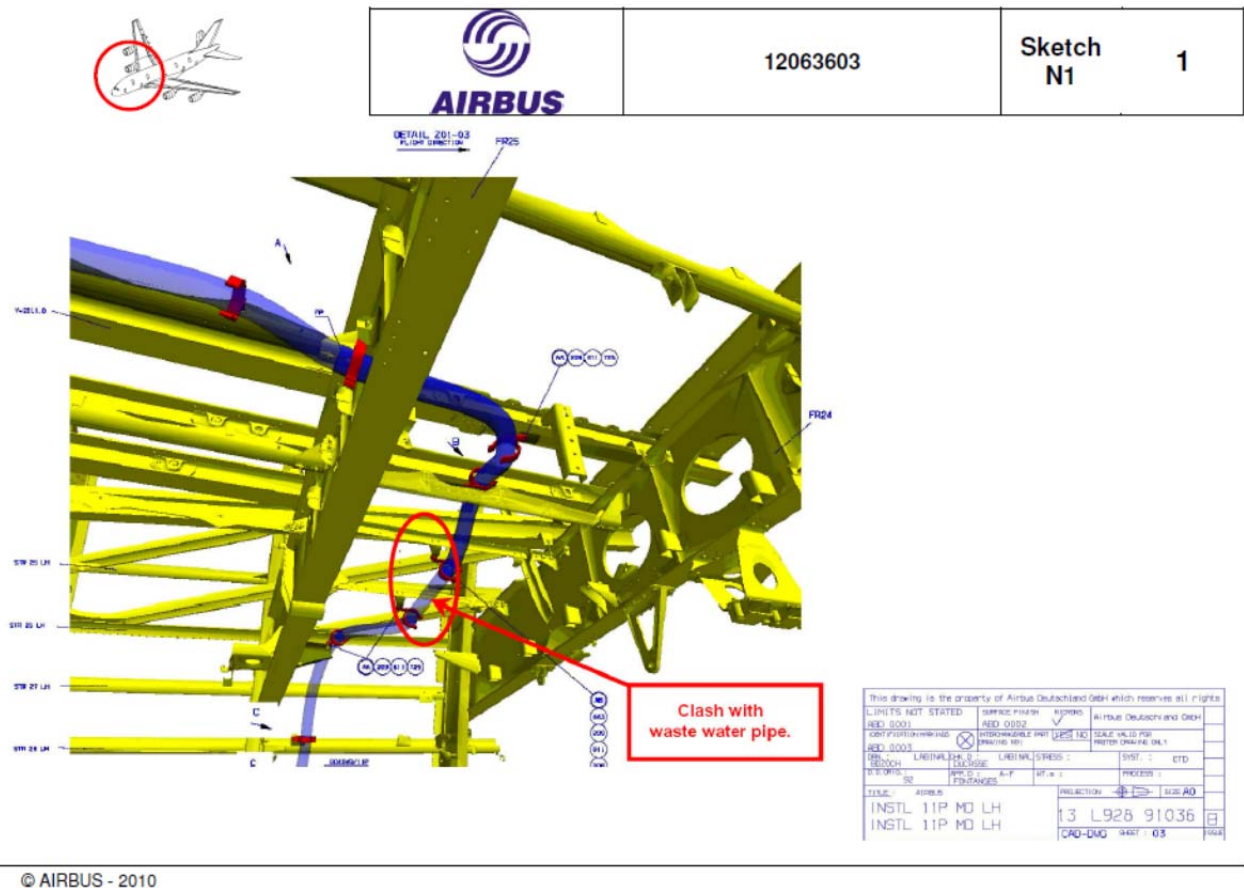


Figure 21-1: Non-Conformity 12063603 describes

The technical solution of this issue is described in DQN TH0072247. The clash has been prevented by deleting a spacer and moving 2 brackets to another position. The Definition Dossier will be updated accordingly.

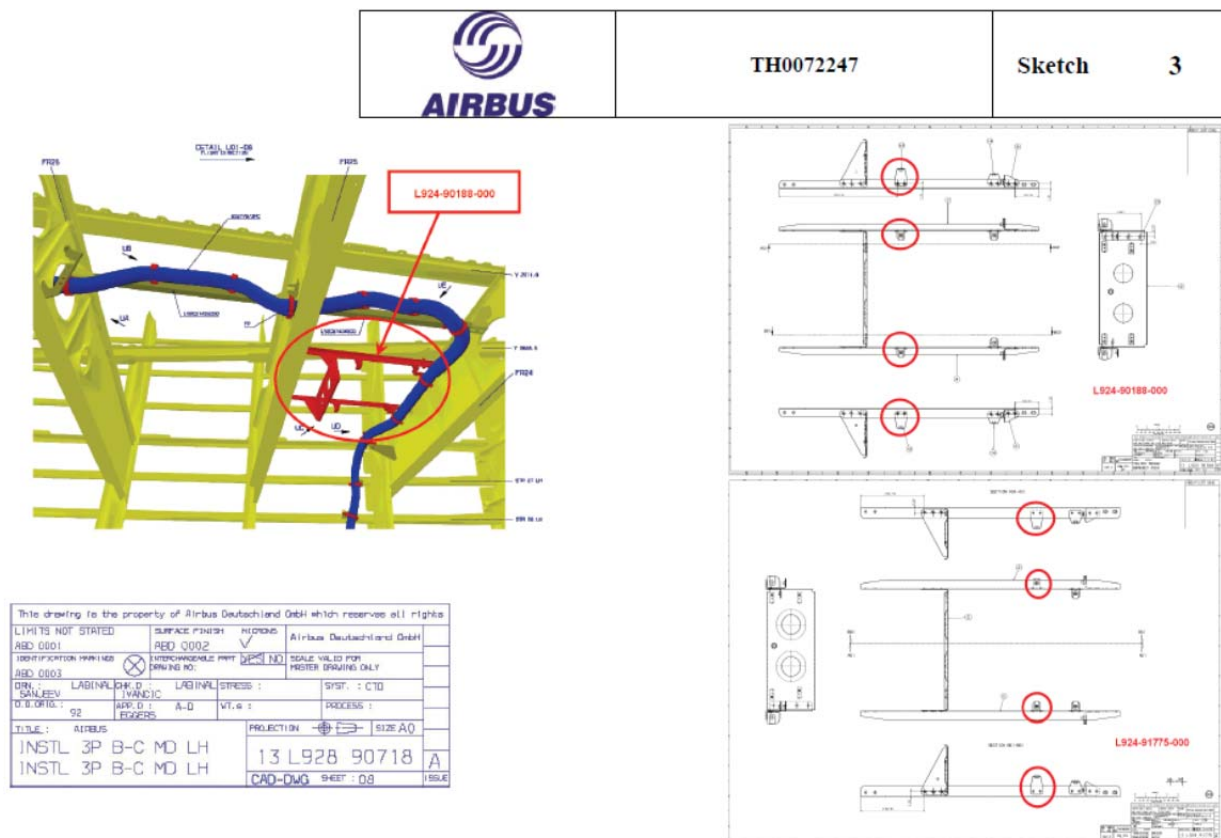


Figure 21-2: Solution from DON TH0072247

Further information regarding this example can be retrieved from the long-term archive ZAMIZ.

21.3 Request for Design Rework (RDR)

Similar to the DQN process but based on a common agreement of DO and ME instead of a NC, a RDR can be created due to different reasons. In case of urgent decisions or if deadlines for Design data Sets release are not met, RDR is quite a faster process to release design changes to avoid delays or out of sequence work in manufacturing.

RDR Solution comes in replacement of CMP (Croquis de Mise au Point) and forward BM (Beanstandungsmeldung), and is therefore a so called “forward change” with the same approach like the two replaced procedures: enabling the Design Office to provide a quick solution of design to the Manufacturing Engineering.



Figure 21-3: RDR flowchart (AP5342.2)

Figure 21-3 shows the main steps of the RDR process. A prerequisite for a RDR solution is an open MOD/MP. Next to the similarity to the DQN processes, even the marking in the DMU looks similar.

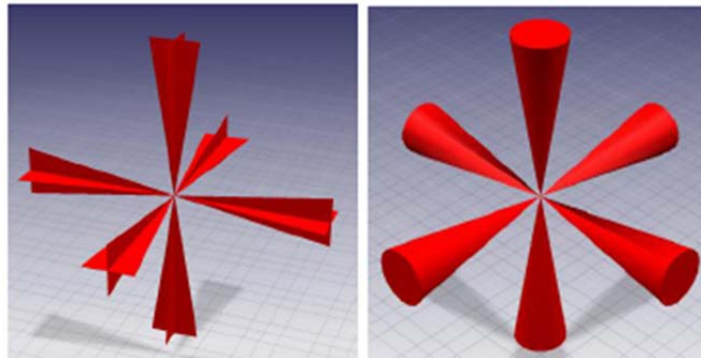


Figure 21-4: DMU-Marking of Task 100 “Analysis ongoing”, Left: DQN, Right: RDR

21.4 Handling the “Results” of NC (Concessions)

The NC can be an error on the part of the worker, e.g. the holes for the bolts are drilled in the wrong size, the result being that only two and not four repair bolts could be used. This would lead to a decision to say “All right, it is not really an error, the bolt is all right, but it can only be repaired twice.” As this leads to a restriction it is advisable to inform the customer accordingly.

The main restrictions related to manufacturing are²³:

- Manufacturing Instructions
- Employment protections / safety regulations
- 80T - Site Germany-specific detailed descriptions of manufacturing processes and methods
- 75T - Site Germany-specific MATERIAL SPECIFICATIONS
- Assembly Handbook

The concession would receive a so-called suffix, in this case one “R” (Restriction) and one “C” (Customer) to be clearly recorded in the documentation of the changes (to the type design).

²³ Refer to Speech “Manufacturing Processes,” held by Maxi Jahnke (TBXQD) for Technical Approver Education.

With this example in mind, the concession is the permission to use or release a product that does not conform to specified requirements.

The process of “Manage Concession” is described in A2406.2 and the Means & Methods are in the document “Classification Guidelines for Concession” (refer to AM2223). The most useful aim of the AM2223 is to give guidance for decisions regarding structural elements.

In the event of a manufacturing failure a concession shall be created to describe deviations of the aircraft’s construction status from the design documents or specifications.

The technical decision to accept a non-conforming item with or without repair by concession shall be taken only by the Design Office.

The Design Office is responsible for examining the non-conformity and making the decision.

Based on the technical assessment, a decision shall be made on whether to accept by concession or reject the non-conforming item. Where necessary, any further work to be performed and/or “limitations imposed on the part” (special spares, inspections, etc.) shall be defined.

Once the Design Office has decided to accept a non-conforming item by concession, they shall classify the concession with or without suffixes. One or more suffixes shall be allocated to a concession when the item needs to be traced. In all other cases no suffix is required. Depending on the level of reporting and the traceability required, one or more of the following suffixes may be allocated:

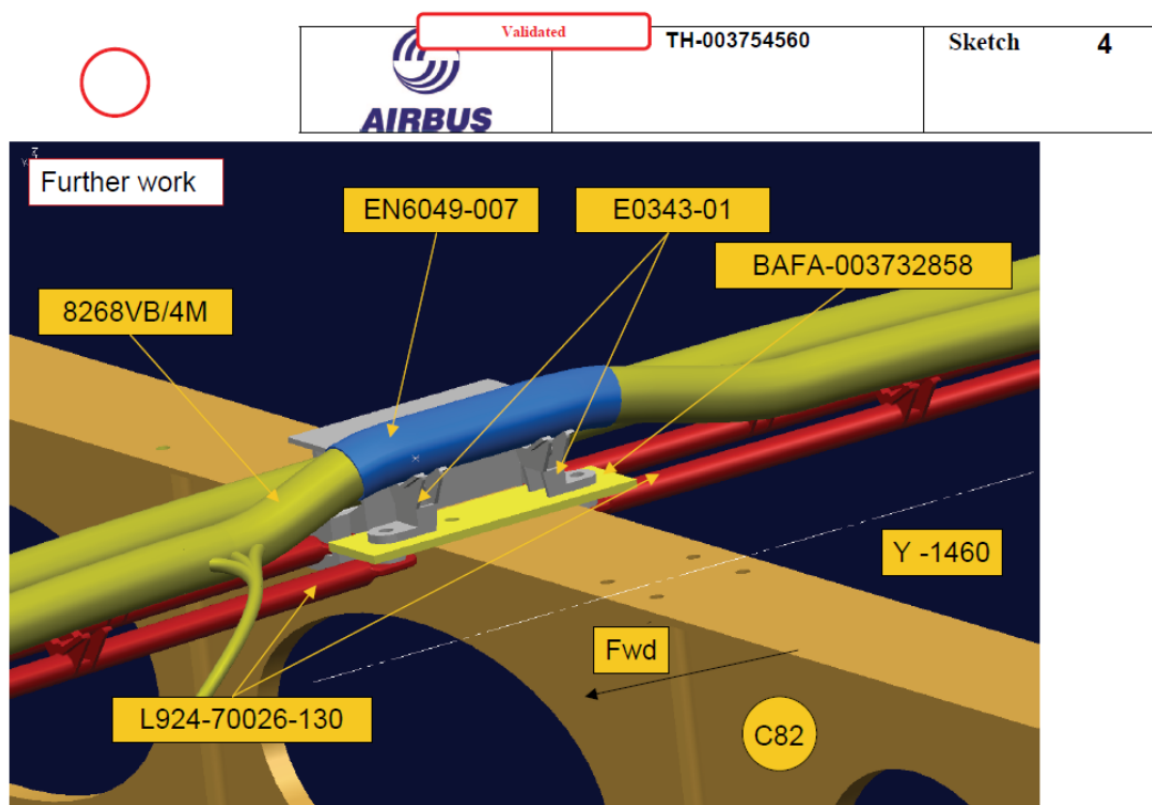
Suffix	Meaning	Definition	Type
R	Restrictions	Concession imposes a limitation on the use, maintenance or life of the item	Permanent
C	Customer	Concession needs to be brought to the attention of the Customer	
P	Performance	Concession may limit the performance of the item to its initial design intent	
T	Traceability	Concession needs to be traced internally	
Q	Quality	Concession needs to be highlighted at a later stage - requires no work/inspections to be performed	Temporary
FW	Further Work	Concession requires work/inspections to be performed	

Table 21-1: Concession Suffixes

Non-Conformities specifically covered in and permitted by applicable in-service documents with limitations but accepted by concession without limitations shall be classified suffix C. In this case, the C suffix classification is used to alert the Customer that the limitations imposed by the in-service documents are not applicable for the non-conformity described on the concession.

Customer representatives shall be provided with an English version of all individual suffix R and C concessions raised against their aircraft at an early stage in accordance with the decision of the Quality organization. R and C concessions, which also have an FW suffix, shall not be distributed to the customer.

Concessions (except concessions with the suffix “R”) must be released finally by Technical Approvers according to AP1020. Concessions with an “R” suffix shall be subject to an airworthiness assessment by authorized signatories (Compliance Verification Engineers – CVE) from or acting on behalf of the local Office of Airworthiness.



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Figure 21-6: Solution from Concession TH-003754560

Further information regarding this example can be retrieved from the long term archive ZAMIZ.

21.5 Non-Compliance Process and Deviation to TDD92 / SIDP92

TDD92/87 represents the reference document for aircraft certification. This means that all design work in the Electrics domain must comply with TDD92/87 in order to secure (type) certification, MOD approval and finally the certificate of Airworthiness for each single MSN.

In exceptional cases, design cannot fulfill a TDD (or in A350 also SIDP) requirement and by mean of "Deviation" an alternative solution will be accepted by the requirement owner. This is the basic principle of "Deviation" for all Airbus programs.

For A350XWB and future programs, a deviation is also required, if the technique is not in line with SIDP. This is based on the fact that this formally "recommended" design principles became mandatory in A350 XWB.

The reason is that in the past, the strict separation between requirements (TDD) on the one hand and solutions (SIDP) on the other hand was not realized the way it is in the A350 program and the whole content of the SIDP was only recommended.

Due to the development of the organization, the authority's requirements and the complexity of the products the deviation process has received some changes, too.

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During the development of A380, the design organization of Airbus was divided by NatCos having their individual Design Organization Approval (DOA). Due to this, each Issue of a TDD has to be accepted by each NatCo and sometimes different Issues are used in parallel. Today, there is only one DOA existing for all the Airbus design organization allowing a more centralized and harmonized way of working concerning TDD/SIDP and the related Deviation process.

Even if the basic principle of requirement and the compliance management is the same, the following rules are first applicable for A350XWB.

AP5884 "Management of Non-Compliances to Specification Baseline" describes all the activities of investigating any non-compliance, its impacts and proposed corrective actions. A Deviation is only one possible way to solve the non-compliance, another solution could be the change of design or the requirement itself.

If the investigation results in a Deviation as the only way to solve the non-compliance, the process described in AP5370 "Deviation Management Process" is applicable for all programs.

Input document is the description of discrepancy for which a template exists (Requirement Discrepancy Form: FM0901580_RDF). The technical approver acts in case of a deviation request as local design authority and has to check and validate the request before the originator is allowed to send it to the requirement owner.

Different actions are necessary depending on the classification (major/minor) of the Deviation and the impacted document(s). A Deviation is major if its acceptability leads to restrictions like a limitation on the use, maintenance or life of the item. Otherwise it is considered as minor. In case of minor deviations, the related solution and the deviation itself can be approved/authorized by a delegated technical approver.

The reason for a deviation is always a design problem or challenge, but it shall never be used to accept bad solutions.

For A350 two detailed notes exist that describe the non-compliance management (V00PP1000154) and the deviation process (V00PP1000762).

22 Delivery Preparation

22.1 Certification (MOD Closure)

The steps necessary to attain MOD Closure are as follows:

- MOD Closure Stage 0 – Design intent complete
- MOD Closure Stage 3a – MOD closed but not all documents are updated
- MOD Closure Stage 3 – MOD closed and all documents are updated

The MOD Closure Stage 3 must be reached within four weeks after 3a. The Stages 2 (Release for Power On) and 2a (Release for First Flight) are no longer in use. Deadline for design activities is MOD closure stage 0. At that stage, all design data must have been released for the first time, but could be impacted by DQN/RDR if “first time right” is missed.

The MAS closes this process step and the MAS Closure is performed only once. This allows us then to use the MOD in the production organization, or in other words, the Fulfill Domain for serial production.

Please be aware that MAS for a MOD can only be obtained if all Means of Compliance (MoC) are fulfilled in agreement with the Authority, at least with the AMC (Acceptable Means of Compliance).

Certification can take place if we show all required data for Closure Stage 3(a). Therefore all relevant information from the MP level can be compiled into the Technical Dossier (TD).

The TD is prerequisite for MAS and leads up to the signing of the MAS unless we have only one MP in the MOD.

If the MOD contains one MP only, the generation of the TD can be omitted (refer to “Change Process - Technical Change Documents (TRS, TD, MAS)” (AP5130)).

In the case of minor changes AIRBUS is allowed to sign the MAS on behalf of the Authority. For major changes the Airworthiness Authority (AA) signs an approval letter which will be referenced on the MAS by the DAE who signs it. In case of a new type certificate or significant major change (both applied with EASA Form 31), all change information are forwarded to the Airworthiness Authorities using a certification summary which provides all the results of the certification activities performed. The summary is signed by a DAE giving the statement of conformity. The final approval will then be given by the EASA and, if applied, by the FAA.

But what is a major or minor change? By its technical content, a change has different impacts to the aircraft. Depending on the impact, the change is classified and takes different processes concerning the way of certification. Figure 22-1 shows the decision tree a change has to pass through on its way to certification.

Nevertheless, whatever classification decision done by Airbus, Authorities could overrule the decision.

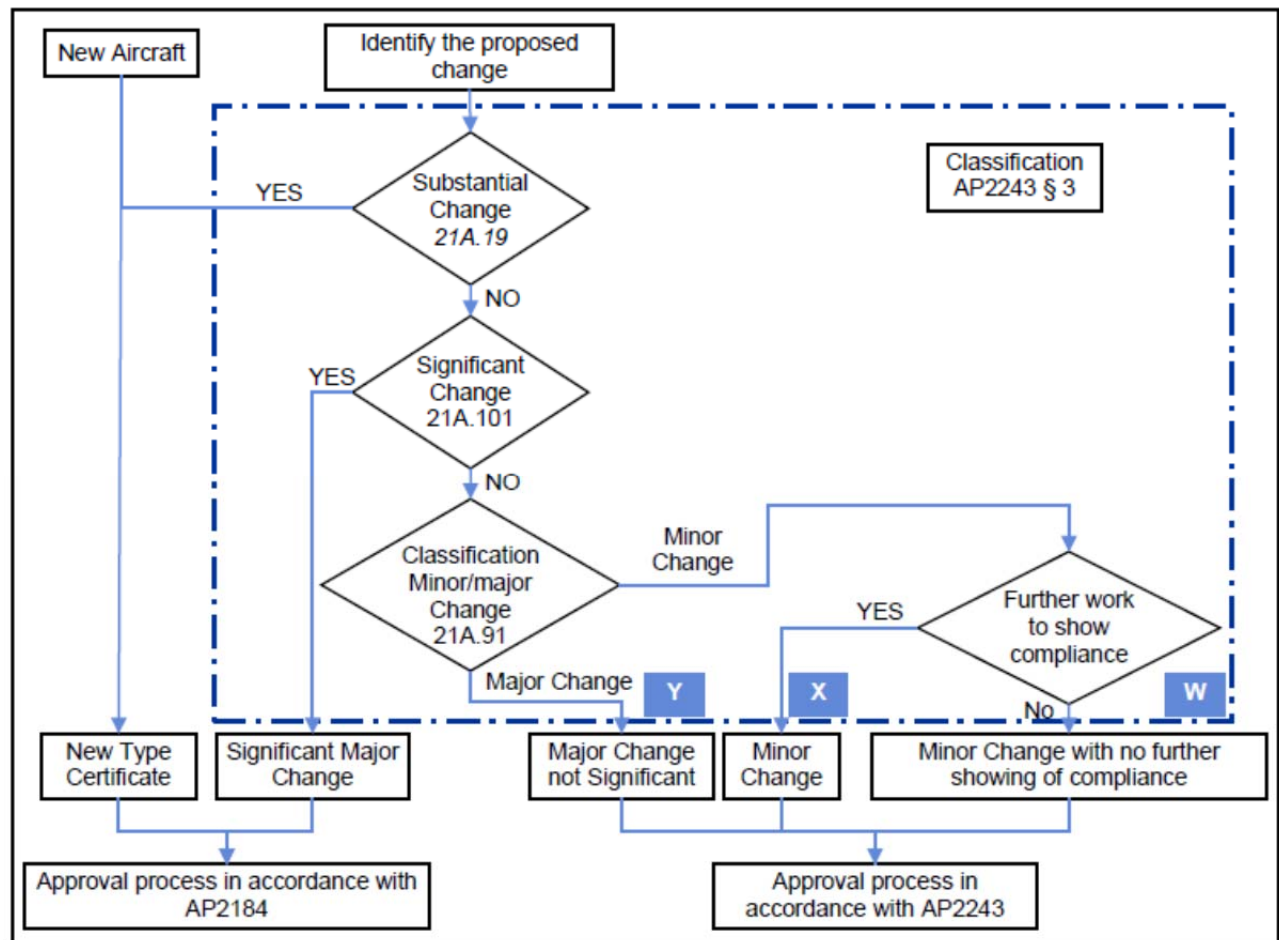


Figure 22-1: Decision tree of classification (AP2243.1)

The letters Y, X and W shown on Figure 22-1 are the values used to identify the classification of a MOD in ACC2. In case of classification “W”, no CVE/DAE is involved. In that case Technical Approver signature on the ECN attests the classification by matching the classification criteria and acts as airworthiness approval of the change under AIRBUS DOA. As the Technical Approver is also performing the initial classification, no doubt in classifying a MOD as “W-MOD” is acceptable it is always recommended to get consultation by DCS/CVE or DAE.

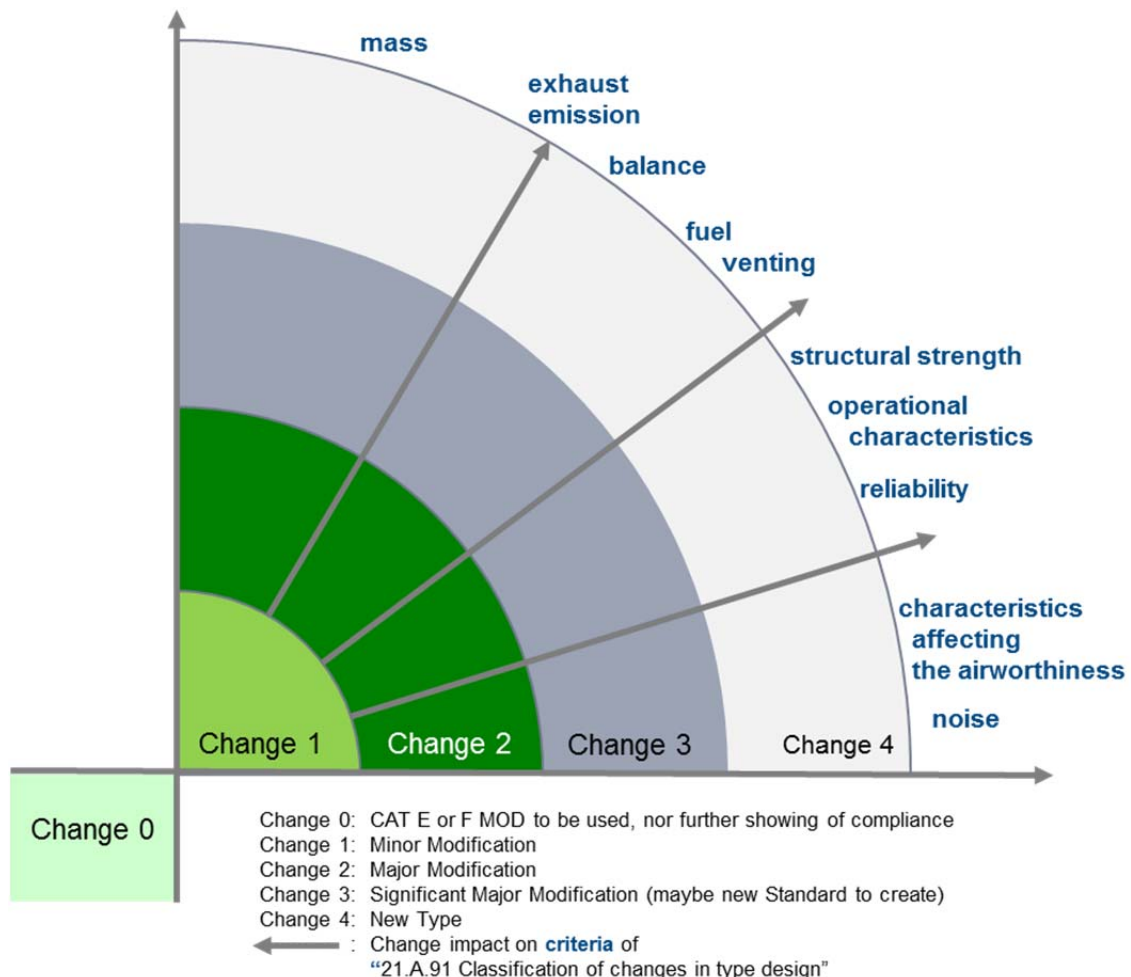


Figure 22-2: Classification criteria of a change

The Authority knows 3 different classifications of a change: Minor, Major and Significant (21.A.91, 21.A.101).

In Airbus Design Organization, we know two additional kinds of minor changes. Figure 22-1 illustrates how the classification is going based on what criteria.

The appendices of AP2243 provide the detailed classification guidelines integrated in Airbus' classification procedure.

The basic steps to approve a change are similar, but not the documents created for approval and the related responsibilities.

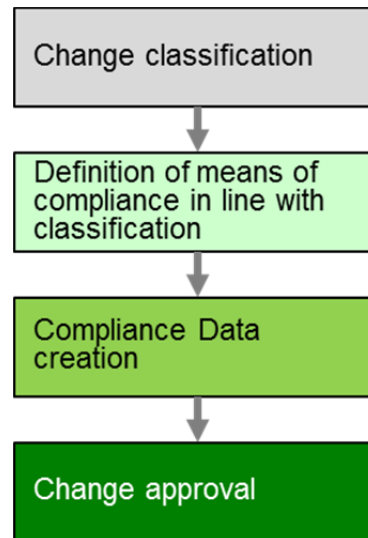


Figure 22-3: Four steps to approval

If a change is classified, the certification is planned and documented in different ways referring to Figure 22-2 Changes:

Change 0: Change approval by classification decision based on approved internal criteria

Change 1: Modification Approval Sheet (MAS)

Change 2: Major Change Certification Plan (MCCP)

Change 3: Certification Plan

Change 4: Type Certification Plan

For the detailed description of who is signing what document, please refer to AP2243. At the end of the approval process stands an approved MOD that we call a “Type MOD” in case of type certification relevance, a “Major MOD” in case of major change, a “Minor MOD” in case of minor change and a “Specific MOD” (CAT E or F) in case of a change with no further showing of compliance.

When we say that the change can be “approved by Airbus”, we mean that because of the privileges given by the Design Organization Approval (DOA), an Airbus employee can act on behalf of the EASA. This Airbus employee in the most cases is the DAE.

Having a look on the DAE stamp next to his signature shows that it is still the EASA which stands indirectly for the approval:



Figure 22-4: DOA Stamp

Another conclusion of design approval process and the involved airworthiness functions is that the CVE function is a prerequisite to get a DOA by showing the implementation of an Independent

Checking. The main document the CVE is signing during MOD approval is the Independent Check Sheet (IPS), which lists all the documents showing compliance to the applicable airworthiness requirements.

The DAE itself is already a result of the DOA, we could say that he is the personalized privilege.

This task is the responsibility of the design organization and the corresponding approval (DOA).

For the delivery of the A/C we also require the attestation of the CAs. This means that all relevant GTR / GTI and FTI have taken place.

The CA is released by means of the EASA Form 1, which will be signed by Quality (Certifying staff). The major prerequisite is the generation of the CAIR, which is a prerequisite for the AIR.

The AIR requires the corresponding CAIR for all CA from which selected data are transferred to the AIR. The AIR is prerequisite for the EASA Form 52, which attests on aircraft level.

The AIR documents the compliance of the A/C with the design documents and is compiled and under the responsibility of the Production Organization.

Before the A/C goes through the Flight Line to the Delivery Center every Non-Conformity (NC) with the expected status leads to a correction process. This can lead to changes in the design organization, meaning the Definition Dossier.

Changes to the DDS, if necessary, will be carried out by means of the DQN.

Deviation in the production organization without impact on the design leads to concessions. Concessions take the decision to use a part even with an error or give it to scrap.

All NC must be incorporated to reach the goal that all delta are "zero."

After transfer of title and the issue of the EASA Form 52 and on condition the EASA Form 53, which covers maintenance work in the production organization, the aircraft can be delivered to the customer. These tasks are the responsibility of the production organization.

22.2 Attestation-Related Documents

22.2.1 Constituent Assemblies - CA (EASA Form 1)

A CA is an Assembly with some special conditions as described in "Cascade Components - Rules and Data" (AP2135). One important attribute of a CA is that an EASA Form 1 is mandatory for delivery and / or integration into the Top-Level CA, the A/C itself. This can be done inside AIRBUS, the basis of which is defined in IR-21 (21A.163(c)).

An EASA Form 1 essentially states that this particular CA is manufactured in conformity to approved design data, meaning AIRBUS has MAS on all MOD inside this CA. As long as the CA is not delivered across borders, the CA can be attested with non-approved design data. Additionally, every CA needs a Constituent Assembly Inspection Report (CAIR). Further information is provided in the document "CA Inspection Report" (refer to AP2104).

22.2.2 Aircraft Statement of Conformity (EASA Form 52)

The possibility to obtain a Statement of Conformity (EASA Form 52) is a privilege granted by the Authority for a holder of a POA as defined in IR-21 (21A.163(b)).

For Details of the EASA Form 52 refer to IR-21 ("Statement of Conformity," 21A.130(a)). As mentioned earlier, an EASA Form 52 requires the AIR (Aircraft Inspection Report, refer to AP5106) and

the AIR requires the condensed data from all CAIR of **all** CA. So, if one CA with its CAIR fails, e.g. a harness is wrong in some way, delivery of the A/C is not possible.

With the EASA Form 52 we hereby declare that:

- the aircraft is fully consistent with the logical model registered at TC (Type Certification),
- the aircraft has passed the flight tests with good results,
- the aircraft is in a condition for safe operation.

The last point means that

EVERY CHANGE TO THE TYPE DESIGN HAS BEEN COVERED BY THE MODIFICATION SYSTEM.

That also means that the A/C and the documentation (e.g. AWL) must be exactly the same, otherwise the A/C stays on the ground and delivery is not possible.

In short, all changes must fulfill all EASA AND AIRBUS requirements / documentations.

This means that AIRBUS has successfully stated that:

- all recordable Concessions
- approval of additional Changes
- Certified Type Design
- Aircraft Technical Acceptance
- Determination of Aircraft Conformity

are in a status to declare that this A/C is in condition for safe operation.

After that, the Statement of Conformity (EASA Form 52) can be signed by Quality / Certifying Staff and the Airworthiness Certificate [FLORIO2006; Chapter 8; Page 141 ff.] to perform the ferry flight (of customer).

In the words of the FAA, for example, the Airworthiness Certificate²⁴ "... is an FAA document which grants authorization to operate an aircraft in flight ... and a registered owner or owner's agent of an aircraft may apply for an airworthiness certificate."

22.2.3 Release to Service (EASA Form 53)

After Attestation with the EASA Form 52 the A/C formally leaves the Production Organization and no more work can be done by the PO.

This would lead to the situation that maintenance jobs on the A/C could only be carried out by Maintenance Organization staff. To avoid this, the EASA Form 53 gives the privilege to the PO to perform maintenance work on a new A/C after the EASA Form 52 has been issued.

This is described in IR-21.163 (d) "Privileges":

²⁴ Refer for more details regarding "Airworthiness Certificate" to the FAA web site (http://www.faa.gov/aircraft/air_cert/airworthiness_certification/aw_overview/).

“... Pursuant to the terms of approval issued under 21A.135, the holder of a production organization approval may: ...

(d) Maintain a new aircraft it has produced and issue a certificate of release to service (EASA Form 53) in respect of that maintenance.”

22.3 Handling of Requests during Handover

Handover to the customer assumes that everything has gone well. The PACC was signed, the flight test was successful, PATR, e-TLB, PATM lead to the PATCC, AIR was closed (signed), QLB was closed, Folio 5 was closed and open points transferred to e-TLB.

Customer Acceptance & A/C Delivery is part of the FCO process and is linked to the “AIRBUS Operational Process Map (AOPM)” (refer to A01) related to:

- “Test Certify & Deliver”
- “Produce & Integrate Product and Services”
- “Obtain Payment and Transfer Title”

The “Production Organization Manual” (refer to A30) and “Design Organization Manual” (refer to A20) in relation to “Individual Aircraft Certification Procedure” (refer to AP5061).

All procedure and golden rules to handle those situations accordingly are described in the document “Manage Customer Aircraft acceptance” (refer to A2940).

23 Configuration Conformity Management & Documentation

23.1 Definition of Outstanding Work

The goal of CCMD is to calculate the outstanding work, to ensure that if it is “zero,” that the given configuration of the product (seen as a set of key characteristics) is equal to the configuration that the customer, Airbus and at least the authorities are willing to accept (at least in the sense of air-worthiness).

So, “... conformity is obtained when the key characteristics of an object are in line with the expected conditions initially expressed²⁵.” In other words the outstanding work is the difference between that which should be done and that which has really been done, thus leading to no “delta” left open if no outstanding work is left.

The calculation of “Outstanding Work” is done by means of so-called “Deltas.” A “delta” is defined as the difference between the input and the output of a process step. In our case this is normally a list of MOD/MP with their corresponding work orders.

The difference between the input to the process step, i.e. what should be done, and output, what actually is done in this process step, should be “zero.” This means all work is performed and nothing is left to do which is the basic idea of a “delta.”

23.2 “Delta” Calculation

To cover the entire process, different types of deltas are defined between Contractual Definition Freeze (CDF) and A/C delivery.

The official definition as described within A2845 lists four deltas and two documents which are tracking all of them (CAIR & AIR):

- Delta 0: Difference between the Target Configuration (MSN Definition) and the industrial configuration. This decision is the responsibility of program management. This delta is the only delta which is our decision, normally a change in the work share or a difference between the tool ACC2 (Aircraft Configuration Control) and the German tool TASKSY (Sub-Module MODA).
- Delta 1: Outstanding work in the Design Organization under Engineering responsibility. Normally the difference between industrial configuration (described in a list of MOD/MP) and the DDS on CA basis.
- Delta 2: Manufacturing Engineering outstanding work (under the responsibility of Manufacturing Engineering) as the difference between DDS and prepared work orders.
- Delta 3: Manufacturing outstanding work (under the responsibility of Manufacturing) as the difference between opened tasks provided with the prepared work orders and the tasks that are declared complete.
- CAIR: Summary of delta 0 – 3 in the CAIR measured against the specified target (under the responsibility of Manufacturing Quality).

²⁵ Refer to CCMD Training material, Department EDCC, Configuration Management, 2007.

- AIR: Summary of all CAIR-data in the AIR measured against the specified target of the A/C (under the responsibility of Manufacturing Quality).

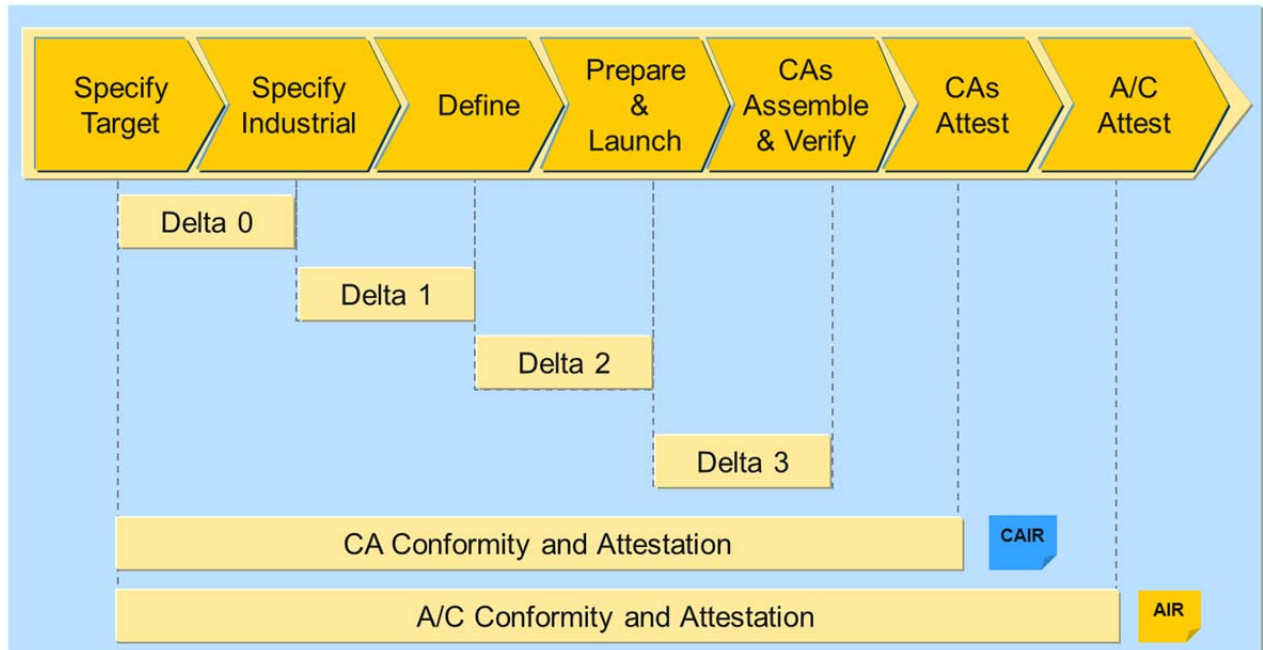


Figure 23-1: Overview, CCMD - Delta Definition

24 Aging Aircraft / In-Service

Changes to the A/C after delivery can only be made by means of Service Bulletins (SB). The existence of a MOD regarding the ACMR is mandatory for SB.

Service Bulletins can be “mandatory” or “not mandatory” and shall be offered to the customer. An Inspection Service Bulletin (ISB) is used on the basis of an SB for preparing changes during normal inspection.

A Service Information Letter (SIL) does not give airworthiness-relevant information to customers.

These activities are the responsibility of the maintenance organization. This domain has important impact on the design and the production organization to incorporate knowledge from in-service. The “Supportability Engineering” (refer to AP2605) is one important part to gain quality of the product during its life cycle.

24.1 Service Bulletins (SB)

A Service Bulletin describes a change to be implemented on an in-service aircraft.

The SB process is in accordance with the DOM (refer to A20, Chapter 3.3.1.0) and the “Produce Service Bulletin” (refer to A2095.1).

Service Bulletins may be necessary for a number of reasons:

During production, or as a result of an aircraft inspection, AIRBUS discovers a design defect. The result of such defects might be that changes have to be incorporated on an aircraft that has already been delivered. This can take place within the prescribed servicing interval or, if safety factors are impacted, grounding of the entire fleet might be necessary.

The airworthiness authorities propose amended requirements to AIRBUS. These requirements can be the result of an aircraft accident investigation. In this case, it might be necessary for AIRBUS to implement the changes in all or part of the fleet while still operating.

AIRBUS also issues Service Bulletins in the event of conversions (e.g. conversion of a passenger version to a cargo carrier – Pax to Freighter). In this case, Service Bulletins contain all existing Definition Dossiers for the Pax version plus additional Definition Dossiers concerning conversion.

Service Bulletins are also issued if customers desire changes on already delivered aircraft. This might concern defects discovered by the airline or changes proposed by AIRBUS retrospectively to the airline as an option.

24.2 Technical Adaptations (TA)

The TA is a technical answer, approved under the authority of EASA Design Organization Approval, to an operator request.

It states the compliance with the aircraft applicable certification basis.

The Technical Adaptation process (refer to A5009, “Approval of Technical Adaptation (TA) for Airbus In-Service A/C”) was put in place to provide directly an AIRBUS operator with airworthiness-approved data, in place of the simple technical advice, which was generally sent by AIRBUS with a No Technical Objection (NTO) document.

A Technical Adaptation can be:

- a repair, ruled by Part 21 Subpart M, or

- an instruction, ruled by Part 21.A61,
- a temporary minor design change, ruled by Part 21 Subpart D.

A TA cannot be used for an adaptation related to STC (Supplemental Type Certificate) / modified configuration not under AIRBUS responsibility. Further limitations for TA used to provide an approved solution are summed up in the related document.

AIRBUS receives requests to evaluate the technical impact of an adaptation to the certified aircraft configuration (adaptation to aircraft definition), an adaptation to maintenance procedures or an adaptation to operational procedures.

When found technically acceptable and in compliance with the applicable certification basis, the response is provided by AIRBUS to the requester as a Technical Adaptation (TA).

Providing a response as a Technical Adaptation is one of the AIRBUS DOA privileges and the TA is issued with the statement of "Approved under the authority of EASA-approved Design Organization Number EASA.21J.031 and as per EASA rules Part 21 Subpart M & D."

Any Technical Adaptation shall fall in several different categories. For details the document A5009 provides further assistance.

When there is "**no additional work to show compliance**," Part 21 does not request an independent check of the showing of compliance to applicable certification basis. That means no CVE signature on the Technical Adaptation is necessary. Only a Repair Technical Engineer (RTE) will approve the technical content of the Technical Adaptation. The DAE will approve the Technical Adaptation finally for airworthiness.

The role of the Repair Technical Engineer (RTE), which is also a delegated function of the design assurance system (AP1020.3), is in many cases the final technical checking function before the DAE approves the TA. For further information on RTE, please consult AP2433.1. RTE's have different responsibilities according to the delegated level 1, 2 (1st Independent Signature) or 3 (2nd Independent Signature).

25 Appendix

25.1 Extract of ATA chapter (based on A350)

ATA	Description
Airframe Systems	
20	Standard Practices
21	Air Conditioning
23	Communications
24	Electrical Power
25	Equipment / Furnishings
26	Fire Protection
27	Flight Controls
28	Fuel
29	Hydraulic Power
30	Ice and Rain Protection
35	Oxygen
36	Pneumatic
38	Water / Waste
44	Cabin Systems
47	Inerting System
Structure	
52	Doors
53	Fuselage
56	Windows
57	Wing
Miscellaneous	
89	Flight Test Installation / General and Sensors
92	Electrical and Electronic Common Installation

25.2 Reference Documents

Reference Document	Title
A00	Airbus Code
A01	Airbus Quality Manual
A1000	Requirements for Procedural Documentation and Design Standards
A1001	Document Retention Requirements
A1002	Airbus Project Management Policy and General Requirements
A1006	Control of Nonconforming Items
A1069	A350 cDMU as Master Directive
A1079	Structures, Systems Installation and Cabin Installation Definition Dossier Top Level Rules
A1350.3	DARE – Electrics Development Policy
A20	Design Organization Manual (DOM)
A2000	Manage Quality
A2015.2	Generate Customized Definition
A2095.1	Produce Service Bulletin
A2240	Design Structure and System Installation
A2406.1	Managing Nonconforming Items
A2406.2	Accepting Non Conformity Items by Concessions
A2838	Manage Design Non Conformance
A2842.3	Produce and Manage Catalogues
A2843.3	Manage Full Change Process
A2845	Manage Configuration Conformity and Attestation
A45	Airbus Maintenance Manual
A5000	Manage Procedural Documentation
A5009	Technical Adaptation (TA)
ABD0088	Standard Models – CAD Modeling Rules
AM2022	Configuration Baselines
AM2037	DMU Design Review with DVise
AM2150	Requirements Based Engineering Method
AM2211	Product Structure

AM2212	Functional Item Number (FIN)
AM2223	Classification and Guidance for Concessions
AM2388	Guidance on General Requirements for Systems and Cabin Items Designers
AM2389	Functional Analysis Methodology
AM2616	Safety and Reliability Handbook
AM5011	Electrical Installation
AM5014	Schematic Editor SEE XP
AM5061.0	CIRCE: Overall View of CIRCE-C/I/G/P
AM5061.6	Configuration Management into CIRCE
AP1007	Configuration Management
AP1020	Rules and Guidance for Signature and Release of Technical Documentation
AP1034	Requirement Based Engineering Policy
AP2020	Continued Airworthiness
AP2041	Customer Acceptance and Aircraft Delivery
AP2054	Develop new Aircraft (DnA): Process Definition
AP2078	Change Process during Concept / Definition Phase
AP2104	Constituent Assembly Inspection Report (CAIR)
AP2135	Cascade Components – Rules and Data
AP2146	Transfer of Development Aircraft and Items between Design and Production Organisations
AP2161	Aircraft Functions Definition Process
AP2183	Managing Quality Audits
AP2184	Type Certification Procedures
AP2190	General Requirements for Aerostructure and Material Suppliers
AP2243	Classify and Approve Changes to the Type Design
AP2266	Digital Mock-Up Rules and Life Cycle
AP2270	Final Assembly Line & Major Component Assembly Line Industrial Process
AP2296	Aircraft Products and Services Validation and Verification Process
AP2579	Steering Configuration
AP2585	Management of DOA-POA Arrangements and Direct Delivery Authorizations in Airbus
AP2600	Developing Aircraft Definition Model
AP2601	Design Principles

AP2605	Supportability Engineering
AP2618.0	Frontier Design Process
AP2641	Product Structure
AP2750	Product Safety Process
AP2750.10	In-Service Occurrences (ISO)
AP2761	Special Instruction Management
AP2763.2	Release Process: Equipment
AP2763.3.1	Electrical Schematic Diagram Release Process for A380
AP2763.5	Release Process: Standard Item FINs
AP2766.1	Create and validate Structure, System and Cabin Integration Product Definition with configures DMU
AP5061	Individual Aircraft Certification Procedure
AP5067	Approval of Flight Conditions Document for
AP5070	Issuance of Permit to Fly for Development Aircraft Issuance of Permit to Fly for In-Service A/C
AP5106	Aircraft Inspection Report (AIR) Requirements
AP5130	Change Process – Technical Change Documents (TRS, TD, MAS)
AP5156	Manage Interfaces Between Cabin, System & Structure from Request to Attestation of Installation in Aircraft
AP5157	Change Process – Electrical Repercussions Elaboration
AP5175	Create Catalogue and Manage Standard Offer Data
AP5204	Change Process - Modification Closure
AP5215	Flight Clearance
AP5365	Provide Operators with Operators Information Transmission (OIT)
AP5366	Provide Operators with Alert Operators Transmission (AOT)
AP5617	Space Allocation Design Process
AP5763.0	Release a Design Data Set
AP5763.3.2	Electrical Schematic Diagram Release Process for A400M, A350 and Future Programs
AP5801	Flight Test Installation Validation Process
AP5997	The Electrics Process
AP5997.1	Create Specific Electrical Functional Design
AP5997.2	Electrical Definition and Installation Process
M20091	CR/MP Concept and Rules

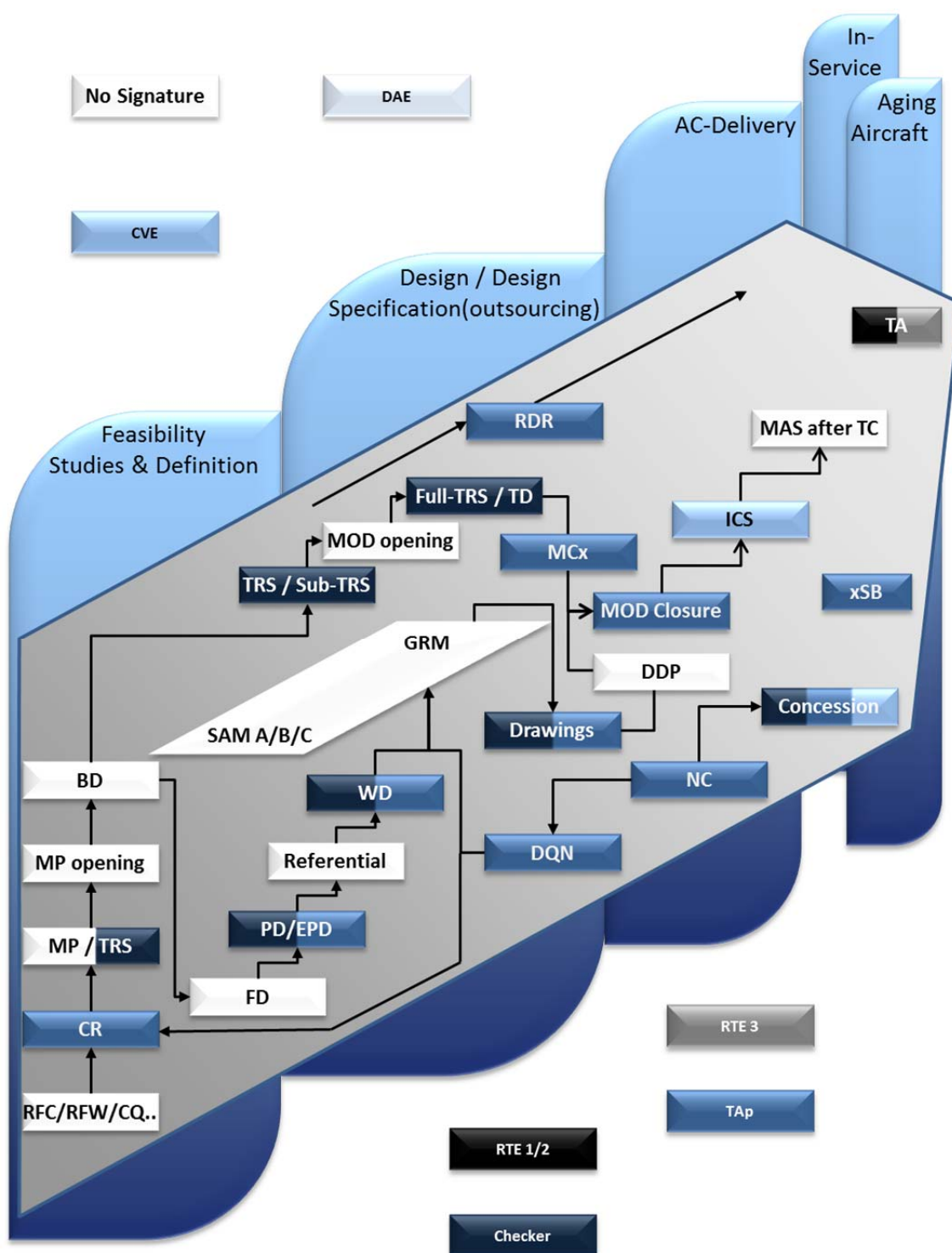
M20196	Manage Fleets
M2277	Method and Rules for Creating Electrical Definition Dossier
M2350	DARE – Prerequisites and Maturity
M2807	Criteria to identify “Significant Events”, “Engineering Occurrences” and In-Service Occurrences”
M2832	cDMU Quality Assurance Process Including Roles and Responsibilities
M2857	Cooperation Between Design Organization and Production Organization According to EASA Part 21 G
M2900	Procedural Document Writing Rules & Recommendations
M5112	e-Tool
SIDP20A001V	Design Principles Applicable to Systems Installation Bonding
SIDP92A001V	Design Principles Applicable to the Electrical and Optical System Installation

25.3 Use of Electrics Technical Notes, Technical Reports and Technical Memos

Document	Reference	Template	Essence	Case of Use	Release		
					Compiled	Checked	Approved
Technical Note	ABD0045 (serial a/c) M2902 (A350XWB)	FM0900476	Explanation of an existing drawing set	Manage additional / complementary information that are part of an existing definition dossier. These information contribute to the definition either of a specific design solution or the aircraft itself. They are used when information cannot be given directly and fully on a drawing due to the volume of text involved. Technical Notes have to refer to the drawing set they are related to, containing MOD/MP. They are an explanation for manufacturing and must not be used to change the design of an aircraft (like a DQN etc.). For details please see mentioned references.	Designer	Technical Checker	Technical Approver
					Author(s)	Approval	Authorization
Technical Report	no procedure / method AM2280	FM0200016	Summary of a special event / investigation	Contains technical information relating to a <u>special event</u> / investigation (e.g. DMU / hardware review, technical study). Technical Reports contain the results of mentioned events as a guideline for further processing (e.g. actions to be performed). Technical Reports must not be used to change the design of an aircraft (like a DQN etc.) or any design requirements (e.g. SIDP). Release on signature table as described within AM2280 ("Engineering Templates")	Designer	Technical Approver	Technical Authorizer
Technical Memorandum	no procedure / method AM2280	FM0601351	Announcement about dealing with a technical issue	Contains technical information relating to a single subject. Technical Memos can be limited in time. They can be used as an announcement about dealing with a technical issue. Technical Memos must not be used to change the design of an aircraft (like a DQN etc.). Release on signature table as described within AM2280 ("Engineering Templates")	Designer	Technical Approver	Technical Authorizer

ATA92/97 Technical Notes, Technical Reports and Technical Memos have to be created by using given templates stored within FileNet on Airbus People. Release of these documents have to be performed according to AP1020 signature rules. If Airworthiness is impacted Airworthiness Signatures have to be involved. It is not allowed to change Design principles, procedures, processes or methods by using the documents mentioned above.

25.4 Simplified Electric Process covering a “Change” with related Responsibilities, Documents & Levels



25.5 Requirements for Electrics Technical Signatories

Electrical Process	Design Org. Signatures	Rules and Procedures	Configuration	Preparation/ Check of Definition Dossier	Airbus	Airworthiness
Forward Change Process CR/RFQ MP/ISAM TRS (Sub-Full)/ MOD-opening PD / EPD VKE / Referential SFM / GRM VKH / VKM WD Treatments / Drawings MOD-closure TD / MAS	Responsibilities CVE, DAE, DCS, CE Technical Authority Technical Approver Technical Checker Displ Superior	Top Level Documents Airbus Code (A00) AQM (A01) COM (A10) EOM (A10-E) DOM (A20) POM (A30) MOM (A40)	TRS Content Sub-Full-TRS Evaluation TRS Involvement Sheet	Formal Title Block ISO 123-24 Identification Marking FA [...]	Work Share Report 6 [...]	Product Safety / Liability Safety Culture Document retention
Correction Processes NC; DQM, WQCN Concession; Deviation Permit to Fly; NTCOTA BSI BM; AM/Letge; CM; stop order (stop & proceed; SB; SI)	Document Types Level 1 Level 2 Level 3 Airworthiness Independent Checking	ACMR MOD-System; MP; CIN MOD-categories MOD-Documentation Validities MOD-Relations [...]	MOD CIN-Stack MOD-Stack MP-Stack Lead Times Engineering Deadlines Data Flow (A-CE/A-D) MP-CA MOD-Leader MOD/MP-Breakdown Domestic - Anti-MOD Standard, Option PGE PSS; CRS, ERS, [WRS] DAP A-counter B-counter	Content DD compliance SIDP compliance ABD compliance Correct informations e.g. section, deck, FIN SI [...]	Strategies Subcontracting "Airbus Crown Jewels" [...]	FAR / EASA CS AMC EASA Form 1 EASA Form 52 EASA Form 53
DARE Maturity Gates [...]	Signatures Compiled Checked Approved Authorized Means of Compliance Airworthiness Signature	ABD ABD0004 ABD100 ABD 200 ABD0031 [...]	Product Structure CI-LO-DS CA Views [...]	Delivery Report SFM / GRM VBE / BGE [...]	ATAI/BAABI Common understanding	MC MC0 – MC9
DnAdnE MO-M14 Feasibility Phase Concept Phase Definition Phase Development Phase Quality Gates	Airbus Approvals DOA POA MOA MTOA	ADSR / BOH Numbering Specs, Stop & Proceed ECN [...]	CCMD Attestation Definition and benefits Delta 0-5	Check Lists [...]	Manufacturing 80T / AIFS BU-Release Timeline [...]	GTR Types of GTR (5-8) ABD0200.2.3 GTI
FCO Purchase Agreement Industrial Delivery [...]	Tools TAKSY CIRCE CATIA SAP PRIMES BCAWD [...]	Processes / Methods AP; AM; A; M TOH; AA; PP; QVA System Requirements / Design Principles SIRD / SWRD/ TDD 82A001 TDD 20A001/4/6/9 SIDP 92A001 SIDP 20A001/4/6/9	Change Forward changes Backward changes MOD 9999 MOD 9998 MOD 9997 [...]	Programs Differences (processes, rules...) Numbering (e.g. MOD, MP, ...	Approval MC MOD-closure MAS	Certification / Attestation Type Certification Certification Base CR)

26 Directories

26.1 Figures

Figure 1-1: Overview, Airbus Design Office Sites (as of September 2014)	16
Figure 3-1: Overview, Structure of ATA Numbers	22
Figure 3-2: CM processes and disciplines allocated to the three Airbus domains (AP1007)	23
Figure 3-3: Product Structure, Overview of Views, Extracted Views and Deliverables (AP2641)...	24
Figure 3-4: Three Levels of Product Structure	25
Figure 3-5: CI/LO/DS – transfer CIN-Validity to Effectivity (Effectivity for FIN01 not shown)	26
Figure 3-6: Package related CIN and corresponding LO	27
Figure 3-7: Change Process Flowchart taken from A1080	28
Figure 4-1: Overview, Simplified DnA Process	29
Figure 4-2: Overview, Simplified FCO Process	29
Figure 4-3: Overview, Simplified DARE Process (refer to A1350.0, Issue A, Chapter 3.1, “Overall Planning”)	30
Figure 6-1: Overview, Airbus Documentation System (as of February 2010)	39
Figure 8-1: Design, Airworthiness and Independent Monitoring Function.....	45
Figure 8-2: Way of delegation (source: DOM A20, Issue C Rev. 7).....	46
Figure 8-3: Design organizations requirements forwarded by Airbus	47
Figure 9-1: EWIS for Type Certification and changes to type design	53
Figure 9-2: Risk of smoke and fire shall be additionally prevented by EZAP results	54
Figure 9-3: EWIS Risk Assessment	55
Figure 10-1: Overview, RBE, Requirement Cascade	58
Figure 10-2: Overview, V&V Flow	59
Figure 10-3: Overview, Simplified V & V Flow (refer to AP2296)	60
Figure 10-4: Overview, TLElecRD – Position in the TLxxRD Process	62

Figure 11-1: Simplified Electric Process covering a “Change” with related Responsibilities, Documents & Levels.....	64
Figure 12-1: Overview, Change Process Phases	65
Figure 15-1: Block Diagram (BD) illustration example	72
Figure 15-2: Functional Diagram (FD) illustration example	72
Figure 16-1: Principle Diagram (PD) illustration example	77
Figure 16-2: Extended Principle Diagram (EPD), logical wires of PD where wire numbers and other information are added to.....	78
Figure 16-3: Wiring Diagram (WD) example showing missing information of the EPD (Team B)...	79
Figure 16-4: ESD workflow.....	79
Figure 18-1: Example, ATA-921.....	87
Figure 18-2: Example, ATA-922.....	88
Figure 18-3: Example, ATA-923.....	89
Figure 18-4: Example of Parts, ATA-924	90
Figure 18-5: Example drawing, ATA-924	90
Figure 18-6: Example of parts, ATA-925.....	91
Figure 18-7: Example drawing, ATA-925	92
Figure 18-8: ATA-928 Example Full3D	93
Figure 18-9: ATA-928 Example picture drawing)	94
Figure 18-10: ATA-929 Example – Bundle with annotations.....	95
Figure 21-1: Non-Conformity 12063603 describes	108
Figure 21-2: Solution from DQN TH0072247	109
Figure 21-3: RDR flowchart (AP5342.2).....	110
Figure 21-4: DMU-Marking of Task 100 “Analysis ongoing”, Left: DQN, Right: RDR	110
Figure 21-5: Non-Conformity 12081826 description	112
Figure 21-6: Solution from Concession TH-003754560	113

Figure 22-1: Decision tree of classification (AP2243.1)	116
Figure 22-2: Classification criteria of a change	117
Figure 22-3: Four steps to approval	118
Figure 22-4: DOA Stamp.....	118
Figure 23-1: Overview, CCMD - Delta Definition.....	123

26.2 Tables

Table 3-1: Example for ATA Chapter 92 of A350 (refer to BAABI - A350).....	23
Table 6-1: Examples illustrating the difference between Certification and Attestation.....	41
Table 10-1: Overview of virtual circuit / Dummy codes for a circuit.....	56
Table 20-1: Overview of GTR types	99
Table 20-2: Different Types of Special Instructions.....	103
Table 21-1: Concession Suffixes.....	111

26.3 Abbreviation and Acronyms

A4A	Airlines for America
A/C	Aircraft
AA	Airworthiness Authority
ABD	Airbus Directive
AC	Advisory Circular
ACJ	Advisory Circular Joint
ACMR	Aircraft Configuration Management Rules
AD	Airworthiness Directive
ADAP	As-Designed and As-Planned
ADD	Aircraft Description Document
ADF	As DeFined

AIR	Aircraft Inspection Report
AM	Airbus Method
AMC	Acceptable Means of Compliance
AOPM	AIRBUS Operational Process Map
AOT	Alert Operators Transmission
AP	Airbus Processes
APS	Airbus PDM Services
ATA	Air Transport Association
AVEE	Air Vehicle, Engines and Equipment
AWL	Aircraft Wiring List
AWM	Aircraft Wiring Manual
BAABI	Basic Approved ATA Breakdown Index
BD	Block Diagram
BDF	Bases de Donn� Fabrication
BDH	Bases de Donn� Historique
BDO	Bases de Donn� Objectif
BILD	Bay Installation Layout Dossier
BIRD	Bay Installation Requirement Dossier
BM	Beanstandungsmeldung
BOM	Bill of Material
CA	Constituent Assembly
CAA	Civil Aviation Agency
CAIR	Constituent Assembly Inspection Report
CAM	Certification Airworthiness Manager
CAMO	Continuing Airworthiness Management Organization
CAP	Civil Aviation Publication
CASA	Construcciones Aeron�uticas S.A.

CC	Configuration Component
CCD	Catia Cadam Drafting
CCMD	Configuration Conformity Management & Documentation
CDF	Contractual Definition Freeze
cDTLL	contractual Document Type Level List
cDMU	Configured Digital Mock-Up Unit
CE	Chief Engineer
CI	Configuration Item
CIN	Change Identification Number
CILAT	Current Injection and Length Assumption Table
CIRCE	Conception Informatisée des Câbles Electriques
CM	Configuration Management
CMP	Croquis de Mise au Point
CoC	Center of Competence
CoE	Center of Excellence
CofA	Certificate of Airworthiness
CustoCoRe	Customization Cost Reduction
CQ	Customer Query
CR	Change Request
CRD	Comment Response Document
CRD	Cabin Requirement Dossier
CRI	Certification Review Item
CRS	Cost Repercussion Sheet
CS	Certification Specification
CSDM	Concession Support Decision Manual
CVE	Compliance Verification Engineer
DA	Deliverable Assembly

DAC	Design Assurance Coordinator
DAE	Designated Airworthiness Engineer
DAID	Design Assurance Interface Document
DAN	Design Assurance Nominees
DARE	Develop and Ramp-up Excellence
DAS	Design Assurance System
DASCCL	Design Assurance System Compliance Checklist
DCS	Designated Certification Specialist
DDP	Declaration of Design and Performance
DDS	Design Data Set
DG	Decision Gates
DnA	Design new Aircraft
DMU	Digital Mock-Up
DnE	Design new Electric
DOA	Design Organization Approval
DOID	Design Organization Interface Document
DOM	Design Organization Manual
DOORS	Dynamic Object Oriented Requirements System
DOS	Design Organization Signatories
DQC	Design Quality Checks
DQN	Design Query Note
DS	Design Solution
DTLL	Document Type Level List
EAPAS	Enhanced Airworthiness Program for Aging Systems
EASA	European Aviation Safety Agency
EC	European Commission (Regulations)
ECN	Engineering Change Note

EDA	European Defense Agency
EDP	Electronic Data Processing
EDR	Electrical Design Requirements
EIRD	Equipment Installation Requirement Dossier
EIS	Entry into Service
EMAR	European Military Airworthiness Requirement
ENGO	Engineering Occurrence
EPAC	Entity per Aircraft Change
EPD	Extended Principle Diagram
ERHCD	Electrical Routing Harness Concept Dossier
ERS	Embodiment Repercussion Sheet
ESD	Electrical Schematic Diagram
ESDCR	Electrical Schematic Diagram Common Repository
ESN	Electrical Structure Network
EWIS	Electrical Wiring Interconnection System
EZAP	Enhanced Zonal Analysis Procedures
e-TLB	Electronic Technical LogBook
FA	Functional Analysis
FAA	Federal Aviation Agency
FAR	Federal Aviation Requirements
FCD	Flight Conditions Document
FCO	Fulfill Customer Order
FD	Functional Diagram
FIN	Functional Item Number
FLT	Fiche de liaison technique
FTE	Flight Test Engineer
FTI	Flight Test Instruction

FTR	Flight test Request
FW	Further Work
GCO	Ground Test Requirement Change Order
GILDA	Gestion Intégrée de la Liasse Division Avions
GLM	General List of Modification
GM	Guidance Material
GRFC	Global Request for Change
GRM	Geometrical Reference Mock-Up
GTI	Ground Test Instructions
GTR	Ground Test Requirements
HdO	Head of Design Organization
HoV	Head of Version
HPD	Harness Pre-definition
ICA	Instructions of Continued Airworthiness
ICAO	International Civil Aviation Organization
ICS	Independent Check Sheet
IFE	In-Flight Entertainment
IR	Implementing Rule
ISB	Inspection Service Bulletin
ISO	In-Service Occurrence
IT	Information Technology
JAA	Joint Aviation Agency
JAR	Joint Aviation Requirements
LO	Link Object
MANO	Manufacturing Occurrence
MAP	Mise Au Point
MAS	Modification Approval Sheet

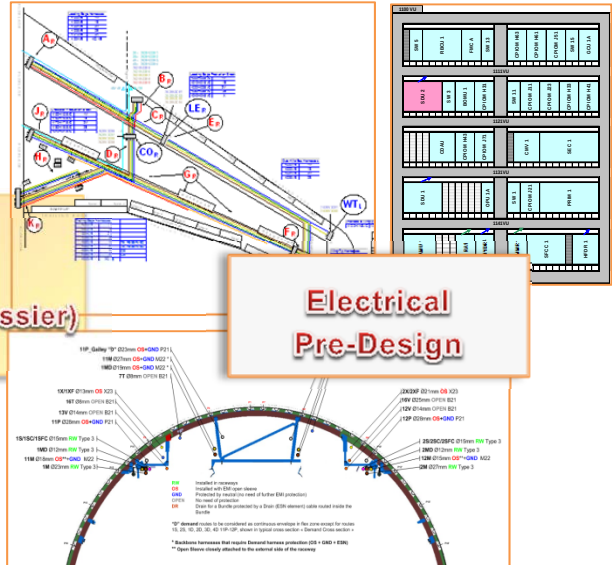
MBB	Messerschmitt-Bölkow-Blohm
MC	Means of Compliance
ME	Manufacturing Engineering
MG	Maturity Gate
MOA	Maintenance Organization Approval
MOD	Modification
MOM	Maintenance Organization Manual
MP	Modification Proposal
MSCN	Manufacturer Specification Change Notices
MSN	Manufacturer Serial Number
NAA	National Aviation Authority
NC	Non Conformity
NPA	Notice of Proposed Amendments
NSDW	Non Specific Design Work
NTO	No Technical Objection
OIT	Operator Information Telex
OSI	Open System Interconnection
PACC	Production Aircraft Test Certificate
PATCC	Production Aircraft Test Completion Certificate
PATM	Production Aircraft Test Manual
PATR	Production Aircraft Test Requirement
PD	Principle Diagram
PDM	Product Data Management
PLM	Product Life Cycle Management
PN	Part Number
PO	Production Organization
POA	Production Organization Approval

POE	Point of Embodiment
POM	Production Organization Manual
PRIMES	
PRS	Production Repercussion Sheet
PSP	Product Safety Process
PtF	Permit to Fly
PTS	Purchaser Technical Specification
QLB	Quality Logbook
RAS	Repair Approval Sheet
RBE	Requirement-Based Engineering
RCN	Requirement Change Note
RDR	Request for Design Rework
RE	Requirements Engineering
RFC	Request for Change
RFE	Rear Fuselage Empennage
RFW	Request for Work
RiC	Referential in Configuration
SAM	Space Allocation Model
SB	Service Bulletin
SCG	System Configuration Guide
SCN	Specification Change Note
SDW	Specific Design Work
SE	System Engineering
SEE XP	SEE Electrical Expert
SFAR	Special Federal Aviation Requirement
SFM	System Frontier Model
SI	Special Instruction

SIDP	System Installation Design Principles
SIRD	System Installation Requirement Dossier
SNS	Standard Numbering System
SRD	System Requirement Dossier
SSCI	System Structure Cabin Integrator
STC	Supplemental Type Certification
SWRD	System Wiring Requirement Dossier
TA	Technical Authority
TA	Technical Adaptation
TASKSY	Technisch Administratives Konstruktionssystem
TAp	Technical Approver
TC	Type Certification
TFE	Test Flight Engineer
TD	Technical Dossier
TDCA	Technical Delegation Capability Assessment
TDD	Technical Design Directive
TDU	Technical Description Unit
TLCabRD	Top Level Cabin Requirement Dossier
TLElecRD	Top Level Electrics Requirements Document
TLSysRD	Top Level System Requirement Dossier
TLSR	Top Level System Requirement
TLxxRD	Top-Level Requirements Dossier
VB	Virtual Bundle
VE	Virtual Equipment
VU	Virtual Unit
TRS	Technical Repercussion Sheet
V&V	Validation & Verification

WD	Wiring Diagram
WRS	Weight Repercussion Sheet
ZAMIZ	Zentrale Archivierung Mit Interaktivem Zugriff

ERHCD (Electrical Routing & Harness Concept Dossier)



Electrical Pre-Design



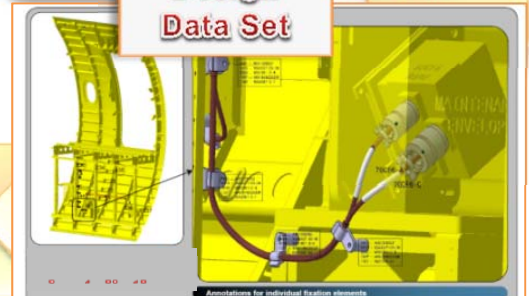
3D Definition Model



FULL 3D SET



Design Data Set



Virtual Units

EWIS (Electrical Wiring Interconnection System)



Manufacturing Harness and Installation